Numerical

Appunti studiabili e pratiche

Parte I: edizione per lo studio, con spiegazioni, esempi e approfondimenti su importance sampling e scelta della proposta.

Parte II: appunti unificati e practical completi. Le due edizioni contengono materiale complementare e sono conservate integralmente; alcuni argomenti ricorrono in entrambe.

Materiale personale non ufficiale. Le attribuzioni del corso seguono l'identificazione dell'utente.

Parte I

Edizione per lo studio

Teoria, esempi e approfondimenti.

Unofficial Unified Notes and Exercises

Numerical and Statistical Methods for Finance

Unofficial Unified Notes and Exercises

Study edition

About these notes

Original note

These notes are unofficial and have been reorganised for study purposes. Some parts of the original material have been omitted, while additional explanations, remarks, and examples have been added. In some cases, the order of the topics has also been changed to make the exposition more coherent.

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Chapter 1

Empirical data and estimators

This chapter introduces the empirical tools used to replace unknown distributional quantities with quantities computed from the observed sample. The first object is the empirical cdf, which counts how much observed mass lies below a given threshold.

DEFINITION – Definition

ECDF. For $y_1, \dots, y_n$ univariate i.i.d data $\hat{F}_n(y) = \frac{\#\{i : y_i \leq y\}}{n}$.

PROPOSITION – Proposition

ECDF

1. Non negative: $\hat{F}_n(y) \geq 0$
2. Increasing
3. Right continuous
4. $\hat{F}_n(y) = 1$ for all $y > y_{(n)}$, $y_{(n)} := \max y_i$
5. $\lim_{y \to -\infty} \hat{F}_n(y) = 0$ and $\lim_{y \to +\infty} \hat{F}_n(y) = 1$

PROPOSITION – Proposition

The ECDF has the following properties:

- $\hat{F}_n(y) = \frac{1}{n} \sum_i \mathbf{1}_{(-\infty, y]}(Y_i)$, Y_i r.v.
- If Y_i is r.v. Then $Z_i := \mathbf{1}_{(-\infty, y]}(Y_i) \sim Ber(p)$, $p = \mathbb{P}(Z_i = 1) = \mathbb{P}(Y_i \leq y) = F(y)$.
- If Y_i is r.v. Then $Z_i := \sum_{i=1}^n \mathbf{1}_{(-\infty, y]}(Y_i) \sim Bin(n, F(y))$, $\mathbb{E}(Ber) = p$, $Var(Ber) = p(1-p)$.
- $\mathbb{E}[\hat{F}_n(y)] = \mathbb{E}\left[\frac{1}{n} \sum_i \mathbf{1}_{(-\infty, y]}(Y_i)\right] = \frac{1}{n} \sum_i \mathbb{E}[\mathbf{1}_{(-\infty, y]}(Y_i)] = \frac{1}{n} \sum_i F(y) = F(y)$, hence $\hat{F}_n(y)$ is unbiased.
- $Var(\hat{F}_n(y)) = \frac{1}{n^2} \sum_i Var[\mathbf{1}_{(-\infty, y]}(Y_i)] = \frac{1}{n} F(y)(1 - F(y))$, so for $n \to +\infty$ $\frac{1}{n} Var(\hat{F}_n(y)) = 0$.
- $\hat{F}_n(y)$ is unbiased and $Var(\hat{F}_n(y)) \xrightarrow[n \to +\infty]{} 0$ hence $\hat{F}_n(y)$ is consistent.

DEFINITION – Definition

EPDF. $y_1, \dots, y_n$ univariate data, $\hat{p}_n : \mathbb{R} \to [0, 1]$ $\hat{p}_n(y) = \frac{1}{n} \sum_i \mathbf{1}_{\{y_i = y\}} = \frac{\#\{i : y_i = y\}}{n}$.

The empirical pdf is useful for discrete empirical calculations, but as a density estimator it has the following limitations.

PROBLEM – Problem

1. it is not continuous
2. It is zero almost everywhere
3. $\int_{-\infty}^{+\infty} \hat{p}_n(y)\, dy \neq 1$

Despite these problems, epdf is still useful for estimating empirical moments because expectations can be approximated by averages over the observed values.

- $\mathbb{E}[\Psi(Y)] = \int_{-\infty}^{+\infty} \Psi(y) f(y)\, dy$ if y_i are distinct (Hence $\hat{p}_n(y) = \frac{1}{n}\mathbf{1}_{\{y_i = y\}}$) Then $\hat{E}_n(\Psi(Y)) = \sum_{i=1}^n \hat{p}_n(y_i)\Psi(y_i) = \frac{1}{n}\sum_i \Psi(y_i)$.
- If $\Psi(y) = y$ Then $\hat{E}_n[Y] = \frac{1}{n}\sum_{i=1}^n y_i = \bar{y}$.
- Then $\widehat{Var}_n[Y] = \hat{E}_n[(Y - \hat{E}_n(Y))^2] = \hat{E}_n[(Y-\bar{y})^2] = \sum_{i=1}^n \hat{p}_n(y_i)(y_i - \bar{y})^2 = \frac{1}{n}\sum_{i=1}^n (y_i - \bar{y})^2$.

1.1 Histogram estimator

The histogram estimator improves on the empirical pdf by grouping the observations into bins and distributing the observed mass uniformly inside each bin.

Let $Y_1, \dots, Y_n$ data; f true density; n sample size; $[a, b]$ interval; m bins; $a = t_1 < \cdots < t_{m+1} = b$; $h_k = t_{k+1} - t_k$ width of the bin; $T_k = (t_k, t_{k+1}]$; $k = 1, \dots, m$; $n_k = \#\{i : Y_i \in T_k\}$ count of data in T_k.

DEFINITION – Definition

The histogram estim. $\hat{p}_H(y) = \sum_{k=1}^m \frac{n_k}{n \cdot h_k}\mathbf{1}_{\{y \in T_k\}} = \frac{n_\ell}{n h_\ell}$ if $y \in T_\ell$ for some $\ell = 1, \dots, m$.

The Area is $h_k \cdot \frac{n_k}{n \cdot h_k} = \frac{n_k}{n}$. Therefore the total area is $\sum_{k=1}^m \frac{n_k}{n} = 1$.

$\mathbb{E}[\hat{p}_H(y)] = \frac{1}{n h_\ell}\mathbb{E}(n_\ell) = \frac{1}{n h_\ell}\mathbb{E}\left[\sum_{i=1}^n \mathbf{1}_{\{Y_i \in T_\ell\}}\right] = \frac{1}{n \cdot h_\ell}\sum_{i=1}^n \mathbb{P}(Y_i \in T_\ell) = \frac{1}{n \cdot h_\ell}\sum_{i=1}^n (F(t_{\ell+1}) - F(t_\ell)) = \frac{F(t_{\ell+1}) - F(t_\ell)}{h_\ell}$.

$F(t_{\ell+1}) - F(t_\ell) = \int_{t_\ell}^{t_{\ell+1}} f(u)\, du$ hence $\mathbb{E}[\hat{p}_H(y)] = \frac{1}{h_\ell}\int_{t_\ell}^{t_{\ell+1}} f(u)\, du$.

Since $\frac{F(t_{\ell+1}) - F(t_\ell)}{h_\ell} \neq f(y)$ then the estimator $\hat{p}_H(y)$ is biased and bias $= \frac{F(t_{\ell+1}) - F(t_\ell)}{h_\ell} - f(y)$ generally.

$$\text{Moreover } Var(\hat{p}_H(y)) = \frac{1}{n^2 h_\ell^2} Var(n_\ell) = \frac{1}{n^2 h_\ell^2} Var\left(\sum_{i=1}^n \mathbf{1}(Y_i \in T_\ell)\right) = \frac{1}{n^2 h_\ell^2}\sum_{i=1}^n Var(\mathbf{1}(Y_i \in T_\ell))$$

$$= \frac{n}{n^2 h_\ell^2}\mathbb{P}(Y \in T_\ell)[1 - \mathbb{P}(Y \in T_\ell)] = \frac{1}{n h_\ell^2}[F(t_{\ell+1}) - F(t_\ell)]\{1 - (F(t_{\ell+1}) - F(t_\ell))\},$$

where $n_\ell \sim Bin(n, p_\ell)$, $p_\ell = \mathbb{P}(Y \in T_\ell)$.

REMARK – Remark

Bias-Variance Trade-off

Smaller bin	=	bias ↓	variance ↑
Larger bin	=	bias ↑	variance ↓

SOLUTION – Solution

We need more refined density estimator s.a. Kernel density est.

1.2 Quantile and QQ plot

After estimating densities, it is useful to compare the sample distribution with a chosen theoretical distribution through quantiles.

Histograms are useful for detecting broad feature s.a symmetry, outliers, kurtosis; but depend on the bin choice.

QQ plot help us to compare the empirical data with a theoretical distribution that we have chosen.

We compare $\left(F^{-1}\left(\frac{i-0.5}{n}\right), y_{(i)}\right)$, $i = 1, \dots, n$, on $\hat{x}$ theoretical quantiles, $\hat{y}$ empirical quantiles.

We can see heavy tail, asymmetry, but the main reason is comparison with a reference distribution.

1.3 Moments

Moments can be seen as functionals of the distribution, so the empirical strategy is to replace the unknown cdf with the empirical cdf.

Since we know $Y_1, \dots, Y_n$ and $\hat{F}_n(y) = \frac{\#\{i : Y_i \leq y\}}{n}$, the moments could be written as $\theta = t(F)$.

And since we don't know F we use the estimator $\hat{\theta} = t(\hat{F}_n)$ Then $\hat{\mu} = \frac{1}{n}\sum y_i = \bar{y}$, $\hat{\sigma} = \frac{1}{n}\sum (y_i - \bar{y}) \dots$ the estimators are unb. and efficient.

REMARK – Remark

We prefer to work with F saying $\mathbb{E}(\Psi(Y)) = \int \Psi(y)\, dF(y)$ ψ test function rather than f that could #.

Chapter 2

Resampling

Resampling techniques are used to approximate uncertainty when exact formulas for the sampling distribution are difficult to derive.

Resampling is useful when a statistic is easy to compute from data but its sampling distribution or other properties are hard to derive. (like bias, variance, se, MSE)

The two main resampling methods considered here are

- The Jackknife based on systematically deleting one obs. at a time
- the bootstrap based on repeatedly sampling with replacement from the observed data, or from fitted parametric model

2.1 Jackknife

The Jackknife recomputes the estimator after removing one observation at a time, so it studies how sensitive the statistic is to each data point.

Let $y = (y_1, \dots, y_n)$.

DEFINITION – Definition

Jackknife sample $y_{[i]} = (y_1, \dots, y_{i-1}, y_{i+1}, \dots, y_n)$ Thus, there are n jackknife sample of size $n-1$.

DEFINITION – Definition

Jackknife replicates $\hat{\theta}_1, \dots, \hat{\theta}_n$ are obtained applying the estimator to the jackknife sample.

EXAMPLE – Example

If $\theta = \mathbb{E}(Y)$ then $\hat{\theta} = \bar{y}$ then $\hat{\theta}_{[i]} = \frac{1}{n-1} \sum_{j \neq i} y_j$.

DEFINITION – Definition

Average Jackknife replicates $\hat{\theta}_{[\cdot]} = \frac{1}{n} \sum_{i=1}^{n} \hat{\theta}_{[i]}$.

DEFINITION – Definition

Jackknife estimate of bias $\widehat{bias}_{Jack} = (n-1)(\hat{\theta}_{[\cdot]} - \hat{\theta})$.

DEFINITION – Definition

Bias corrected jackknife estimator $\hat{\theta}_{Jack} = \hat{\theta} - \widehat{bias}_{Jack} = \hat{\theta} - (n-1)(\hat{\theta}_{[\cdot]} - \hat{\theta}) = n\hat{\theta} - (n-1)\hat{\theta}_{[\cdot]}$.

DEFINITION – Definition

Jackknife variance $\widehat{Var}_{Jack} = \frac{n-1}{n}\sum_{i=1}^{n}(\hat{\theta}_{[i]} - \hat{\theta}_{[\cdot]})^2$.

DEFINITION – Definition

Jackknife standard error $\hat{se}_{Jack} = \sqrt{\widehat{Var}_{Jack}}$.

DEFINITION – Definition

Jackknife estimate of mean squared error $\widehat{MSE}_{Jack} = \widehat{Var}_{Jack} + (\widehat{bias}_{Jack})^2$.

REMARK – Remark

If n is very small, in particular if $n = 1$, the Jackknife is not informative.

2.2 Bootstrap

The Bootstrap replaces an analytical sampling distribution with the empirical distribution of many recomputed statistics.

Bootstrap approximates a statistic's sampling distribution by repeatedly resampling with replacement from observed data and recomputing the statistic.

2.2.1 Non Parametric

In the non parametric version, we resample directly from the original empirical distr.

DEFINITION – Definition

Bootstrap sample: sample $y^* = (y_1^*, \dots, y_n^*)$ of size n drawn WITH REPLACEMENT from the original observed data.

DEFINITION – Definition

Bootstrap replication: $\hat{\theta}^*$ value of the statistic or estimator computed from a bootstrap sample.

DEFINITION – Definition

Bootstrap bias: $bias_{\hat{F}_n}(\hat{\theta}^*) = \mathbb{E}_{\hat{F}_n}(\hat{\theta}^*) - \hat{\theta}_{\hat{F}_n}$ is approximated by $\widehat{bias}_N = \hat{\theta}^*_{(\cdot)} - \hat{\theta}_{\hat{F}_n}$ where $\hat{\theta}^*_{(\cdot)} = \frac{1}{N}\sum_{i=1}^{N}\hat{\theta}^*_i$ and $\hat{\theta}_{\hat{F}_n}$ is the estimator computed on the original sample.

DEFINITION – Definition

$\widehat{Var}_N(\hat{\theta}^*) = \frac{1}{N-1}\sum_{i=1}^{N}(\hat{\theta}^*_{(i)} - \hat{\theta}^*_{(\cdot)})^2$ and $\hat{se}_N = \sqrt{\widehat{Var}_N}$.

2.2.2 Parametric

In the parametric version, we resample by simulating data from a fitted parametric model.

The unknown cdf is assumed to belong to a parametric family $\{F_\psi : \psi \in \Psi\}$.

1. Estimate the parameter $\hat{\psi}$ from the original sample
2. Set $\hat{F} = F_{\hat{\psi}}$

3. Generate bootstrap samples from $\hat{F}$ instead of from the empirical cdf $\hat{F}_n$
4. Compute the bootstrap replications and estimate bias and se.

So

1. Estimate ψ from the original sample
2. generate N bootstrap samples of size n from $F_{\hat{\psi}}$
3. compute $\hat{\theta}^*$ for each sample
4. estimate bias and standard error

2.3 Bootstrap confidence intervals

Once bootstrap replications are available, their empirical percentiles can be used to build confidence intervals.

DEFINITION – Definition

Bootstrap percentile. For a non parametric bootstrap if $\hat{\theta}^*_1, \dots, \hat{\theta}^*_N$ are bootstrap replications let $\hat{\theta}^{*(\alpha)}$ be the 100α th percentile of the ordered bootstrap values.

DEFINITION – Definition

The percentile interval is $(\hat{\theta}^{*(\alpha/2)}, \hat{\theta}^{*(1-\alpha/2)})$ This interval does not correct for bias and asymmetry. We need a more refined interval called BCa (bias corrected and accelerated).

DEFINITION – Definition

The bias correction term is $\hat{z}^* = \Phi^{-1}\left(\frac{\#\{i:\hat{\theta}^*_i<\hat{\theta}\}}{N}\right)$ where

- Φ^{-1} is the quantile function of normal distribution that converts the percentile position of $\hat{\theta}$ inside the bootstrap distribution into a standard normal score or z-score.
- the proportion $\frac{\#\{i:\hat{\theta}^*_i<\hat{\theta}\}}{N}$ tell us where the original estimate $\hat{\theta}$ lies inside the bootstrap distr.

- If $\frac{\#\{i:\hat{\theta}^*_i<\hat{\theta}\}}{N} = 0.5$ hence $\hat{z}^* = \Phi^{-1}(0.5) = 0$ then $\hat{\theta}$ is at the centre of the bootstrap distr.
- If $\frac{\#\{i:\hat{\theta}^*_i<\hat{\theta}\}}{N} > 0.5$ Then $\hat{z}^* > 0$ $\hat{\theta}$ lies in the right of the centre of the distribution.
- If $\frac{\#\{i:\hat{\theta}^*_i<\hat{\theta}\}}{N} < 0.5$ then $\hat{z}^* < 0$ $\hat{\theta}$ lies in the left of the centre of the distribution.
- If $\frac{\#\{i:\hat{\theta}^*_i<\hat{\theta}\}}{N}$ differs substantially from .5 the bootstrap estimator $\hat{\theta}^*$ is not central at original estimate suggesting that the original estimator may be biased.

Thus $\hat{z}^*$ shifts the percentile levels in order to correct for lack of centring of the bootstrap dist. around $\hat{\theta}$.

DEFINITION – Definition

The acceleration term is $\hat{a} = \frac{\sum_{i=1}^n (\hat{\theta}_{[\cdot]} - \hat{\theta}_{[i]})^3}{6\left[\sum_{i=1}^n (\hat{\theta}_{[\cdot]} - \hat{\theta}_{[i]})^2\right]^{3/2}}$ where $\hat{\theta}_{[\cdot]} = \frac{1}{n}\sum_{i=1}^n \hat{\theta}_{[i]}$.

The role of $\hat{a}$ is measure the asymetry or skewness more precisely the acceleration term is computed from the jackknife replication so it reflects how the estimator changes when one observation is deleted at a time:

- if $\hat{a} \simeq 0$ then the estimator react roughly symmetric way to these deletions
- if $\hat{a} \neq 0$ then the the estimator react in a asymmetric or nonlinear way

Thus $\hat{a}$ adjust the shape of sampling dist. In particular, when the estimator has a skewed dist. it modifies the lower and upper percentile levels asymmetrically.

DEFINITION – Definition

The BCa interval is $(\hat{\theta}^{*(\alpha_1)}, \hat{\theta}^{*(\alpha_2)})$ where $\alpha_1 = \Phi\left(\hat{z}^* + \frac{\hat{z}^* + z_{\alpha/2}}{1+\hat{a}(\hat{z}^* + z_{\alpha/2})}\right)$, $\alpha_2 = \Phi\left(\hat{z}^* + \frac{\hat{z}^* + z_{1-\alpha/2}}{1+\hat{a}(\hat{z}^* + z_{1-\alpha/2})}\right)$.

So in summary $\hat{z}^*$ corrects for bias, $\hat{a}$ corrects for acceleration. If both $\hat{z} = 0$ and $\hat{a} = 0$ BCa interval is the percentile interval.

REMARK – Remark

We can define ϕ as a generic CDF and we can use t-Student quantiles.

REMARK – Remark

If the sample size is n, the number of distinct bootstrap samples is $\binom{2n-1}{n-1} = \frac{(2n-1)!}{n!(n-1)!}$. Thus $N \leq \binom{2n-1}{n-1}$, in practice $N \geq 400$ is sufficient for estimating s.e. while for bootstrap confidence one usually takes $N \geq 2000$.

Chapter 3

Simulating Data

Simulation starts from the need to generate random inputs that drive the stochastic model.

PROBLEM – Problem

We need to simulate a trajectory of a stochastic process; therefore we need a way to simulate its innovations.

EXAMPLE – Example

Simulate $(X_t)_t$ of $X_t = X_{t-1} + \bar{Z}_t$ where $\bar{Z}_t \sim N(0,1)$ is the innovation.

SOLUTION – Solution

The general strategy is to start from uniforms and then transform them until the desired distribution is obtained. We will see that a solution involves generating uniform r.v. on $[0,1]$, then transforming these uniforms into known variables, and finally generating sample from a chosen target distribution that can be written as $\pi(x) = \frac{\gamma(x)}{Z}, \quad \gamma(x) > 0, \quad Z = \int_{-\infty}^{+\infty} \gamma(x)\,dx$ where γ is an unnormalized density and Z is the normalising constant. In many applications γ is known but Z is difficult or impossible to compute explicitly.

3.1 Generating uniform random variates

The first building block of simulation is a generator for numbers that behave like independent uniforms.

PROBLEM – Problem

Exact real valued random numbers are difficult to generate;

SOLUTION – Solution

generating pseudo-uniform random numbers.

DEFINITION – Definition

A sequence of pseudo-random $u_1, u_2, \ldots$ is a deterministic sequence whose space sequence mimic those of random numbers from a desired distribution.

A useful generator should be fast, reliable and its output should look random. The most popular method is Linear Congruential Generator (LCG).

DEFINITION – Definition

Linear Congruential Generator: $x_{n+1} = (ax_n + b) \bmod m$ where x_0 is the seed, m is the modulus, b the shift.

PROPOSITION – Proposition

If $b = 0$ LCG is multiplicative; if $b \neq 0$ it is mixed.

DEFINITION – Definition

Pseudo-uniform are $u_n = \frac{x_n}{m}, \qquad u_n \in [0,1]\ \forall n$ where $n \in \mathbb{N}$ and $x_0, a, b \in \{0, \dots, m-1\}$.

PROPOSITION – Proposition

The sequence $(u_n)_{n\geq 0}$ is periodic with period $T \leq m$. Ideally one chooses parameters giving the full period $T = m$. The choice of m is linked to precision for example $m \simeq 2^{32}$ for single precision.

PROBLEM – Problem

If $a \equiv b \mod m$ hence $a - b = km$, $k \in \mathbb{N}$ then LCG's poles and plotting (u_k, u_{k+1}) we may see a lattice structure.

SOLUTION – Solution

Modern implementation typically use more sophisticated generator s.a Mersenne Twister but that's beyond the scope of this course.

In this course we properly initialize generators s.a rng.uniform $(0,1)$.

```
rng = np.random.default_rng(10)
u = rng.uniform(0,1)
v = np.random.uniform(0,1)
```

3.2 Transformation methods

Once uniforms are available, transformation methods turn them into variables with the desired distribution.

Given a pseudo-uniform generator we can write a target r.v. Y as a transformation of a variable that is easy to simulate s.a uniform hence $Y = g(U), \qquad U \sim Uni(0,1)$.

THEOREM – Theorem

Probability integral Transform: Let X be r.v with cdf F_X and let $U \sim Uni(0,1)$ Then $F_X^{-1}(U)$ has the same distribution of X.

Proof

Let $Y = F_X^{-1}(U)$; $\mathbb{P}(Y \leq y) = \mathbb{P}(F_X^{-1}(U) \leq y) = \mathbb{P}(U \leq F_X(y)) = F_X(y)$. ■

The following algorithm summarizes inverse transform sampling for continuous random variables.

Algorithm 1: Inverse transform sampling continuous r.v.

Algorithm

1. Input the number of samples n and cdf F_X
2. for $i = 1, \ldots, n$ do
3. Generate $U_i \sim Uni(0, 1)$
4. Set $X_i = F_X^{-1}(U_i)$
5. end for
6. Output: $X_1, \ldots, X_n$ with cdf F_X

EXAMPLE – Example

Expo dist. $X \sim Exp(\lambda)$ $F_X = 1 - e^{-\lambda x}$, $U = 1 - e^{-\lambda X}$, $1 - U = e^{-\lambda X}$, $x = -\frac{\ln(1-U)}{\lambda}$.
So $F_X^{-1}(U) = -\frac{1}{\lambda}\ln(1-U)$ so we 1) Generate U_i Then set $X_i = -\frac{1}{\lambda}\log(1-U_i)$ or $X_i = -\frac{1}{\lambda}\log(U_i)$.

REMARK – Remark

Since $1 - U \overset{d}{=} U$ we may use $x = -\frac{1}{\lambda}\ln(U)$.

PROBLEM – Problem

In the discrete case F_X is not invertible due to jumps.

SOLUTION – Solution

We could use the generalized inverse cdf.

DEFINITION – Definition

Generalized inverse cdf: $F_X^{-1}(u) = \inf\{x \in \mathbb{R} : F_X(x) \geq u\}$ with u from $U \sim Uni(0, 1)$.

For discrete random variables, the most direct approach is to search the first cdf value that exceeds the simulated uniform.

Algo: Naive discrete transform

Algorithm

1. Input n, F
2. For $i = 1, \ldots, n$ do
3. $U_i \sim Uni(0, 1)$
4. find $k_i = \min\{k : F_X(x_k) \geq U_i\}$
5. Set $X_i = x_{k_i}$

PROBLEM – Problem

This algo is NOT efficient. To find S_k it has to search for $F_X(s_1), \dots, F_X(s_n)$. If $S = \{s_1, \dots, s_n\}$ in the worst case the traversal cost is almost $O(nk)$ for n simulation.

SOLUTION – Solution

Let's work with ordered samples.

Sorting the uniforms makes the discrete inverse transform more efficient because the cdf levels are crossed only once.

Algo 2. Inverse Transform sampling discrete r.v.

Algorithm

1. Input the number of sample n, the pmf $\pi_1, \dots, \pi_k$ and sorted uniforms $U^{(1)} \leq \dots \leq U^{(n)}$
2. set $m = 1$, $q = \pi_1$
3. For $i = 1, \dots, n$ do
4. while $q < U^{(i)}$ do
5. Set $m = m + 1$ and $q = q + \pi_m$
6. endwhile
7. Set $X_i = s_m$
8. end for.

EXAMPLE – Example

Poisson $\mathbb{P}(X = K) = \frac{\lambda^K e^{-\lambda}}{K!}, \qquad K = 0, 1, \dots$

- Sample $U \sim Uni(0, 1)$
- Set $m = 0$, $q = \mathbb{P}(X = 0) = e^{-\lambda}$
- while $q < U$ update $m = m + 1$ and $q = q + \frac{\lambda^m e^{-\lambda}}{m!}$
- return m

PROBLEM – Problem

For some distribution F_X^{-1} is not available in a closed form s.a. standard gaussian.

SOLUTION – Solution

Use other transformation.

When the inverse cdf is not available, a different transformation can be used.

Algo 3: Pseudocode for transformation method

Algorithm

1. Input: the number of samples n.
2. for $i = 1, \ldots, n$ do
3. Generate X_i from a distribution we can simulate
4. Set $Y_i = g(X_i)$
5. end for

REMARK – Remark

If $X \sim \pi(x)$ and $Y = g(x)$ then in one dimension $p_Y(y) = \pi(g^{-1}(y)) \left| \frac{\partial g^{-1}(y)}{\partial y} \right|$. For a n dimensional transformation $p_Y(y) = \pi(g^{-1}(y)) |\det J_{g^{-1}}(y)|$ where $J_{g^{-1}}$ is the Jacobian matrix of an inverse map.

EXAMPLE – Example

Box Muller Transformation. If $U_1, U_2 \sim Uni(0,1)$ independent Then $Z_1 = \sqrt{-2\log U_1}\cos(2\pi U_2)$, $Z_2 = \sqrt{-2\log U_1}\sin(2\pi U_2)$ are independent standard normal r.v.

Chapter 4

Rejection Sampling

4.1 Introduction to rejection sampling

PROBLEM – Problem

π does not admit inverse cdf and cannot be written as a transformation of another pdf.

As said before, the target distribution can be written as $\pi(x) = \frac{\gamma(x)}{z}$, $\gamma(x) > 0$, $z = \int_{-\infty}^{+\infty} \gamma(x)\,dx$, where γ is an unnormalised density and z is the normalising constant. We are in the case in which γ is known, but z is unknown.

SOLUTION – Solution

Rejection sampling is based on a simple geometric idea. Instead of sampling directly from the target density, we look at the region under the graph of the unnormalised density $\gamma(x)$.

If we can generate points uniformly in this region, then the horizontal coordinates of those points automatically follow the desired target distribution. This is possible thanks to the Fundamental Theorem of Simulation.

EXAMPLE – Example

We want to simulate $\gamma(x) = e^{-x^2/2}$. We can easily compute every $\gamma(x)$ for $x \in \mathbb{R}$, but $z = \int_{-\infty}^{+\infty} \gamma(x)\,dx$ is non-trivial. So we can plot $\gamma(x)$ and complete the simulation by sampling uniformly under its graph. Then the horizontal coordinates of those points are distributed according to $\pi(x) = \frac{\gamma(x)}{z}$.

THEOREM – Theorem

Fundamental Theorem of Simulation.

Let π be the target density and suppose that it can be written as $\pi(x) = \frac{\gamma(x)}{z}$, where γ is an unnormalised density and $z = \int_{-\infty}^{+\infty} \gamma(x)\,dx$ is the normalising constant.

Assume that $\gamma(x) \geq 0$ and define $A = \{(x,u) \in \mathbb{R}^2 : 0 \leq u \leq \gamma(x)\}$. If (X,U) is uniformly distributed on A, then the marginal density of X is $\pi(x) = \frac{\gamma(x)}{z}$.

Proof

The area of A is $|A| = \int_{-\infty}^{+\infty} \gamma(x)\,dx = z$. Since (X,U) is uniformly distributed on A, its joint density is $q(x,u) = \frac{1}{|A|} = \frac{1}{z}$, with $(x,u) \in A$. For fixed x, the vertical section has length $\gamma(x)$. Hence $q_X(x) = \int_0^{\gamma(x)} \frac{1}{z}\,du = \frac{\gamma(x)}{z} = \pi(x)$. Therefore the marginal density of X is exactly the target density π. ■

REMARK – Remark

The same result can also be proved by using the factorisation of the joint density into conditional density times marginal density. Since (X, U) is uniformly distributed on A, its joint density is $q(x, u) = \frac{1}{|A|} = \frac{1}{z}$, for $(x, u) \in A$.

Now fix x. Conditional on $X = x$, the variable U is uniformly distributed on the vertical section $\{u : 0 \le u \le \gamma(x)\}$, whose length is $\gamma(x)$. Therefore $q(u \mid x) = \frac{1}{\gamma(x)}$. Using the identity $q(x, u) = q(u \mid x) q_X(x)$, we obtain $q(x, u) = \frac{1}{\gamma(x)} q_X(x)$.

But we also know that $q(x, u) = \frac{1}{z}$. Comparing the two expressions gives $\frac{q_X(x)}{\gamma(x)} = \frac{1}{z}$. Hence $q_X(x) = \frac{\gamma(x)}{z} = \pi(x)$.

So this proof is equivalent to the previous one: instead of integrating the joint density over u, it first writes the joint density as conditional density times marginal density.

4.2 The rectangle method

A first simple version of rejection sampling is the rectangle method, which is applicable only if π is contained in $[a, b]$. Thus, assume that $\pi(x) \le p^*$, for all $x \in [a, b]$.

If $(X', U') \sim \text{Unif}\big([a, b] \times [0, p^*]\big)$, and we accept X' whenever $U' \le \pi(X')$, then the accepted values have density π.

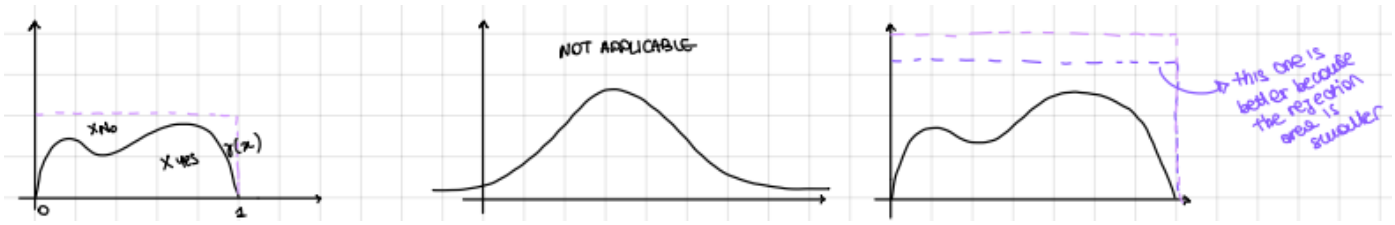


Figure 4.1: The rectangle method can be inefficient when the box leaves a large amount of empty area above the target.

4.3 The real rejection sampler

The box method may be inefficient. A better idea is to use a proposal density q, whose shape is closer to the target.

DEFINITION – Definition

Let q be a proposal density. Assume there exists a constant $M > 0$ such that $\gamma(x) \le Mq(x)$ for all x. Then Mq is called an envelope for γ.

REMARK – Remark

If M is chosen such that $\pi(x) \le Mq(x)$, then M is called scaling factor.

REMARK – Remark

Using Mq is often much better than using a rectangle, because the envelope can adapt to the shape of the target density.

If π has support contained in $[a, b]$, then the rectangle method is a particular case of the envelope method, obtained by taking $q(x) = \frac{1}{b-a} \mathbf{1}_{[a,b]}(x)$, that is, the uniform density on $[a, b]$.

In this case, $Mq(x) = \frac{M}{b-a} \mathbf{1}_{[a,b]}(x)$, so the envelope is constant on $[a, b]$. If we choose $M = (b - a)p^*$, then $Mq(x) = p^*$ for all $x \in [a, b]$, and therefore the envelope coincides exactly with the rectangle used in the rectangle method.

In the general envelope method, instead, $q(x)$ is not necessarily constant. Therefore $Mq(x)$ is not flat: it has the same shape as the proposal density q. Hence it can be high where the target density is high and low where the target density is low. For this reason, it may leave much less empty area above the target than the rectangle method, as shown in the figure.

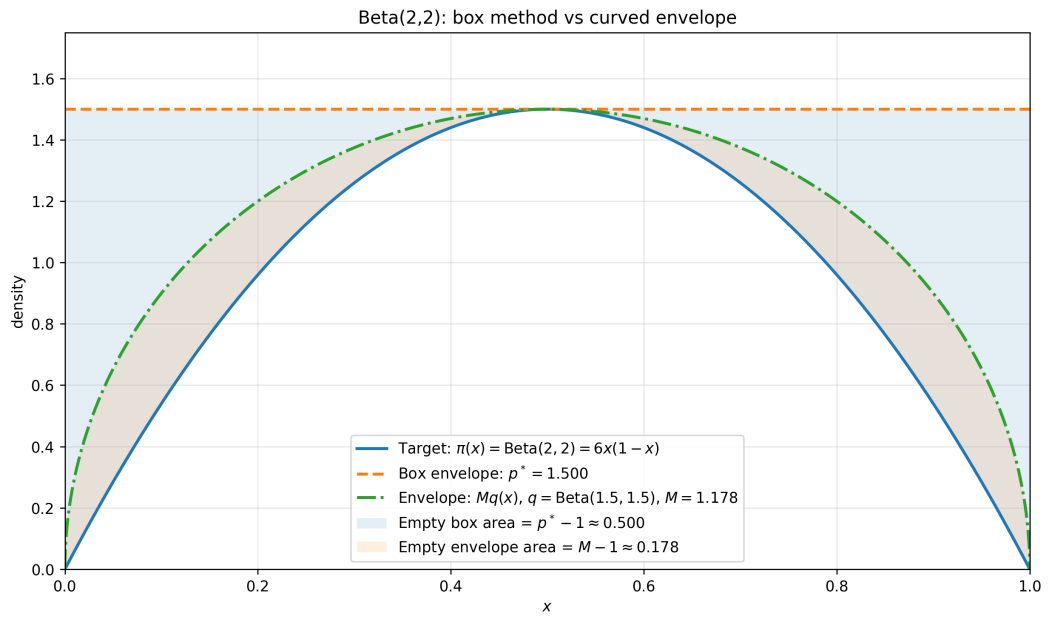


Figure 4.2: Box method and envelope method for a Beta$(2,2)$ target density GENERATED BY CHATGPT.

Algorithm: Rejection Sampling for continuous random variables

Algorithm

1. Input: the number of iterations n and scaling factor M.
2. For $i = 1, \dots, n$ do:
3. Generate $X_i' \sim q(x')$.
4. Generate $U \sim \text{Unif}(0,1)$.
5. If $U \leq \frac{\pi(X_i')}{Mq(X_i')}$, then accept X_i'.
6. Return the accepted samples.

REMARK – Remark

If we want n accepted samples, we can use a while loop with $i < n$, and update $i = i + 1$ only when a sample is accepted.

REMARK – Remark

For an unnormalised target, where π is not known, we can substitute $\pi(x)$ with $\gamma(x)$. In this case M has to be chosen such that $Mq(x)$ is an envelope for $\gamma(x)$, that is $\gamma(x) \leq Mq(x)$ for all x.

4.4 Acceptance rate

PROPOSITION – Proposition

For a normalised density $\pi(x)$, with M fixed such that $\pi(x) \leq Mq(x)$ for all x, the acceptance rate is $\hat{a} = \frac{1}{M}$.

Proof

The acceptance rate is $\hat{a} = \mathbb{P}(\text{accept})$. By the rejection sampling rule, $\hat{a} = \mathbb{P}\left(U \leq \frac{\pi(X')}{Mq(X')}\right)$.

Conditioning on $X' = x'$, we have $\mathbb{P}(\text{accept} \mid X' = x') = \mathbb{P}\left(U \leq \frac{\pi(x')}{Mq(x')}\right)$. Since $U \sim \text{Unif}(0,1)$, this gives $\mathbb{P}(\text{accept} \mid X' = x') = \int_0^{\frac{\pi(x')}{Mq(x')}} du = \frac{\pi(x')}{Mq(x')}$.

Therefore, $\mathbb{P}(\text{accept}) = \int \mathbb{P}(\text{accept} \mid X' = x')q(x')\,dx'$. Hence $\mathbb{P}(\text{accept}) = \int \frac{\pi(x')}{Mq(x')}q(x')\,dx' = \frac{1}{M}\int \pi(x')\,dx' = \frac{1}{M}$. ■

PROPOSITION – Proposition

For an unnormalised density $\gamma(x)$, where $\pi(x)$ is not known and $z = \int \gamma(x)\,dx$, with M chosen such that $\gamma(x) \leq Mq(x)$ for all x, the acceptance rate is $\hat{a} = \frac{z}{M}$.

Proof

Using the unnormalised acceptance probability, we have $\hat{a} = \mathbb{P}(\text{accept}) = \int \mathbb{P}(\text{accept} \mid X' = x')q(x')\,dx'$.

Since $\mathbb{P}(\text{accept} \mid X' = x') = \frac{\gamma(x')}{Mq(x')}$, we obtain $\hat{a} = \int \frac{\gamma(x')}{Mq(x')}q(x')\,dx' = \frac{1}{M}\int \gamma(x')\,dx' = \frac{z}{M}$. ■

PROPOSITION – Proposition

There is a difference between the pointwise acceptance probability $a(x) = \frac{\pi(x)}{Mq(x)}$ and the acceptance rate $\hat{a}$. The first one is the probability of accepting a proposed value when the proposal is exactly x, while the second one is the average acceptance probability over all possible proposals. Therefore, $\mathbb{E}[a(X')] = \hat{a}$.

REMARK – Remark

In the unnormalised case, γ does not integrate to one. Since $\gamma(x) \leq Mq(x)$, by integrating both sides we get $z = \int \gamma(x)\,dx \leq \int Mq(x)\,dx = M$. Hence $M \geq z$, and the acceptance rate is $\hat{a} = \frac{z}{M}$.

PROPOSITION – Proposition

The number of proposals needed until the first acceptance is geometrically distributed.

Each proposed sample is either accepted or rejected. Therefore each proposal can be interpreted as a Bernoulli trial, where success means acceptance.

Let T be the number of proposals needed until the first accepted sample. In the normalised case, the success probability is $p = \frac{1}{M}$, hence $T \sim \text{Geom}\left(\frac{1}{M}\right)$.

In the unnormalised case, the success probability is $p = \frac{z}{M}$, hence $T \sim \text{Geom}\left(\frac{z}{M}\right)$.

REMARK – Remark

The constant M must be large enough to make $Mq(x)$ an envelope for the target.

In the normalised case, since π and q are both densities, from $\pi(x) \leq Mq(x)$ we get $1 = \int \pi(x)\,dx \leq \int Mq(x)\,dx = M$. Hence $M \geq 1$, and therefore $0 < \frac{1}{M} \leq 1$.

This is necessary because the parameter of a geometric distribution is a probability, so it cannot be greater than 1.

The case $M = 1$ is the ideal case. It means that $Mq(x) = q(x)$ covers $\pi(x)$, and since both π and q integrate to one, this can happen only when $q(x) = \pi(x)$ almost everywhere. In that case every proposal is accepted and $T = 1$ with probability one.

Usually $M > 1$. Then $\frac{1}{M} < 1$, so some proposals are rejected. The larger M is, the smaller the acceptance rate $\hat{a} = \frac{1}{M}$, and the larger the expected number of proposals needed before the first acceptance.

REMARK – Remark

The geometrically distributed random variable is not the number of accepted samples in a fixed number of iterations. It is the number of proposals needed until the first accepted sample.

If instead we fix the total number of proposals, say n, then the number of accepted samples is binomially distributed with parameters n and $\hat{a}$.

4.5 Distribution of accepted values

THEOREM – Theorem

The distribution of the accepted values in the rejection sampling algorithm is the target distribution π.

This result is true both for a normalised target density and for an unnormalised target density. Here we give the version for the unnormalised case, where $\pi(x) = \frac{\gamma(x)}{z}$.

Proof

Let A denote the event that X' is accepted. Then $\mathbb{P}(X' \le x \mid A) = \frac{\mathbb{P}(X' \le x, A)}{\mathbb{P}(A)}$.

For an unnormalised density γ, the denominator is $\mathbb{P}(A) = \frac{z}{M}$, as shown in the acceptance rate proposition.

Now consider the numerator. Since $A = \left\{U' \le \frac{\gamma(X')}{Mq(X')}\right\}$, we have

$$\mathbb{P}(X' \le x, A) = \mathbb{E}_{(X',U')}\left[\mathbf{1}_{\{X' \le x\}}\mathbf{1}_A\right] = \mathbb{E}_{X'}\left[\mathbf{1}_{\{X' \le x\}} \underbrace{\mathbb{E}_{U'}\left[\mathbf{1}_A \mid X' = x'\right]}_{\mathbb{P}_{U'}(A \mid X' = x') = \frac{\gamma(x')}{Mq(x')}}\right] = \int_{-\infty}^{+\infty} q(x')\mathbf{1}_{\{x' \le x\}} \frac{\gamma(x')}{Mq(x')}\, dx'$$

$$= \frac{1}{M}\int_{-\infty}^{x} \gamma(x')\, dx'.$$

Hence $\frac{\text{NUM}}{\text{DEN}} = \left(\frac{1}{M}\int_{-\infty}^{x} \gamma(x')\, dx'\right)\left(\frac{z}{M}\right)^{-1} = \frac{\int_{-\infty}^{x} \gamma(x')\, dx'}{z} = \int_{-\infty}^{x} \frac{\gamma(x')}{z}\, dx' = \int_{-\infty}^{x} \pi(x')\, dx'$.

Therefore $\mathbb{P}(X' \le x \mid A) = \int_{-\infty}^{x} \pi(x')\, dx'$, which means that the accepted values have density π. ■

4.6 Discrete random variables

Given a target pmf $\boldsymbol{\pi} = (\pi_1, \ldots, \pi_K)$ on the state space $\{s_1, \ldots, s_K\}$, and a proposal pmf $\mathbf{q} = (q_1, \ldots, q_K)$, assume that there exists $M > 0$ such that $\pi_j \le Mq_j$, for every $j = 1, \ldots, K$.

Then the discrete rejection sampler works as follows.

Algorithm: Rejection Sampling for discrete random variables

Algorithm

1. Input: the number of iterations n, the target pmf $\boldsymbol{\pi}$, the proposal pmf $\mathbf{q}$, and the scaling factor M.
2. For $i = 1, \ldots, n$ do:
3. Sample an index $J \in \{1, \ldots, K\}$ with probability $\mathbb{P}(J = j) = q_j$.
4. Set $X_i' = s_J$.
5. Generate $U \sim \text{Unif}(0, 1)$.
6. If $U \le \frac{\pi_J}{Mq_J}$, then accept $X_i' = s_J$.

7. Return the accepted samples.

REMARK – Remark

This algorithm is different from the discrete inverse transform method. In the inverse transform method, we generate uniforms and compare them with the cumulative probabilities of the target distribution. For this reason, sorting the uniforms can improve efficiency, because the cumulative probabilities are crossed only once.

In discrete rejection sampling, instead, we do not invert the cdf. We first propose a state from the proposal pmf $\mathbf{q}$, and then we accept or reject this proposed state using an independent uniform random variable. Therefore there is no need to sort the uniforms.

We use two different indices to avoid ambiguity. The index i counts the iterations of the algorithm, while the random index J identifies the proposed state s_J. Thus i tells us which proposal attempt we are considering, while J tells us which value has been proposed at that attempt.

EXAMPLE – Example

Toy example generated.

Consider the discrete state space $\{s_1, s_2, s_3, s_4\} = \{10, 20, 30, 40\}$.
The target pmf is $\boldsymbol{\pi} = (0.10, 0.20, 0.50, 0.20)$, and the proposal pmf is $\mathbf{q} = (0.25, 0.25, 0.25, 0.25)$. We need M such that $\pi_j \le M q_j$, for $j = 1, \dots, 4$.
Equivalently, $M \ge \max_{j=1,\dots,4} \frac{\pi_j}{q_j}$. From the first table, the maximum value of $\frac{\pi_j}{q_j}$ is 2, hence we choose $M = 2$. If the proposed state is s_J, the acceptance probability is $\frac{\pi_J}{M q_J}$. Since $M = 2$, the acceptance probabilities are those in the second table.
Now suppose we run 10 iterations of the algorithm. At each iteration i, we first sample a random index J from the proposal distribution $\mathbf{q}$. Then we set $X_i' = s_J$, generate $U \sim \text{Unif}(0,1)$, and accept if $U \le \frac{\pi_J}{M q_J}$. The accepted values are 30, 10, 30, 20, 40, 30, 20. Thus, in this toy run, we obtained 7 accepted samples out of 10 proposals. The index i counts the iteration of the algorithm, while the random index J identifies which state has been proposed.
Therefore, after rejection sampling, we can use the accepted values to approximate expectations under π. For example, $\mathbb{E}_\pi[X]$ can be estimated by $\frac{1}{N_{\text{acc}}} \sum_{k=1}^{N_{\text{acc}}} X_k^{\text{acc}}$.

J	s_J	π_J	$\frac{\pi_J}{q_J}$
1	10	0.10	0.40
2	20	0.20	0.80
3	30	0.50	2.00
4	40	0.20	0.80

In this toy example, the accepted values are 30, 10, 30, 20, 40, 30, 20.

J	s_J	$\frac{\pi_J}{M q_J}$
1	10	0.20
2	20	0.40
3	30	1.00
4	40	0.40

i	J	X_i'	U	Dec.
1	3	30	0.62	A
2	1	10	0.15	A
3	4	40	0.70	R
4	3	30	0.91	A
5	2	20	0.35	A
6	2	20	0.55	R
7	4	40	0.22	A
8	1	10	0.21	R
9	3	30	0.99	A
10	2	20	0.39	A

Thus $\frac{1}{7}(30+10+30+20+40+30+20) = \frac{180}{7} \simeq 25.71$. The true value is $\mathbb{E}_\pi[X] = 10 \cdot 0.10 + 20 \cdot 0.20 + 30 \cdot 0.50 + 40 \cdot 0.20 = 28$. With only 7 accepted samples the estimate is rough, but as the number of accepted samples increases, the empirical average converges to the target expectation.

4.7 Optimal rejection sample

For a given target π, or unnormalised target γ, and proposal density q, the optimal scaling constant is the smallest value of M such that the envelope condition holds.

For a normalised target, this means $M^* = \sup_x \frac{\pi(x)}{q(x)}$. For an unnormalised target, this means $M^* = \sup_x \frac{\gamma(x)}{q(x)}$.

REMARK – Remark

To maximise the acceptance rate, M should be chosen as small as possible. Indeed, in the normalised case the acceptance rate is $\hat{a} = \frac{1}{M}$, while in the unnormalised case it is $\hat{a} = \frac{z}{M}$. Therefore, the optimal choice is the smallest M satisfying $\gamma(x) \leq Mq(x)$ for all x, namely $M^* = \sup_x \frac{\gamma(x)}{q(x)}$.

4.8 Gaussian target with Cauchy proposal

EXAMPLE – Example

Consider the unnormalised target $\gamma(x) = e^{-x^2/2}$, so that the target density is proportional to a standard Gaussian density. Take as proposal the Cauchy density $q(x) = \frac{1}{\pi(1+x^2)}$. Since we can sample from q, rejection sampling is feasible.

We need to find M such that $\gamma(x) \leq Mq(x)$ for all x. The smallest valid constant is $M^* = \sup_x \frac{\gamma(x)}{q(x)}$.

Now $\frac{\gamma(x)}{q(x)} = \pi(1+x^2)e^{-x^2/2}$. Define $R(x) = \pi(1+x^2)e^{-x^2/2}$. To maximise R, it is enough to maximise $\log R$, where $\log R(x) = \log \pi + \log(1+x^2) - \frac{x^2}{2}$. Then $\frac{d}{dx} \log R(x) = \frac{2x}{1+x^2} - x$.

Setting this equal to zero gives $\frac{2x}{1+x^2} - x = 0 \iff x\left(\frac{2}{1+x^2} - 1\right) = 0 \iff x(x^2-1) = 0$. Thus the critical points are $x = 0, \pm 1$.

Comparing the values, the maximum is attained at $x = \pm 1$, while $x = 0$ is a minimum. Therefore $M^* = R(\pm 1) = \pi(1+1)e^{-1/2} = 2\pi e^{-1/2}$.

The rejection sampler is therefore:

Algorithm

1. Draw $X \sim q$, where $q(x) = \frac{1}{\pi(1+x^2)}$.
2. Draw $U \sim \text{Unif}(0,1)$.
3. Accept X if $U \leq \frac{\gamma(X)}{M^* q(X)} = \frac{1+X^2}{2} e^{(1-X^2)/2}$.

Since $z = \int_{-\infty}^{+\infty} e^{-x^2/2}\, dx = \sqrt{2\pi}$, the corresponding acceptance rate is $\frac{z}{M^*} = \frac{\sqrt{2\pi}}{2\pi e^{-1/2}} = \frac{e^{1/2}}{\sqrt{2\pi}}$.

Chapter 5

Composition and multivariate

5.1 Composition and multivariate distributions

PROBLEM – Problem

How to sample less basic random variables.

The main idea of composition is that, if a distribution can be written in terms of simpler distributions, then we can sample from it by first sampling an auxiliary random object and then sampling from the corresponding simpler distribution.

5.2 Discrete mixtures

DEFINITION – Definition

Discrete mixture. A random variable has a discrete mixture distribution if its pmf or density can be written as $\pi(x) = \sum_{j=1}^{K} w_j q_j(x)$, where each q_j is a distribution and the weights satisfy $w_j \geq 0, \qquad \sum_{j=1}^{K} w_j = 1.$

REMARK – Remark

If we sample an index J from $\mathbb{P}(J = j) = w_j, \qquad j = 1, \ldots, K$, and, conditionally on $J = j$, we generate $X \mid J = j \sim q_j$, then the unconditional distribution of X is $\pi(x) = \sum_{j=1}^{K} w_j q_j(x)$.

Proof

By the law of total probability, $\mathbb{P}(X \in dx) = \sum_{j=1}^{K} \mathbb{P}(X \in dx \mid J = j)\mathbb{P}(J = j)$.

Since, conditionally on $J = j$, we have $X \sim q_j$, then $\mathbb{P}(X \in dx \mid J = j) = q_j(x)\, dx$.

Therefore $\mathbb{P}(X \in dx) = \sum_{j=1}^{K} q_j(x) w_j\, dx$. Hence the unconditional density or pmf of X is $\pi(x) = \sum_{j=1}^{K} w_j q_j(x)$. ■

Algorithm 1: Sampling discrete mixtures

Algorithm

1. Input: the number of samples N.
2. For $i = 1, \ldots, N$ do
3. Generate $J \sim \pi(j)$, with $\mathbb{P}(J = j) = w_j$.
4. Generate $X_i \sim q_J(x)$.
5. end for.

EXAMPLE – Example

A mixture of Gamma distributions: $F_X(x) = \sum_{k=1}^{5} w_k F_{X_k}(x), \qquad X_k \sim \text{Gamma}(3, 1/k), \qquad w_k = \frac{k}{15}.$

To generate one sample:

1. sample $K \in \{1, 2, 3, 4, 5\}$ with probabilities $\mathbb{P}(K = k) = w_k = \frac{k}{15}$;
2. then sample from the chosen component: $X \mid K = k \sim X_k, \qquad X_k \sim \text{Gamma}(3, 1/k)$.

Equivalently, after K has been chosen, we generate $X \sim \text{Gamma}(3, 1/K)$.

REMARK – Remark

In the same way, we can sample from conditional and joint distributions. The idea is always the same: first sample the variable that selects or fixes the distribution, and then sample from the distribution conditional on that value.

5.3 Sampling from conditional densities

A conditional density is a density which depends on the value taken by another variable.

Let X have marginal distribution $\pi(x)$, and let the conditional distribution of Y given $X = x$ be $Y \mid X = x \sim \pi(y \mid x)$.

This means that, once X has been fixed equal to a specific value x, the variable Y is generated from the density $\pi(y \mid x)$.

THEOREM – Theorem

If we generate $X \sim \pi(x), \qquad Y \mid X \sim \pi(y \mid X)$, then the pair (X, Y) has joint density $\pi(x, y) = \pi(x)\pi(y \mid x)$.

REMARK – Remark

The order is important. First we generate X. Once X has been generated, it takes a specific value, say $X = x$. Then Y is generated from the conditional density corresponding to that value: $Y \mid X = x \sim \pi(y \mid x)$.

In compact notation, this is written as $Y \mid X \sim \pi(y \mid X)$.

Algorithm 2: Sampling from conditional densities GENERATED

Algorithm

1. Input: number of samples N, marginal density $\pi(x)$, conditional density $\pi(y \mid x)$.
2. For $i = 1, \ldots, N$ do
3. Generate $X_i \sim \pi(x)$.
4. Given the realised value $X_i = x_i$, generate $Y_i \mid X_i = x_i \sim \pi(y \mid x_i)$.
5. end for.
6. Output: samples $(X_1, Y_1), \ldots, (X_N, Y_N)$ from the joint density $\pi(x, y) = \pi(x)\pi(y \mid x)$.

EXAMPLE – Example

Consider the conditional density $\pi(y \mid x) = \mathcal{N}(y; x, 1)$.

Suppose that $X \sim \mathcal{N}(0,1)$, and $Y \mid X = x \sim \mathcal{N}(x,1)$.

Then the marginal density of X is $\pi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$, and the conditional density of Y given $X = x$ is $\pi(y \mid x) = \frac{1}{\sqrt{2\pi}} e^{-(y-x)^2/2}$.

Therefore the joint density is $\pi(x,y) = \pi(x)\pi(y \mid x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \frac{1}{\sqrt{2\pi}} e^{-(y-x)^2/2}$.

Algorithm 3: Gaussian conditional sampler GENERATED

Algorithm

1. Input: number of samples N.
2. For $i = 1, \ldots, N$ do
3. Generate $X_i \sim \mathcal{N}(0,1)$.
4. Generate $\varepsilon_i \sim \mathcal{N}(0,1)$.
5. Set $Y_i = X_i + \varepsilon_i$.
6. end for.
7. Output: samples $(X_1, Y_1), \ldots, (X_N, Y_N)$ from the joint density $\pi(x,y)$.

Indeed, conditionally on $X_i = x_i$, we have $Y_i = x_i + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0,1)$, and therefore $Y_i \mid X_i = x_i \sim \mathcal{N}(x_i, 1)$.

Hence $(X_i, Y_i) \sim \pi(x,y)$.

5.4 Sampling from joint distributions

As introduced before, a two-dimensional joint density can be factorised as $\pi(x,y) = \pi(y \mid x)\pi(x)$.

Hence, to sample $(X,Y) \sim \pi(x,y)$, one may first sample $X \sim \pi(x)$, and then sample $Y \mid X = x \sim \pi(y \mid x)$.

More generally, by repeated factorisation, a d-dimensional joint density can be written as

$\pi(x_1, \ldots, x_d) = \pi(x_1)\pi(x_2 \mid x_1)\pi(x_3 \mid x_1, x_2) \cdots \pi(x_d \mid x_1, \ldots, x_{d-1})$.

Therefore, simulation can be performed sequentially: $X_1 \sim \pi(x_1)$, $X_2 \mid X_1 = x_1 \sim \pi(x_2 \mid x_1)$,

$X_3 \mid X_1 = x_1, X_2 = x_2 \sim \pi(x_3 \mid x_1, x_2)$, and so on, until $X_d \mid X_1 = x_1, \ldots, X_{d-1} = x_{d-1} \sim \pi(x_d \mid x_1, \ldots, x_{d-1})$.

Algorithm 4: Sampling from joint distributions GENERATED

Algorithm

1. Input: number of samples N, dimension d, marginal density $\pi(x_1)$, and conditional densities $\pi(x_j \mid x_1, \ldots, x_{j-1})$, $j = 2, \ldots, d$.
2. For $i = 1, \ldots, N$ do
3. Generate $X_{i,1} \sim \pi(x_1)$.
4. For $j = 2, \ldots, d$ do
5. Given the realised values $X_{i,1} = x_{i,1}, \ldots, X_{i,j-1} = x_{i,j-1}$, generate $X_{i,j} \mid X_{i,1} = x_{i,1}, \ldots, X_{i,j-1} = x_{i,j-1} \sim \pi(x_j \mid x_{i,1}, \ldots, x_{i,j-1})$.
6. end for.
7. Set $\mathbf{X}_i = (X_{i,1}, \ldots, X_{i,d})$.
8. end for.
9. Output: samples $\mathbf{X}_1, \ldots, \mathbf{X}_N$ from the joint density $\pi(x_1, \ldots, x_d)$.

REMARK – Remark

Equivalently, the joint density may also be factorised in reverse order: $\pi(x_1, \dots, x_d) = \pi(x_d)\pi(x_{d-1} \mid x_d)\pi(x_{d-2} \mid x_{d-1}, x_d) \cdots \pi(x_1 \mid x_2, \dots, x_d)$.

This does not define a different joint distribution. It only gives a different sampling order. In practice, one chooses the factorisation according to which marginal and conditional distributions are easier to simulate.

EXAMPLE – Example

Consider the joint density $\pi(x, y, z) = \dfrac{e^{-x}}{(1+e^{-x})^2} \cdot \dfrac{1}{\sqrt{2\pi}} e^{-(y-x)^2/2} \cdot \dfrac{1}{\sqrt{2\pi}} e^{-(z-(x+y))^2/2}$.

This can be read as $X \sim \text{Logistic}(0,1)$, $\quad Y \mid X = x \sim \mathcal{N}(x, 1)$, $\quad Z \mid X = x, Y = y \sim \mathcal{N}(x+y, 1)$.

Therefore, the joint density is factorised as $\pi(x, y, z) = \pi(x)\pi(y \mid x)\pi(z \mid x, y)$.

Algorithm 5: Logistic-Gaussian joint GENERATED

Algorithm

1. Input: number of samples N.
2. For $i = 1, \dots, N$ do
3. Generate $X_i \sim \text{Logistic}(0,1)$.
4. Generate $\varepsilon_{i,1} \sim \mathcal{N}(0,1)$.
5. Set $Y_i = X_i + \varepsilon_{i,1}$.
6. Generate $\varepsilon_{i,2} \sim \mathcal{N}(0,1)$.
7. Set $Z_i = X_i + Y_i + \varepsilon_{i,2}$.
8. end for.
9. Output: samples $(X_1, Y_1, Z_1), \dots, (X_N, Y_N, Z_N)$ from the joint density $\pi(x, y, z)$.

The first step can be implemented by inverse cdf sampling. If $U_i \sim \text{Unif}(0,1)$, then $X_i = \log\left(\dfrac{U_i}{1-U_i}\right) \sim \text{Logistic}(0,1)$.

Indeed, conditionally on $X_i = x_i$, we have $Y_i = x_i + \varepsilon_{i,1}$, $\quad \varepsilon_{i,1} \sim \mathcal{N}(0,1)$, and therefore $Y_i \mid X_i = x_i \sim \mathcal{N}(x_i, 1)$.

Moreover, conditionally on $X_i = x_i$ and $Y_i = y_i$, we have $Z_i = x_i + y_i + \varepsilon_{i,2}$, $\quad \varepsilon_{i,2} \sim \mathcal{N}(0,1)$, and therefore $Z_i \mid X_i = x_i, Y_i = y_i \sim \mathcal{N}(x_i + y_i, 1)$.

Hence $(X_i, Y_i, Z_i) \sim \pi(x, y, z)$.

5.5 Sampling from marginal distributions

Suppose that the joint density of (X, Y) is available and can be factorised as $\pi(x, y) = \pi(y \mid x)\pi(x)$.

Since $\pi(y) = \displaystyle\int \pi(x, y)\, dx = \int \pi(y \mid x)\pi(x)\, dx$, one can sample from the marginal density $\pi(y)$ by first sampling from the joint distribution and then retaining only the Y-coordinates.

Algorithm 6: Sampling from a marginal distribution GENERATED

Algorithm

1. Input: number of samples N, marginal density $\pi(x)$, conditional density $\pi(y \mid x)$.
2. For $i = 1, \dots, N$ do
3. Generate $X_i \sim \pi(x)$.
4. Given the realised value $X_i = x_i$, generate $Y_i \mid X_i = x_i \sim \pi(y \mid x_i)$.
5. Keep only Y_i.
6. end for.
7. Output: samples $Y_1, \dots, Y_N$ from the marginal density $\pi(y)$.

REMARK – Remark

In some cases it is convenient to factorise $\pi(x, y) = \pi(y)\pi(x \mid y)$, sample Y first, and then sample $X \mid Y$.

If instead only $\pi(x, y) = \pi(x)\pi(y \mid x)$ is available and $\pi(x)$ is difficult to sample from, this factorisation may be less useful for direct simulation of the marginal density $\pi(y)$.

EXAMPLE – Example

Linear model. To simulate data with a linear relationship, consider $Y = aX + b + \eta, \qquad \eta \sim \mathcal{N}(0, \sigma^2)$.

Take, for example, $\pi(x) = \text{Unif}(x; -10, 10), \qquad \pi(y \mid x) = \mathcal{N}(y; ax + b, \sigma^2)$.

Then the joint density is $\pi(x, y) = \pi(x)\pi(y \mid x)$.

Algorithm 7: Sampling a linear model GENERATED

Algorithm

1. Input: number of samples N, parameters a, b, σ^2.
2. For $i = 1, \dots, N$ do
3. Generate $X_i \sim \text{Unif}(-10, 10)$.
4. Generate $\eta_i \sim \mathcal{N}(0, \sigma^2)$.
5. Set $Y_i = aX_i + b + \eta_i$.
6. end for.
7. Output: samples $(X_1, Y_1), \dots, (X_N, Y_N)$ from the joint density $\pi(x, y)$.

Indeed, conditionally on $X_i = x_i$, we have $Y_i = ax_i + b + \eta_i, \qquad \eta_i \sim \mathcal{N}(0, \sigma^2)$, and therefore $Y_i \mid X_i = x_i \sim \mathcal{N}(ax_i + b, \sigma^2)$.

If we keep only $Y_1, \dots, Y_N$, then we obtain samples from the marginal density $\pi(y)$.

5.6 Sampling multivariate densities and Gaussian variables

A multivariate density is simply a joint density in d dimensions: $\pi(x_1, \dots, x_d)$.

The previous factorisation ideas still apply when the conditional densities are known. However, an important special case is the multivariate Gaussian distribution.

Let $x \in \mathbb{R}^d$. The density of a d-dimensional Gaussian random vector with mean vector $\mu \in \mathbb{R}^d$ and covariance matrix $\Sigma \in \mathbb{R}^{d \times d}$ is $\pi(x) = (2\pi)^{-d/2} |\det \Sigma|^{-1/2} \exp\left\{ -\frac{1}{2}(x - \mu)^T \Sigma^{-1} (x - \mu) \right\}$, where Σ is symmetric and positive definite.

More generally, let $X = (X_1, \dots, X_d)^T \sim \mathcal{N}(\mu, \Sigma)$.

A useful special case is obtained from a standard Gaussian vector $Z = (Z_1, \dots, Z_d)^T \sim \mathcal{N}(0, I_d)$, where $Z_j \sim \mathcal{N}(0, 1)$, $j = 1, \dots, d$, independently.

THEOREM – Theorem

Let $Z \sim \mathcal{N}(0, I_d)$. If L is a matrix such that $\Sigma = LL^T$, then $X = \mu + LZ$ has multivariate Gaussian distribution $X \sim \mathcal{N}(\mu, \Sigma)$.

Proof

Since $Z \sim \mathcal{N}(0, I_d)$, we have $\mathbb{E}[Z] = 0, \qquad \mathrm{Var}(Z) = I_d$.

Define $X = \mu + LZ$.

Then $\mathbb{E}[X] = \mathbb{E}[\mu + LZ] = \mu + L\mathbb{E}[Z] = \mu$.

Moreover, $\mathrm{Var}(X) = \mathrm{Var}(\mu + LZ) = L\mathrm{Var}(Z)L^T = LI_dL^T = LL^T = \Sigma$.

Therefore $X \sim \mathcal{N}(\mu, \Sigma)$. ■

REMARK – Remark

In practice, the matrix L is often computed by the Cholesky decomposition: $\Sigma = LL^T$, where L is lower triangular. Then one generates independent standard normal variables and applies the linear transformation $X = \mu + LZ$.

Algorithm 8: Sampling a multivariate Gaussian GENERATED

Algorithm

1. Input: number of samples N, dimension d, mean vector μ, covariance matrix Σ.
2. Compute L such that $\Sigma = LL^T$ using the Cholesky decomposition.
3. For $i = 1, \ldots, N$ do
4. Generate d independent standard normal variables $Z_{i,1}, \ldots, Z_{i,d} \sim \mathcal{N}(0, 1)$.
5. Form the vector $Z_i = (Z_{i,1}, \ldots, Z_{i,d})^T$.
6. Set $X_i = \mu + LZ_i$.
7. end for.
8. Output: samples $X_1, \ldots, X_N$ from the multivariate Gaussian distribution $\mathcal{N}(\mu, \Sigma)$.

Chapter 6

Monte Carlo Integration

6.1 Vanilla Monte Carlo estimator

PROBLEM – Problem

Use a generated sample to approximate expectations and integrals.

Let π be a probability distribution and let $\varphi(x)$ be a test function. We are often interested in estimating $\pi(\varphi) = \mathbb{E}_\pi[\varphi(X)] = \int \varphi(x)\pi(x)\,dx$.

PROPOSITION – Proposition

Let $X_1, \dots, X_N \sim \pi$ i.i.d. Then $\mathbb{E}_\pi[X] = \int x\pi(x)\,dx \approx \frac{1}{N}\sum_{i=1}^N X_i$.

DEFINITION – Definition

The empirical approximation of π is $\pi^N(x) = \frac{1}{N}\sum_{i=1}^N \delta_{X_i}(x)$, where δ_{X_i} is the Dirac mass centred at X_i.

REMARK – Remark

If the state space is discrete, then π^N corresponds to the empirical pmf $\pi^N(x) = \frac{\#\{i:X_i=x\}}{N}$. If the state space is continuous, π^N is not an ordinary density, but a sum of Dirac masses.

DEFINITION – Definition

Vanilla Monte Carlo estimator. Let $X_1, \dots, X_N \sim \pi$ i.i.d. The quantity $\pi(\varphi) = \mathbb{E}_\pi[\varphi(X)] = \int \varphi(x)\pi(x)\,dx$ is approximated by the vanilla Monte Carlo estimator $\pi^N(\varphi) := \frac{1}{N}\sum_{i=1}^N \varphi(X_i)$.

Proof

$$\int \varphi(x)\pi(x)\,dx \approx \int \varphi(x)\pi^N(x)\,dx = \int \varphi(x)\frac{1}{N}\sum_{i=1}^N \delta_{X_i}(x)\,dx = \frac{1}{N}\sum_{i=1}^N \int \varphi(x)\delta_{X_i}(x)\,dx = \frac{1}{N}\sum_{i=1}^N \varphi(X_i). \qquad \blacksquare$$

REMARK – Remark

The key property of the Dirac mass is $\int \varphi(x)\delta_a(x)\,dx = \varphi(a)$. Hence, when we integrate φ with respect to the empirical measure, the integral becomes the average of the values of φ computed on the simulated sample.

REMARK – Remark

We can define $g(x) := \varphi(x)\pi(x)$. Then $\int \varphi(x)\pi(x)\,dx = \int g(x)\,dx$ is the area under the curve g.

A deterministic quadrature rule, such as the composite midpoint rule, divides the domain into intervals I_j. If x_j is the midpoint of I_j and Δ_j is its length, then $\int \varphi(x)\pi(x)\,dx \approx \sum_j \varphi(x_j)\pi(x_j)\Delta_j$.

Monte Carlo replaces the deterministic grid points x_j with random samples $X_1, \dots, X_N \sim \pi$. Since the samples are generated from π, regions where $\pi(x)$ is large are sampled more often. Thus the weights $\pi(x_j)\Delta_j$ are replaced by empirical frequencies, and the quadrature approximation becomes $\pi^N(\varphi) = \frac{1}{N}\sum_{i=1}^N \varphi(X_i)$.

REMARK – Remark

Monte Carlo estimators can also approximate ordinary integrals of the form $I = \int_D f(x)\,dx$, where D is the domain of integration.

To use Monte Carlo, choose a probability density π whose support covers the region where f is nonzero.

Then $I = \int_D f(x)\,dx = \int_D \frac{f(x)}{\pi(x)}\pi(x)\,dx = \mathbb{E}_\pi\left[\frac{f(X)}{\pi(X)}\right]$, where $X \sim \pi$.

Therefore, if $X_1, \dots, X_N \sim \pi$ i.i.d., then $\hat{I}_N = \frac{1}{N}\sum_{i=1}^N \frac{f(X_i)}{\pi(X_i)}$ is a Monte Carlo estimator of I.

The density π must be chosen properly: $\pi(x) > 0 \quad \text{whenever} \quad f(x) \neq 0$. Otherwise, some parts of the integral would never be sampled.

For $D = [0,1]$, one can choose $\pi = \text{Unif}(0,1)$. For $D = [a,b]$, one can choose $\pi = \text{Unif}(a,b)$. For $D = \mathbb{R}$, no uniform density exists, so one may choose a Gaussian or a Cauchy density. If f has heavy tails, a Cauchy density is often preferable to a Gaussian density. For $D = \mathbb{R}_+$, one may choose an Exponential or Gamma density. If f has heavy positive tails, a Lognormal or Pareto density may be more appropriate.

If we want to use uniforms on $[0,1]$ for an integral over a different domain, we must first apply a valid change of variables and include the Jacobian correction.

6.2 Properties of the vanilla Monte Carlo estimator

PROPOSITION – Proposition

Unbiasedness. If $X_1, \dots, X_N \sim \pi$ i.i.d., then the vanilla Monte Carlo estimator $\pi^N(\varphi) = \frac{1}{N}\sum_{i=1}^N \varphi(X_i)$ is unbiased for $\pi(\varphi)$.

Proof

We have $\mathbb{E}[\pi^N(\varphi)] = \mathbb{E}\left[\frac{1}{N}\sum_{i=1}^N \varphi(X_i)\right] = \frac{1}{N}\sum_{i=1}^N \mathbb{E}[\varphi(X_i)] = \frac{1}{N}\sum_{i=1}^N \int \varphi(x)\pi(x)\,dx = \int \varphi(x)\pi(x)\,dx = \pi(\varphi)$.
Therefore, $\pi^N(\varphi)$ is unbiased. ■

PROPOSITION – Proposition

Consistency. If $X_1, \dots, X_N \sim \pi$ i.i.d., then $Var[\pi^N(\varphi)] = \frac{1}{N}Var_\pi[\varphi(X)]$. Hence, if $Var_\pi[\varphi(X)] < +\infty$, then the estimator $\pi^N(\varphi)$ is consistent.

Proof

By independence, $Var[\pi^N(\varphi)] = Var\left[\frac{1}{N}\sum_{i=1}^N \varphi(X_i)\right] = \frac{1}{N^2}\sum_{i=1}^N Var[\varphi(X_i)] = \frac{1}{N^2} N Var_\pi[\varphi(X)] = \frac{1}{N} Var_\pi[\varphi(X)]$.

Since $Var_\pi[\varphi(X)] = \int (\varphi(x) - \pi(\varphi))^2 \pi(x)\, dx$, if this quantity is finite, then $Var[\pi^N(\varphi)] \to 0$ as $N \to +\infty$. Since the estimator is also unbiased, it is consistent. ■

REMARK – Remark

The quantity $\pi(\varphi)$ is a number, not a random variable. The random object is the estimator $\pi^N(\varphi) = \frac{1}{N}\sum_{i=1}^N \varphi(X_i)$. Therefore, when studying the Monte Carlo error, the relevant variance is $Var[\pi^N(\varphi)]$.

REMARK – Remark

In practice, the exact variance $Var[\pi^N(\varphi)] = \frac{1}{N} Var_\pi[\varphi(X)]$ depends on the unknown quantity $\pi(\varphi)$, because $Var_\pi[\varphi(X)] = \int (\varphi(x) - \pi(\varphi))^2 \pi(x)\, dx$.

Replacing $\pi(\varphi)$ with its Monte Carlo estimate $\pi^N(\varphi)$ yields the empirical estimator $\sigma^2_{\varphi,N} = \frac{1}{N^2}\sum_{i=1}^N \left(\varphi(X_i) - \pi^N(\varphi)\right)^2$.

This leads to the asymptotic normal approximation $\frac{\pi^N(\varphi) - \pi(\varphi)}{\sigma_{\varphi,N}} \xrightarrow{d} \mathcal{N}(0,1), \qquad N \to +\infty$.

Also, $Std[\pi^N(\varphi)] = \frac{\sigma_\varphi}{\sqrt{N}}$, where $\sigma^2_\varphi = Var_\pi[\varphi(X)]$. This shows the typical Monte Carlo convergence rate $O(N^{-1/2})$.

REMARK – Remark

Monte Carlo can also be used to estimate the probability of an event A.

Take $\varphi(x) = \mathbf{1}_A(x)$. Then $\mathbb{P}(X \in A) = \int \mathbf{1}_A(x)\pi(x)\, dx = \pi(\mathbf{1}_A)$. The Monte Carlo estimator is $\pi^N(\mathbf{1}_A) = \frac{1}{N}\sum_{i=1}^N \mathbf{1}_A(X_i)$.

Equivalently, $\mathbb{P}(X \in A) = \int \mathbf{1}_A(x)\pi(x)\, dx \approx \int \mathbf{1}_A(x)\pi^N(x)\, dx = \int \mathbf{1}_A(x) \frac{1}{N}\sum_{i=1}^N \delta_{X_i}(x)\, dx = \frac{1}{N}\sum_{i=1}^N \mathbf{1}_A(X_i)$.

Thus the probability of A is estimated by the empirical frequency of simulated samples falling inside A.

Chapter 7

Variance Reduction

PROBLEM – Problem

We want to reduce the variance of M.C. estimators without excessively increasing the computational cost.

SOLUTION – Solution

We propose three methods: Antithetic sampling, Control variates, Stratified sampling

7.1 Antithetic sampling

Intuition. Consider two random variables X and Y with the same distribution. Since $\mathbb{E}[\varphi(X)] = \mathbb{E}[\varphi(Y)] = \pi(\varphi)$, we can estimate $\pi(\varphi)$ by using the paired estimator $\frac{\varphi(X)+\varphi(Y)}{2}$.

Its variance is $Var\left[\frac{\varphi(X)+\varphi(Y)}{2}\right] = \frac{1}{4}\left[Var(\varphi(X)) + Var(\varphi(Y)) + 2Cov(\varphi(X), \varphi(Y))\right]$. If $Cov(\varphi(X), \varphi(Y)) < 0$, then the two terms compensate each other, and the paired estimator has lower variance than a standard estimator based on independent samples.

DEFINITION – Definition

Antithetic random variables. Two random variables X and Y defined on the same probability space are called antithetic if they have the same distribution and their covariance is negative.

EXAMPLE – Example

A classical example is $X = U, \quad Y = 1 - U, \quad U \sim \text{Unif}(0,1)$. Then $X \sim \text{Unif}(0,1)$ and $Y \sim \text{Unif}(0,1)$, but they move in opposite directions. Indeed, $Corr(X, Y) = -1$.

EXAMPLE – Example

For symmetric distributions, such as a Gaussian distribution centred at zero, we can take $Y = -X$. If $X \sim \mathcal{N}(0,1)$, then also $Y = -X \sim \mathcal{N}(0,1)$, but large positive values of X correspond to large negative values of Y.

REMARK – Remark

More generally, suppose that a simulated quantity can be obtained as a function of independent uniforms: $X = h(U_1, \ldots, U_n), \quad U_1, \ldots, U_n \sim \text{Unif}(0,1)$ independently. The antithetic version is obtained by replacing each U_i with $1 - U_i \Rightarrow Y = h(1 - U_1, \ldots, 1 - U_n)$.

X and Y have the same distribution, and if h is monotone, they are negatively correlated.

PROPOSITION – Proposition

If $h(x_1, \dots, x_n)$ is monotone in each of its arguments, and $U_1, \dots, U_n$ are independent random numbers, then $Cov\left(h(U_1, \dots, U_n), h(1-U_1, \dots, 1-U_n)\right) \leq 0$.

DEFINITION – Definition

Antithetic estimator. Let X and Y be two random variables with the same distribution π. The antithetic estimator is $\widehat{\pi}_N^A(\varphi) := \frac{1}{N}\sum_{i=1}^N \frac{\varphi(X_i)+\varphi(Y_i)}{2}$, where $X_i \sim X$ and $Y_i \sim Y$, for $i = 1, \dots, N$.
Notice that this estimator uses $2N$ function evaluations, N from the X_i's and N from the Y_i's.

EXAMPLE – Example

Suppose we want to estimate $I = \mathbb{E}[U^2]$, $\quad U \sim \text{Unif}(0,1)$. Here the function is $h(u) = u^2$.

Let $X = U^2$, $\quad Y = (1-U)^2$. The antithetic estimator based on one pair is $\widehat{\pi}_1^A = \frac{X+Y}{2} = \frac{U^2+(1-U)^2}{2}$.
For example, if $U = 0.8$, then $h(U) = 0.8^2 = 0.64$, $\quad h(1-U) = 0.2^2 = 0.04$. Hence $\widehat{\pi}_1^A = \frac{0.64+0.04}{2} = 0.34$.

The true value is $\mathbb{E}[U^2] = \int_0^1 u^2\, du = \frac{1}{3} \simeq 0.33$.

The theoretical variance of one standard Monte Carlo draw is $Var(X) = Var(U^2) = \mathbb{E}[U^4] - \left(\mathbb{E}[U^2]\right)^2 = \frac{1}{5} - \left(\frac{1}{3}\right)^2 = \frac{4}{45}$.
To compare the variance reduction, we must compare estimators with the same number of function evaluations, as will be clarified in the following remark. The standard Monte Carlo estimator with $2N$ evaluations is $\pi^{2N}(\varphi) = \frac{1}{2N}\sum_{i=1}^{2N} U_i^2$.
Therefore $Var[\pi^{2N}(\varphi)] = \frac{1}{2N}Var(U^2) = \frac{1}{2N}\frac{4}{45} = \frac{2}{45N}$.

The antithetic estimator with N pairs is $\widehat{\pi}_N^A(\varphi) = \frac{1}{N}\sum_{i=1}^N \frac{U_i^2+(1-U_i)^2}{2}$. Hence $Var[\widehat{\pi}_N^A(\varphi)] = \frac{1}{N}Var\left(\frac{U^2+(1-U)^2}{2}\right) = \frac{1}{4N}Var\left(U^2+(1-U)^2\right)$. Now $U^2+(1-U)^2 = 2U^2-2U+1$, so $Var[\widehat{\pi}_N^A(\varphi)] = \frac{1}{4N}Var(2U^2-2U+1) = \frac{1}{N}Var(U^2-U)$.

Moreover, $\quad Var(U^2-U) = \mathbb{E}[(U^2-U)^2] - \left(\mathbb{E}[U^2-U]\right)^2 = \int_0^1 (u^2-u)^2\, du - \left(\int_0^1 (u^2-u)\, du\right)^2 =$

$$= \int_0^1 (u^4 - 2u^3 + u^2)\, du - \left(\int_0^1 u^2\, du - \int_0^1 u\, du\right)^2 = \left(\frac{1}{5} - \frac{2}{4} + \frac{1}{3}\right) - \left(\frac{1}{3} - \frac{1}{2}\right)^2 = \frac{1}{30} - \frac{1}{36} = \frac{1}{180}.$$

Therefore $Var[\widehat{\pi}_N^A(\varphi)] = \frac{1}{180N}$.

The variance reduction is $1 - \dfrac{Var[\widehat{\pi}_N^A(\varphi)]}{Var[\pi^{2N}(\varphi)]} = 1 - \dfrac{\frac{1}{180N}}{\frac{2}{45N}} = 1 - \dfrac{1}{180N}\dfrac{45N}{2} = 1 - \dfrac{1}{8} = \dfrac{7}{8} = 87.5\%$.

Thus, in this example, the antithetic estimator has only $1/8$ of the variance of the standard Monte Carlo estimator with the same number of function evaluations.

PROPOSITION – Proposition

The antithetic estimator is unbiased.

Proof

Since X_i and Y_i have the same distribution π, we have $\mathbb{E}[\varphi(X_i)] = \mathbb{E}[\varphi(Y_i)] = \pi(\varphi)$.

Therefore, $\mathbb{E}[\widehat{\pi}_N^A(\varphi)] = \mathbb{E}\left[\frac{1}{N}\sum_{i=1}^N \frac{\varphi(X_i)+\varphi(Y_i)}{2}\right] = \frac{1}{N}\sum_{i=1}^N \frac{\mathbb{E}[\varphi(X_i)]+\mathbb{E}[\varphi(Y_i)]}{2} = \frac{1}{N}\sum_{i=1}^N \frac{\pi(\varphi)+\pi(\varphi)}{2} = \pi(\varphi)$. ■

PROPOSITION – Proposition

The variance of the antithetic estimator is $Var[\widehat{\pi}_N^A(\varphi)] = \frac{Var(\varphi(X_1))+Cov(\varphi(X_1),\varphi(Y_1))}{2N}$.

Equivalently, $Var[\widehat{\pi}_N^A(\varphi)] = \frac{Var(\varphi(X_1))}{2N}\left[1 + Corr(\varphi(X_1),\varphi(Y_1))\right]$.

Proof

Assume that the pairs (X_i, Y_i), $i = 1, \dots, N$, are independent and identically distributed. The variables inside each pair need not be independent.

We have $Var[\widehat{\pi}_N^A(\varphi)] = Var\left[\frac{1}{N}\sum_{i=1}^N \frac{\varphi(X_i)+\varphi(Y_i)}{2}\right]$.

Since the pairs are independent across i, $Var[\widehat{\pi}_N^A(\varphi)] = \frac{1}{4N^2}\sum_{i=1}^N Var[\varphi(X_i) + \varphi(Y_i)]$.

Hence $Var[\widehat{\pi}_N^A(\varphi)] = \frac{1}{4N^2}\sum_{i=1}^N \left[Var(\varphi(X_i)) + Var(\varphi(Y_i)) + 2Cov(\varphi(X_i),\varphi(Y_i))\right]$.

Since X_i and Y_i have the same distribution, $Var(\varphi(X_i)) = Var(\varphi(Y_i))$.

Therefore $Var[\widehat{\pi}_N^A(\varphi)] = \frac{1}{4N^2}\sum_{i=1}^N \left[2Var(\varphi(X_i)) + 2Cov(\varphi(X_i),\varphi(Y_i))\right]$.

Thus $Var[\widehat{\pi}_N^A(\varphi)] = \frac{1}{2N^2}\sum_{i=1}^N \left[Var(\varphi(X_i)) + Cov(\varphi(X_i),\varphi(Y_i))\right]$.

Since the pairs are identically distributed, $Var[\widehat{\pi}_N^A(\varphi)] = \frac{Var(\varphi(X_1))+Cov(\varphi(X_1),\varphi(Y_1))}{2N}$.

Using $Cov(\varphi(X_1),\varphi(Y_1)) = Corr(\varphi(X_1),\varphi(Y_1))\sqrt{Var(\varphi(X_1))}\sqrt{Var(\varphi(Y_1))}$, and since $Var(\varphi(X_1)) = Var(\varphi(Y_1))$, we obtain $Cov(\varphi(X_1),\varphi(Y_1)) = Corr(\varphi(X_1),\varphi(Y_1))Var(\varphi(X_1))$.

Hence $Var[\widehat{\pi}_N^A(\varphi)] = \frac{Var(\varphi(X_1))+Corr(\varphi(X_1),\varphi(Y_1))Var(\varphi(X_1))}{2N}$.

Therefore $Var[\widehat{\pi}_N^A(\varphi)] = \frac{Var(\varphi(X_1))}{2N}\left[1 + Corr(\varphi(X_1),\varphi(Y_1))\right]$. ■

EXAMPLE – Example

Equivalently, we can compute the same variance of the previous example using the covariance formula

Since $X = U^2$, $Y = (1-U)^2$, we have $Var\left(\frac{X+Y}{2}\right) = \frac{1}{4}\left[Var(X) + Var(Y) + 2Cov(X,Y)\right]$.

Since X and Y have the same distribution, $Var(X) = Var(Y) = Var(U^2) = \frac{4}{45}$.

Moreover, $Cov(X,Y) = Cov(U^2,(1-U)^2) = \mathbb{E}[U^2(1-U)^2] - \mathbb{E}[U^2]\mathbb{E}[(1-U)^2] = \int_0^1 u^2(1-u)^2\,du - \left(\frac{1}{3}\right)^2$

$= \int_0^1 (u^2 - 2u^3 + u^4)\,du - \frac{1}{9} = \frac{1}{3} - \frac{1}{2} + \frac{1}{5} - \frac{1}{9} = \frac{1}{30} - \frac{1}{9} = -\frac{7}{90}$.

Therefore $Var[\widehat{\pi}_N^A(\varphi)] = \frac{1}{N}Var\left(\frac{X+Y}{2}\right) = \frac{1}{N}\frac{1}{4}\left[\frac{4}{45} + \frac{4}{45} + 2\left(-\frac{7}{90}\right)\right] = \frac{1}{N}\frac{1}{4}\left[\frac{8}{45} - \frac{7}{45}\right] = \frac{1}{N}\frac{1}{180} = \frac{1}{180N}$.

REMARK – Remark

To compare the antithetic estimator with the simple Monte Carlo estimator, we must use the same computational cost. Since the antithetic estimator uses $2N$ function evaluations, we compare it with the simple Monte Carlo estimator based on $2N$ independent samples: $\pi^{2N}(\varphi) = \frac{1}{2N}\sum_{i=1}^{2N}\varphi(Z_i)$, where $Z_1, \dots, Z_{2N} \sim \pi$ are independent.

The variance of the simple Monte Carlo estimator is $Var[\pi^{2N}(\varphi)] = \frac{Var(\varphi(Z_1))}{2N}$.

Since Z_1, X_1, and Y_1 have the same distribution, $Var(\varphi(Z_1)) = Var(\varphi(X_1))$.

Therefore $Var[\pi^{2N}(\varphi)] = \frac{Var(\varphi(X_1))}{2N}$. Now $Var[\widehat{\pi}_N^A(\varphi)] - Var[\pi^{2N}(\varphi)] = \frac{Var(\varphi(X_1))}{2N}\left[1 + Corr(\varphi(X_1),\varphi(Y_1))\right] - \frac{Var(\varphi(X_1))}{2N}$.

Hence $Var[\widehat{\pi}_N^A(\varphi)] - Var[\pi^{2N}(\varphi)] = \frac{Var(\varphi(X_1))}{2N}Corr(\varphi(X_1),\varphi(Y_1))$.

Thus, antithetic sampling reduces the variance whenever $Corr(\varphi(X_1),\varphi(Y_1)) < 0$.

REMARK – Remark

Since $-1 \leq Corr(\varphi(X_1), \varphi(Y_1)) \leq 1$, we have $0 \leq 1 + Corr(\varphi(X_1), \varphi(Y_1)) \leq 2$.

Therefore $0 \leq Var[\widehat{\pi}_N^A(\varphi)] \leq 2\frac{Var(\varphi(X_1))}{2N} = \frac{Var(\varphi(X_1))}{N}$.

In the best case, the correlation is -1, and the antithetic estimator can have zero variance. In the worst case, the correlation is 1, and the variance is twice the variance of the simple Monte Carlo estimator with the same number of function evaluations.

REMARK – Remark

Pros and cons.

- The choice of the antithetic variable is often straightforward. For example, if the simulation is based on $U \sim \text{Unif}(0,1)$, the antithetic variable is usually obtained by replacing U with $1-U$.
- If φ is monotone and the antithetic variables are constructed in the standard way, then antithetic sampling is expected to reduce the variance. For example, if $X = h(U)$ and $Y = h(1-U)$, with h monotone, then X and Y tend to move in opposite directions.
- The function φ usually has to be evaluated twice for each pair.
- Variance reduction is guaranteed only when the induced correlation is negative $Corr(\varphi(X_1), \varphi(Y_1)) < 0$
- Antithetic sampling can still be useful even when φ is not monotone, provided that the induced correlation is negative.

7.2 Control Variates

Intuition. Suppose we want to estimate $\pi(\varphi) = \mathbb{E}[\varphi(X)] = \int \varphi(x)\pi(x)\,dx$.

Let us assume that we know another random variable Y such that $\mathbb{E}[Y] = 0$. Then, for any $\beta \in \mathbb{R}$, $\mathbb{E}[\varphi(X) - \beta Y] = \pi(\varphi)$.

Therefore, the term $-\beta Y$ does not change the mean. If Y is correlated with $\varphi(X)$, we choose β with the right sign in order to compensate part of the fluctuations of $\varphi(X)$. Hence the estimator can be stabilized without introducing bias.

DEFINITION – Definition

Control variate. Let Y be a random variable such that $\mathbb{E}[Y] = 0$. For any $\beta \in \mathbb{R}$, the control variate estimator is $\widehat{\pi}_N^{CV}(\varphi) = \frac{1}{N}\sum_{i=1}^N [\varphi(X_i) - \beta Y_i]$. The random variable Y is called the control variate.

PROPOSITION – Proposition

The control variate estimator is unbiased.

Proof

$$\mathbb{E}[\widehat{\pi}_N^{CV}(\varphi)] = \mathbb{E}\left[\frac{1}{N}\sum_{i=1}^N [\varphi(X_i) - \beta Y_i]\right] = \frac{1}{N}\sum_{i=1}^N [\mathbb{E}[\varphi(X_i)] - \beta\mathbb{E}[Y_i]] = \frac{1}{N}\sum_{i=1}^N [\pi(\varphi) - \beta \cdot 0] = \pi(\varphi). \quad \blacksquare$$

PROPOSITION – Proposition

If the pairs $\{(X_i, Y_i)\}_{i=1,\dots,N}$ are iid, then $Var[\widehat{\pi}_N^{CV}(\varphi)] = \frac{1}{N}\left[Var(\varphi(X_1)) + \beta^2 Var(Y_1) - 2\beta Cov(\varphi(X_1), Y_1)\right]$.

Proof

Since the pairs are indep, we have $Var[\widehat{\pi}_N^{CV}(\varphi)] = Var\left[\frac{1}{N}\sum_{i=1}^N (\varphi(X_i) - \beta Y_i)\right] = \frac{1}{N^2}\sum_{i=1}^N Var\left[\varphi(X_i) - \beta Y_i\right] =$

$= \frac{1}{N^2}\sum_{i=1}^N \left[Var(\varphi(X_i)) + \beta^2 Var(Y_i) - 2\beta Cov(\varphi(X_i), Y_i)\right]$.

Since the pairs are identically distributed, $Var[\widehat{\pi}_N^{CV}(\varphi)] = \dfrac{1}{N}\left[Var(\varphi(X_1)) + \beta^2 Var(Y_1) - 2\beta Cov(\varphi(X_1), Y_1)\right]$. ■

REMARK – Remark

The variance depends on β and on the covariance $Cov(\varphi(X_1), Y_1)$. Compared with the standard Monte Carlo estimator, the control variate reduces the variance whenever $\beta^2 Var(Y_1) - 2\beta Cov(\varphi(X_1), Y_1) < 0$.

If $Cov(\varphi(X_1), Y_1) > 0$, then the variance reduction interval is $0 < \beta < \frac{2Cov(\varphi(X_1), Y_1)}{Var(Y_1)}$. If $Cov(\varphi(X_1), Y_1) < 0$, then the variance reduction interval is $\frac{2Cov(\varphi(X_1), Y_1)}{Var(Y_1)} < \beta < 0$. The optimal value $\beta^* = \frac{Cov(\varphi(X_1), Y_1)}{Var(Y_1)}$ is the midpoint of the corresponding interval which will be obtained by minimizing the variance with respect to β.

REMARK – Remark

We do not need Y to have zero mean. It is enough that $\mathbb{E}[Y]$ is known. In this case, we can construct a zero-mean random variable by taking $\widetilde{Y} = Y - \mathbb{E}[Y]$. Then the estimator becomes $\widehat{\pi}_N^{CV}(\varphi) = \frac{1}{N}\sum_{i=1}^N \left[\varphi(X_i) - \beta\left(Y_i - \mathbb{E}[Y]\right)\right]$.

REMARK – Remark

Under finite-variance assumptions, the control variate estimator is consistent and satisfies a central limit thm.

PROPOSITION – Proposition

The optimal value of β is $\beta^* = \frac{Cov(\varphi(X_1), Y_1)}{Var(Y_1)}$.

Proof

Since the variance depends on β, we minimize $g(\beta) = Var(\varphi(X_1)) + \beta^2 Var(Y_1) - 2\beta Cov(\varphi(X_1), Y_1)$.

Differentiating with respect to β, we get $\frac{\partial}{\partial\beta} g(\beta) = 2\beta Var(Y_1) - 2Cov(\varphi(X_1), Y_1)$.

Setting $\frac{\partial}{\partial\beta} g(\beta) = 0$, we obtain $\beta^* = \frac{Cov(\varphi(X_1), Y_1)}{Var(Y_1)}$.

Moreover, $\frac{\partial^2}{\partial\beta^2} g(\beta) = 2Var(Y_1)$, and therefore $\frac{\partial^2}{\partial\beta^2} g(\beta^*) = 2Var(Y_1) > 0$. Hence β^* is a minimum. ■

PROPOSITION – Proposition

With the optimal choice β^*, the variance is $Var[\widehat{\pi}_N^{CV,\beta^*}(\varphi)] = \frac{1}{N} Var(\varphi(X_1))\left[1 - Corr(\varphi(X_1), Y_1)^2\right]$.

Proof

Substituting $\beta^* = \frac{Cov(\varphi(X_1), Y_1)}{Var(Y_1)}$ in the variance formula, we get

$$
\begin{aligned}
Var[\widehat{\pi}_N^{CV,\beta^*}(\varphi)] &= \frac{1}{N}\left[Var(\varphi(X_1)) + \frac{Cov(\varphi(X_1), Y_1)^2}{Var(Y_1)^2} Var(Y_1) \; -2\frac{Cov(\varphi(X_1), Y_1)}{Var(Y_1)} Cov(\varphi(X_1), Y_1)\right] \\
&= \frac{1}{N}\left[Var(\varphi(X_1)) - \frac{Cov(\varphi(X_1), Y_1)^2}{Var(Y_1)}\right] = \frac{1}{N}\left[Var(\varphi(X_1)) - \frac{Var(\varphi(X_1))\, Var(Y_1)\, Corr(\varphi(X_1), Y_1)^2}{Var(Y_1)}\right] \\
&= \frac{1}{N} Var(\varphi(X_1))\left[1 - Corr(\varphi(X_1), Y_1)^2\right].
\end{aligned}
$$

■

REMARK – Remark

Since the standard Monte Carlo estimator has variance $Var[\pi^N(\varphi)] = \frac{1}{N}Var(\varphi(X_1))$, we have

$Var[\widehat{\pi}_N^{CV,\beta^*}(\varphi)] - Var[\pi^N(\varphi)] = \frac{1}{N}Var(\varphi(X_1))\left[\left(1 - Corr(\varphi(X_1), Y_1)^2\right) - 1\right] = -\frac{1}{N}Var(\varphi(X_1))Corr(\varphi(X_1), Y_1)^2 \leq 0.$

Therefore, the optimal variance is always smaller than or equal to the variance of the standard Monte Carlo estimator. The stronger the correlation between $\varphi(X_1)$ and Y_1, the larger the variance reduction.

REMARK – Remark

The variance reduction depends on $Corr(\varphi(X_1), Y_1)^2$. Therefore, the sign of the correlation does not matter. What matters is the absolute size of the correlation.

If $|Corr(\varphi(X_1), Y_1)|$ is close to 1, the variance reduction is large. If it is close to 0, the control variate is almost useless.

REMARK – Remark

Estimation of the optimal coefficient. In practice, we usually do not know $\beta^* = \frac{Cov(\varphi(X_1),Y_1)}{Var(Y_1)}$. Therefore, we replace the covariance and the variance with their empirical versions:

$\widehat{\beta} = \frac{\sum_{i=1}^N (Y_i - \overline{Y})(\varphi(X_i) - \pi^N(\varphi))}{\sum_{i=1}^N (Y_i - \overline{Y})^2}$, where $\overline{Y} = \frac{1}{N}\sum_{i=1}^N Y_i$ and $\pi^N(\varphi) = \frac{1}{N}\sum_{i=1}^N \varphi(X_i)$.

REMARK – Remark

To avoid introducing bias, a safe procedure is to split the sample into two parts.

- The first part is used to estimate β^*.
- The second part is used to compute the control variate estimator.

This cost should be taken into account when comparing the control variate estimator with the standard Monte Carlo estimator.

REMARK – Remark

The idea of control variates can be generalized to multiple random variables. Suppose we have K zero-mean control variates $Y^1, \ldots, Y^K$. Then the estimator becomes $\widehat{\pi}_N^{CV}(\varphi) = \frac{1}{N}\sum_{i=1}^N \left[\varphi(X_i) - \sum_{k=1}^K \beta_k Y_i^k\right]$.

The coefficients $\beta_1, \ldots, \beta_K$ are chosen in order to minimize the variance of the estimator.

EXAMPLE – Example

Suppose we want to compute $I = \int_0^1 e^u\, du$. The exact value is $I = e - 1$. Here $\varphi(u) = e^u$.

A standard Monte Carlo estimator is obtained by taking $U_i \sim \text{Unif}(0,1)$ independently and using $\pi^N(\varphi) = \frac{1}{N}\sum_{i=1}^N e^{U_i}$.

A natural control variate is U, because U is positively correlated with e^U, and its expectation is known: $\mathbb{E}[U] = \frac{1}{2}$.

Therefore, we use the zero-mean control variate $Y = U - \frac{1}{2}$.

The control variate estimator is $\widehat{\pi}_N^{CV,\beta}(\varphi) = \frac{1}{N}\sum_{i=1}^N \left[e^{U_i} - \beta\left(U_i - \frac{1}{2}\right)\right]$.

We now compute the optimal value of β. Since shifting U by its mean does not change the covariance,

$Cov(e^U, Y) = Cov\left(e^U, U - \frac{1}{2}\right) = Cov(e^U, U)$. Now, $Cov(U, e^U) = \mathbb{E}[Ue^U] - \mathbb{E}[U]\mathbb{E}[e^U] \quad = \int_0^1 ue^u\, du - \frac{1}{2}(e-1)$.

Using integration by parts, $\int_0^1 ue^u\, du = [ue^u]_0^1 - \int_0^1 e^u\, du \quad = e - (e-1) = 1.$

Therefore, $Cov(U, e^U) = 1 - \dfrac{e-1}{2} = \dfrac{3-e}{2} \simeq 0.14086$. Moreover, $Var(U) = \frac{1}{12}$.

Hence, $\beta^* = \dfrac{Cov(e^U, U)}{Var(U)} = \dfrac{0.14086}{1/12} \simeq 1.69032$.

Now we compute the variance reduction. First, $Var(e^U) = \mathbb{E}[e^{2U}] - \left(\mathbb{E}[e^U]\right)^2$.

We have $\mathbb{E}[e^{2U}] = \displaystyle\int_0^1 e^{2u}\, du = \frac{e^2-1}{2}$, and $\mathbb{E}[e^U] = e - 1$.

Therefore, $Var(e^U) = \dfrac{e^2-1}{2} - (e-1)^2 \simeq 0.2420$.

Using the optimal control variate formula, $Var[\widehat{\pi}_N^{CV,\beta^*}(\varphi)] = \dfrac{1}{N}\left[Var(e^U) - \dfrac{Cov(e^U,U)^2}{Var(U)}\right] = \dfrac{1}{N}\left[0.2420 - \dfrac{0.14086^2}{1/12}\right]$

$$= \frac{1}{N}\left[0.2420 - 0.2380\right] = \frac{0.0039}{N}.$$

On the other hand, the variance of the standard Monte Carlo estimator is, $Var[\pi^N(\varphi)] = \dfrac{Var(e^U)}{N} = \dfrac{0.2420}{N}$.

Therefore, the variance reduction is $1 - \dfrac{Var[\widehat{\pi}_N^{CV,\beta^*}(\varphi)]}{Var[\pi^N(\varphi)]} = 1 - \dfrac{0.0039/N}{0.2420/N} \simeq 1 - 0.0161 = 0.9839$.

Thus, in this example, the control variate estimator reduces the variance by approximately 98.4%.

In this example, β^* is known analytically, so the comparison is between a standard Monte Carlo estimator using N simulations and a control variate estimator using the same N simulations.

If instead β^* has to be estimated from a pilot sample of size M, then the control variate method uses $M + N$ simulations in total: M to estimate β^* and N to estimate the integral. Therefore, the fair comparison is not with a standard Monte Carlo estimator using only N simulations, but with a standard Monte Carlo estimator using $M + N$ simulations.

REMARK – Remark

Pros and cons.

- The method can give a very large variance reduction when the control variate is strongly correlated with $\varphi(X)$.
- The sign of the correlation is not important, because the optimal variance depends on the squared correlation.
- The choice of the control variate Y is not always straightforward.
- In practice, the optimal coefficient β^* is usually unknown and must be estimated, often by using pilot runs.
- The usefulness of the control variate ultimately depends on its computational cost. If simulating or evaluating Y is expensive, the variance reduction may not compensate for the extra cost.

Lecture 8.05: control variate note

REMARK – Remark

This page is not itself an importance-sampling formula, but it belongs to the same variance-reduction block of the lecture.

We want to estimate $\mathbb{P}(X \le a)$ when the mean of X is known. For $X \sim \text{Unif}(0,1)$, set $\psi(X) = \mathbf{1}_{\{X \le a\}}$, and $Y = X - \mathbb{E}[X] = X - \frac{1}{2}$.

The control-variate estimator is $\frac{1}{N}\sum_{i=1}^N (\psi(X_i) - \beta Y_i)$, with optimal coefficient $\beta^\star = \frac{Cov(\psi(X),Y)}{Var(Y)}$.

For $a \in (0,1)$, $Var(Y) = \frac{1}{12}$, and $Cov(\psi(X), Y) = \int_0^a \left(x - \frac{1}{2}\right) dx = \frac{a^2 - a}{2}$.

Hence $\beta^\star = 6a(a-1) = -6a(1-a)$.

The vanilla estimator has variance $\frac{a(1-a)}{N}$, while the optimally controlled estimator has variance $\frac{a(1-a)}{N}[1 - 3a(1-a)]$.

REMARK – Remark

The variance reduction is largest near $a = 1/2$, where the indicator and X are most strongly negatively correlated.

7.3 Stratified sampling

Intuition. Split the simulation into groups and force the algorithm to sample from each group using conditional expectation with an auxiliary variable Y.

DEFINITION – Definition

Stratified estimator.

Suppose we want to estimate $\pi(\varphi) = \mathbb{E}[\varphi(X)]$ and suppose we have a discrete random variable Y such that

- $p_i = \mathbb{P}(Y = y_i)$ is known, for $i = 1, \dots, k$;
- we can easily simulate from the conditional distribution $X \mid Y = y_i$.

For every stratum i, simulate N_i independent samples from $Z^i := X \mid Y = y_i$. Denote them by $Z^i_1, \dots, Z^i_{N_i}$.

The stratified estimator is $\widehat{\pi}^{STRAT}_N(\varphi) := \sum_{i=1}^k p_i \frac{1}{N_i} \sum_{l=1}^{N_i} \varphi(Z^i_l)$.

REMARK – Remark

For each stratum i, $\frac{1}{N_i} \sum_{l=1}^{N_i} \varphi(Z^i_l)$ is the standard Monte Carlo estimator of the conditional expectation $\mathbb{E}[\varphi(X) \mid Y = y_i]$. Indeed, since $Z^i_l \sim X \mid Y = y_i$, we have $\mathbb{E}[\varphi(Z^i_l)] = \int \varphi(x)\pi(x \mid Y = y_i)\, dx$.

PROPOSITION – Proposition

The stratified estimator is unbiased.

Proof

$\mathbb{E}\left[\widehat{\pi}^{STRAT}_N(\varphi)\right] = \mathbb{E}\left[\sum_{i=1}^k p_i \frac{1}{N_i} \sum_{l=1}^{N_i} \varphi(Z^i_l)\right] = \sum_{i=1}^k p_i \frac{1}{N_i} \sum_{l=1}^{N_i} \mathbb{E}[\varphi(Z^i_l)] = \sum_{i=1}^k p_i \int \varphi(x)\pi(x \mid Y = y_i)\, dx =$
$\sum_{i=1}^k \mathbb{P}(Y = y_i)\mathbb{E}[\varphi(X) \mid Y = y_i] = \mathbb{E}[\varphi(X)] = \pi(\varphi)$, where the last equality follows from the tower property. ■

PROPOSITION – Proposition

Assume that the samples are independent across strata. Then $Var\left[\widehat{\pi}^{STRAT}_N(\varphi)\right] = \sum_{i=1}^k \frac{p_i^2}{N_i} Var\left(\varphi(X) \mid Y = y_i\right)$.

Proof

Since samples are ind. across strata, $Var\left[\widehat{\pi}^{STRAT}_N(\varphi)\right] = Var\left[\sum_{i=1}^k p_i \frac{1}{N_i} \sum_{l=1}^{N_i} \varphi(Z^i_l)\right] =$
$\sum_{i=1}^k p_i^2 Var\left[\frac{1}{N_i} \sum_{l=1}^{N_i} \varphi(Z^i_l)\right]$.

For fixed i, the variables $Z^i_1, \dots, Z^i_{N_i}$ are iid from $X \mid Y = y_i$. Therefore $Var\left[\frac{1}{N_i} \sum_{l=1}^{N_i} \varphi(Z^i_l)\right] =$
$\frac{1}{N_i} Var\left(\varphi(X) \mid Y = y_i\right)$.

Hence $Var\left[\widehat{\pi}^{STRAT}_N(\varphi)\right] = \sum_{i=1}^k \frac{p_i^2}{N_i} Var\left(\varphi(X) \mid Y = y_i\right)$. ■

REMARK – Remark

Under finite conditional variances, the stratified estimator is consistent and satisfies a central limit theorem.

The intuition is the same as in vanilla Monte Carlo: inside each stratum we compute an ordinary Monte Carlo average, and then we combine a finite number of consistent estimators with deterministic weights.

REMARK – Remark

To compare stratified sampling with the standard Monte Carlo estimator using N samples, we use proportional allocation: $N_i = Np_i$. In this way, $\sum_{i=1}^k N_i = \sum_{i=1}^k Np_i = N$, so both estimators have the same simulation cost.

THEOREM – Theorem

Law of total variance.

For any square-integrable random variable Z and conditioning random variable Y, $Var(Z) = \mathbb{E}[Var(Z \mid Y)] + Var(\mathbb{E}[Z \mid Y])$.

PROPOSITION – Proposition

(Strat vs MC est) With proportional allocation $N_i = Np_i$, the variance of the stratified estimator is always smaller than or equal to the variance of the standard Monte Carlo estimator.

Proof

With $N_i = Np_i$, we have $Var\left[\widehat{\pi}_N^{STRAT}(\varphi)\right] = \sum_{i=1}^k \frac{p_i^2}{N_i} Var\left(\varphi(X) \mid Y = y_i\right) = \sum_{i=1}^k \frac{p_i^2}{Np_i} Var\left(\varphi(X) \mid Y = y_i\right)$

$= \frac{1}{N}\sum_{i=1}^k p_i Var\left(\varphi(X) \mid Y = y_i\right) = \frac{1}{N}\mathbb{E}[Var(\varphi(X) \mid Y)]$.

The standard Monte Carlo estimator has variance $Var[\pi^N(\varphi)] = \frac{1}{N}Var(\varphi(X))$.

By the law of total variance, $Var(\varphi(X)) = \mathbb{E}[Var(\varphi(X) \mid Y)] + Var(\mathbb{E}[\varphi(X) \mid Y])$.

Therefore $Var[\pi^N(\varphi)] - Var\left[\widehat{\pi}_N^{STRAT}(\varphi)\right] = \frac{1}{N}Var(\mathbb{E}[\varphi(X) \mid Y]) \geq 0$.

Hence stratification cannot increase the variance. ■

REMARK – Remark

The variance reduction is large when the strata have different conditional means.

Indeed, $Var(\mathbb{E}[\varphi(X) \mid Y])$ measures how much the conditional expectation changes from one stratum to another.

If the conditional means are very different, stratification removes a large part of the variability. If instead $\mathbb{E}[\varphi(X) \mid Y]$ is almost constant, the variance reduction is small.

EXAMPLE – Example

Toy example: iid data with different conditional means. **Generated**

Let $Y \sim Ber\left(\frac{1}{2}\right)$, and let $\varepsilon \sim \mathcal{N}(0,1)$ be independent of Y. Define $X = 10Y + \varepsilon$.

Suppose we generate $(X_1, Y_1), \ldots, (X_N, Y_N)$ iid from the joint distribution of (X, Y). Thus the observations are iid in the usual Monte Carlo sense.

However, conditioning on the value of Y gives two different distributions for X.

If $Y = 0$, then $X = 10 \cdot 0 + \varepsilon = \varepsilon$, hence $X \mid Y = 0 \sim \mathcal{N}(0,1)$.

If $Y = 1$, then $X = 10 \cdot 1 + \varepsilon = 10 + \varepsilon$, hence $X \mid Y = 1 \sim \mathcal{N}(10,1)$.

Therefore $\mathbb{E}[X \mid Y = 0] = 0, \qquad \mathbb{E}[X \mid Y = 1] = 10.$

So iid does not mean that the conditional means across strata must be equal. It only means that every draw (X_j, Y_j) comes independently from the same unconditional joint distribution.

Take $\varphi(x) = x$. Then $\mathbb{E}[\varphi(X) \mid Y = 0] = 0, \qquad \mathbb{E}[\varphi(X) \mid Y = 1] = 10.$

Moreover, $Var(X \mid Y = 0) = 1, \qquad Var(X \mid Y = 1) = 1.$

By the law of total variance, $Var(X) = \mathbb{E}[Var(X \mid Y)] + Var(\mathbb{E}[X \mid Y])$.

Here $\mathbb{E}[Var(X \mid Y)] = 1$.

Instead, $\mathbb{E}[X \mid Y] = 10Y$.

Since $Y \sim Ber\left(\frac{1}{2}\right)$, we have $Var(Y) = \frac{1}{4}$, and therefore $Var(\mathbb{E}[X \mid Y]) = Var(10Y) = 100Var(Y) = 25$.

Hence $Var(X) = 1 + 25 = 26$.

The vanilla Monte Carlo estimator has variance $Var[\pi^N(\varphi)] = \frac{26}{N}$.

With proportional stratification, $N_0 = N_1 = \frac{N}{2}$.

Hence $Var[\widehat{\pi}_N^{STRAT}(\varphi)] = \frac{1}{N}\mathbb{E}[Var(X \mid Y)] = \frac{1}{N}$.

Thus $Var[\pi^N(\varphi)] - Var[\widehat{\pi}_N^{STRAT}(\varphi)] = \frac{25}{N}$.

The reduction is large because the two strata have very different conditional means.

REMARK – Remark

The point is not that the original simulations are non-iid. The pairs (X_j, Y_j) are iid.

The point is that, once we condition on Y, we split the same iid population into subpopulations with different conditional distributions. Stratification removes the variability due to moving randomly between these subpopulations.

PROPOSITION – Proposition

(Control variates vs stratified sampling) Suppose that Y is the same discrete random variable used to build the strata, and suppose that $p_i = \mathbb{P}(Y = y_i)$ is known. Then $\mathbb{E}[Y]$ is known, so Y can also be used as a control variate.

Stratified sampling with proportional allocation has variance smaller than or equal to the optimal control variate estimator based on Y: $Var\left[\widehat{\pi}_N^{STRAT}(\varphi)\right] \leq Var\left[\widehat{\pi}_N^{CV,\beta^*}(\varphi)\right]$.

Proof

With proportional allocation, $Var\left[\widehat{\pi}_N^{STRAT}(\varphi)\right] = \frac{1}{N}\mathbb{E}[Var(\varphi(X) \mid Y)]$. With the optimal control variate coefficient, $Var\left[\widehat{\pi}_N^{CV,\beta^*}(\varphi)\right] = \frac{1}{N}\left[Var(\varphi(X)) - \frac{Cov(\varphi(X),Y)^2}{Var(Y)}\right] = \frac{1}{N}\left[\mathbb{E}[Var(\varphi(X) \mid Y)] + Var(\mathbb{E}[\varphi(X) \mid Y]) - \frac{Cov(\varphi(X),Y)^2}{Var(Y)}\right]$.

Hence $Var\left[\widehat{\pi}_N^{CV,\beta^*}(\varphi)\right] - Var\left[\widehat{\pi}_N^{STRAT}(\varphi)\right] = \dfrac{1}{N}\left[Var(\mathbb{E}[\varphi(X) \mid Y]) - \dfrac{Cov(\varphi(X),Y)^2}{Var(Y)}\right]$.

By the tower property, $Cov(\varphi(X), Y) = Cov(\mathbb{E}[\varphi(X) \mid Y], Y)$.

Indeed, $Cov(\varphi(X), Y) = \mathbb{E}[\varphi(X)Y] - \mathbb{E}[\varphi(X)]\mathbb{E}[Y] = \mathbb{E}\left[Y\mathbb{E}[\varphi(X) \mid Y]\right] - \mathbb{E}\left[\mathbb{E}[\varphi(X) \mid Y]\right]\mathbb{E}[Y] = Cov(\mathbb{E}[\varphi(X) \mid Y], Y)$.

By Cauchy–Schwarz, $Cov(\mathbb{E}[\varphi(X) \mid Y], Y)^2 \leq Var(\mathbb{E}[\varphi(X) \mid Y])Var(Y)$.

Therefore $\frac{Cov(\varphi(X),Y)^2}{Var(Y)} \leq Var(\mathbb{E}[\varphi(X) \mid Y])$. Thus $Var\left[\widehat{\pi}_N^{CV,\beta^*}(\varphi)\right] - Var\left[\widehat{\pi}_N^{STRAT}(\varphi)\right] \geq 0$. ■

REMARK – Remark

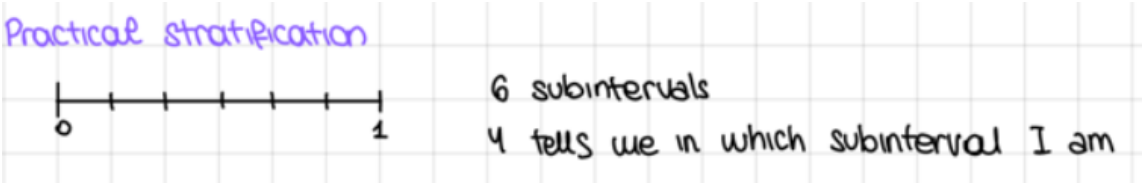
The reason is that control variates remove only the part of $\varphi(X)$ explained linearly by Y, while stratification removes the whole conditional mean effect $\mathbb{E}[\varphi(X) \mid Y]$.

Thus, with the same discrete Y, stratification is at least as good as the optimal linear control variate based on Y.

REMARK – Remark

In practice, it may be difficult to find a useful auxiliary variable Y.

A common practical strategy is to stratify the uniforms used in the simulation. In this case the auxiliary r.v. Y tells us in which subinterval of $[0,1]$ the uniform sample lies.



DEFINITION – Definition

Stratified uniforms.

Instead of sampling directly $U \sim \text{Unif}([0,1])$, we split $[0,1]$ into k subintervals $A_i = \left[\frac{i-1}{k}, \frac{i}{k}\right], \qquad i = 1, \dots, k.$

Then we define the auxiliary random variable $Y = i \iff U \in A_i$.

Thus, stratifying the uniforms means that, instead of letting the simulation choose the stratum randomly, we force the simulation to take one or more observations from each stratum.

More generally, suppose that we take m observations from each stratum and that we have k strata. Since all intervals have the same length, $\mathbb{P}(Y = i) = \frac{1}{k}$.

Moreover, $U \mid Y = i \sim \text{Unif}\left(\left[\frac{i-1}{k}, \frac{i}{k}\right]\right)$.

Therefore, if $V_\ell^i \sim U \mid Y = i, \qquad i = 1, \dots, k, \qquad \ell = 1, \dots, m$, the stratified estimator is $I^{\text{STRAT}} = \frac{1}{k}\sum_{i=1}^{k} \frac{1}{m}\sum_{\ell=1}^{m} \varphi(V_\ell^i)$.

Here i indexes the stratum, while ℓ indexes the observation inside that stratum.

In the particular case in which we force the simulation to take one observation from each stratum, we have $m = 1$. If the total number of observations is N, then necessarily $k = N$.

Hence the auxiliary variables are fixed as $Y_1 = 1, \quad Y_2 = 2, \quad \dots, \quad Y_N = N$.

For each stratum j, we generate $V_j \sim U \mid Y = j$, that is $V_j \sim \text{Unif}\left(\left[\frac{j-1}{N}, \frac{j}{N}\right]\right)$.

Therefore the general stratified estimator collapses to $I^{\text{STRAT}} = \frac{1}{N}\sum_{j=1}^{N} \varphi(V_j)$.

Indeed, the integral can be decomposed as $\int_0^1 \varphi(u)\,du = \sum_{j=1}^{N} \int_{(j-1)/N}^{j/N} \varphi(u)\,du$.

Since each subinterval has length $1/N$, one observation V_j inside the j-th stratum gives $\int_{(j-1)/N}^{j/N} \varphi(u)\,du \approx \frac{1}{N}\varphi(V_j)$.

Thus $\int_0^1 \varphi(u)\,du \approx \sum_{j=1}^{N} \frac{1}{N}\varphi(V_j) = \frac{1}{N}\sum_{j=1}^{N} \varphi(V_j)$.

So the sum over j is already the sum over all strata; a second summation appears only in the general case with m observations inside each stratum.

PROPOSITION – Proposition

Consider the integral $I = \int_0^1 \varphi(u)\,du$.

The vanilla Monte Carlo estimator is $I^N = \frac{1}{N}\sum_{j=1}^{N} \varphi(U_j), \qquad U_j \sim \text{Unif}([0,1])$.

The stratified estimator is $I^{STRAT} = \frac{1}{N}\sum_{j=1}^{N} \varphi(V_j), \qquad V_j \sim \text{Unif}\left(\left[\frac{j-1}{N}, \frac{j}{N}\right]\right)$.

Both estimators are unbiased and $Var(I^{STRAT}) \leq Var(I^N)$.

Proof

For the vanilla estimator, $\mathbb{E}[I^N] = \mathbb{E}\left[\frac{1}{N}\sum_{j=1}^N \varphi(U_j)\right] = \frac{1}{N}\sum_{j=1}^N \mathbb{E}[\varphi(U_j)] = \int_0^1 \varphi(u)\,du$.

For the stratified estimator, the V_j's are independent but not identically distributed. Indeed, $V_j \sim \text{Unif}\left(\left[\frac{j-1}{N}, \frac{j}{N}\right]\right)$, hence the density of V_j is equal to N (from 1/(b-a) density of the uniform) on the interval $\left[\frac{j-1}{N}, \frac{j}{N}\right]$.

Therefore $\mathbb{E}[\varphi(V_j)] = N\int_{(j-1)/N}^{j/N} \varphi(v)\,dv$.

Thus $\mathbb{E}[I^{STRAT}] = \frac{1}{N}\sum_{j=1}^N \mathbb{E}[\varphi(V_j)] = \frac{1}{N}\sum_{j=1}^N N\int_{(j-1)/N}^{j/N} \varphi(v)\,dv = \int_0^1 \varphi(v)\,dv$.
So both estimators are unbiased.

To compare the variances, assume without loss of generality that $\int_0^1 \varphi(u)\,du = 0$. Indeed, if the integral is not zero, we can replace φ by $\widetilde{\varphi}(u) = \varphi(u) - I$, and the variances of the estimators do not change.

For the vanilla estimator, $Var(I^N) = Var\left[\frac{1}{N}\sum_{j=1}^N \varphi(U_j)\right] = \frac{1}{N^2}\sum_{j=1}^N Var(\varphi(U_j)) = \frac{1}{N}Var(\varphi(U))$.

Since $\mathbb{E}[\varphi(U)] = 0$, we have $Var(\varphi(U)) = \mathbb{E}[\varphi(U)^2] = \int_0^1 \varphi(u)^2\,du$.

Hence $Var(I^N) = \frac{1}{N}\int_0^1 \varphi(u)^2\,du$.

For the stratified estimator, $Var(I^{STRAT}) = Var\left[\frac{1}{N}\sum_{j=1}^N \varphi(V_j)\right] = \frac{1}{N^2}\sum_{j=1}^N Var(\varphi(V_j))$, where we used independence across strata.

Moreover, $Var(\varphi(V_j)) \le \mathbb{E}[\varphi(V_j)^2]$.

Therefore $Var(I^{STRAT}) \le \frac{1}{N^2}\sum_{j=1}^N \mathbb{E}[\varphi(V_j)^2] = \frac{1}{N^2}\sum_{j=1}^N N\int_{(j-1)/N}^{j/N} \varphi(v)^2\,dv = \frac{1}{N}\int_0^1 \varphi(v)^2\,dv$.

Since $Var(I^N) = \frac{1}{N}\int_0^1 \varphi(v)^2\,dv$, we obtain $Var(I^{STRAT}) \le Var(I^N)$. ■

EXAMPLE – Example

GENERATED Consider $I = \int_0^1 e^u\,du = e - 1$.

The vanilla Monte Carlo estimator is $I^N = \frac{1}{N}\sum_{i=1}^N e^{U_i}, \qquad U_i \sim \text{Unif}(0,1)$.

The stratified estimator is $I^{\text{STRAT}} = \frac{1}{N}\sum_{i=1}^N e^{V_i}, \qquad V_i \sim \text{Unif}\left(\left[\frac{i-1}{N}, \frac{i}{N}\right]\right)$.

Thus we take one observation from each subinterval. For example, V_1 is sampled from the first interval, V_2 from the second one, and so on. Hence every part of $[0,1]$ is represented in the estimator.

For the vanilla estimator, $Var(I^N) = \frac{1}{N}Var(e^U), \qquad U \sim \text{Unif}(0,1)$.

Now $\mathbb{E}[e^U] = \int_0^1 e^u\,du = e - 1$, and $\mathbb{E}[e^{2U}] = \int_0^1 e^{2u}\,du = \frac{e^2-1}{2}$.

Therefore, $Var(e^U) = \mathbb{E}[e^{2U}] - \mathbb{E}[e^U]^2 = \frac{e^2-1}{2} - (e-1)^2 \simeq 0.2420$.

Hence $Var(I^N) = \frac{0.2420}{N}$.

Now we compute the variance of the stratified estimator.

Since the variables $V_1, \dots, V_N$ are independent, $Var(I^{\text{STRAT}}) = Var\left(\frac{1}{N}\sum_{i=1}^N e^{V_i}\right) = \frac{1}{N^2}\sum_{i=1}^N Var(e^{V_i})$.

Set $a_i = \frac{i-1}{N}, \qquad b_i = \frac{i}{N}$. Since $V_i \sim \text{Unif}([a_i, b_i])$ and the length of the interval is $1/N$, the density of V_i on $[a_i, b_i]$ is N. Hence $\mathbb{E}[e^{V_i}] = N\int_{a_i}^{b_i} e^u\,du = N(e^{b_i} - e^{a_i})$, and $\mathbb{E}[e^{2V_i}] = N\int_{a_i}^{b_i} e^{2u}\,du = \frac{N}{2}(e^{2b_i} - e^{2a_i})$.

Therefore, $Var(e^{V_i}) = \frac{N}{2}(e^{2b_i} - e^{2a_i}) - N^2(e^{b_i} - e^{a_i})^2$.

So $Var(I^{\text{STRAT}}) = \frac{1}{N^2}\sum_{i=1}^N \left[\frac{N}{2}(e^{2b_i} - e^{2a_i}) - N^2(e^{b_i} - e^{a_i})^2\right]$.

Using $a_i = \frac{i-1}{N}, \qquad b_i = \frac{i}{N}$, we get $\sum_{i=1}^N \frac{N}{2}(e^{2b_i} - e^{2a_i}) = \frac{N}{2}(e^2 - 1)$, because the sum is telescopic.

Moreover, $e^{b_i} - e^{a_i} = e^{(i-1)/N}\left(e^{1/N} - 1\right)$. Hence $\sum_{i=1}^N (e^{b_i} - e^{a_i})^2 = \left(e^{1/N} - 1\right)^2 \sum_{i=1}^N e^{2(i-1)/N}$.

The last sum is geometric: $\sum_{i=1}^N e^{2(i-1)/N} = \frac{e^2-1}{e^{2/N}-1}$.

Therefore, $Var(I^{\text{STRAT}}) = \frac{1}{N^2}\left[\frac{N}{2}(e^2 - 1) - N^2\left(e^{1/N} - 1\right)^2 \frac{e^2-1}{e^{2/N}-1}\right]$.

Equivalently, $Var(I^{\text{STRAT}}) = \frac{e^2-1}{2N} - (e^2 - 1)\frac{\left(e^{1/N}-1\right)^2}{e^{2/N}-1}$.

Since $e^{2/N}-1=(e^{1/N}-1)(e^{1/N}+1)$, we can also write $Var(I^{\text{STRAT}})=(e^2-1)\left[\frac{1}{2N}-\frac{e^{1/N}-1}{e^{1/N}+1}\right]$.

For large N, $Var(I^{\text{STRAT}})\sim\frac{e^2-1}{24N^3}$.

Therefore the stratified estimator has a much smaller variance than the vanilla Monte Carlo estimator, whose variance is of order $1/N$. The intuition is that vanilla Monte Carlo may generate random clusters and leave some parts of the interval poorly sampled, while stratified sampling forces the simulation to cover the whole interval $[0,1]$.

For stratified sampling:

- with the same discrete random variable Y, stratification is always at least as good as the optimal linear control variate based on Y;
- stratification can give a large variance reduction when the strata have different conditional means;
- it may be difficult to find a useful stratifying variable Y;
- stratifying uniforms is simple and inexpensive;
- in uniform stratification, Y only labels the subinterval; in practice we often do not simulate Y, because we directly force one or more samples in every stratum;
- if we take one sample for each of N strata, the estimator has only one summation, because the summation already runs over all strata;
- a double summation appears only if we take more than one observation inside each stratum;
- the method is computationally cheap once conditional simulation from each stratum is available.

7.4 Option Pricing Example

We now apply variance reduction techniques to a simple option pricing problem.

If $Z\sim\mathcal{N}(0,1)$, then $X=\exp(\mu+\sigma Z)$ is lognormally distributed, with $\mathbb{E}[X]=\exp(\mu+\sigma^2/2)$ and $\text{Var}(X)=\exp(2\mu+\sigma^2)(\exp(\sigma^2)-1)$.

In the Black–Scholes model, under the risk-neutral measure, the terminal asset price is $S_T=S_0\exp(((r-\sigma^2/2)T+\sigma\sqrt{T}Z))$, with $Z\sim\mathcal{N}(0,1)$.

For a European call option with strike K, the payoff is $\varphi(Z)=(S_T(Z)-K)^+=\max\{S_T(Z)-K,0\}$.

Therefore the option price is $C_0=e^{-rT}\mathbb{E}[\varphi(Z)]$.

If we define $I=\mathbb{E}[\varphi(Z)]$, then $C_0=e^{-rT}I$.

The standard Monte Carlo estimator of I is $\widehat{I}_N=\frac{1}{N}\sum_{i=1}^N\varphi(Z_i)$, where $Z_i\sim\mathcal{N}(0,1)$.

Thus the corresponding estimator of the option price is $\widehat{C}_{0,N}=e^{-rT}\widehat{I}_N$.

The variance is $\text{Var}(\widehat{I}_N)=\text{Var}(\varphi(Z))/N$. Hence variance reduction methods try to estimate the same quantity I, but using an unbiased estimator with smaller variance.

Antithetic variables. Since $Z\overset{d}{=}-Z$, we can use the pair Z_i and $-Z_i$ in the same simulation.

The antithetic estimator of I is $\widehat{I}_N^A=\frac{1}{N}\sum_{i=1}^N\frac{\varphi(Z_i)+\varphi(-Z_i)}{2}$.

It is unbiased because $\mathbb{E}[\varphi(-Z)]=\mathbb{E}[\varphi(Z)]$. Therefore $\mathbb{E}[(\varphi(Z)+\varphi(-Z))/2]=I$.

The corresponding estimator of the call price is $\widehat{C}_{0,N}^A=e^{-rT}\widehat{I}_N^A$.

The variance reduction comes from the covariance term: $\text{Var}((\varphi(Z)+\varphi(-Z))/2)=\frac{1}{2}\text{Var}(\varphi(Z))+\frac{1}{2}\text{Cov}(\varphi(Z),\varphi(-Z))$.

Therefore, if $\text{Cov}(\varphi(Z),\varphi(-Z))<0$, then the antithetic estimator has smaller variance.

This is natural here because $Z\mapsto S_T(Z)$ is increasing and the call payoff is increasing in S_T. Hence, when Z gives a large payoff, $-Z$ tends to give a smaller payoff. Their average is therefore more stable than a single payoff.

Control variates. A natural control variate is the terminal asset price, because its expectation is known.

Under the risk-neutral measure, $\mathbb{E}[S_T]=S_0e^{rT}$.

Therefore we define the centred control variate $Y=S_T-S_0e^{rT}$, so that $\mathbb{E}[Y]=0$.

The control variate estimator is $\widehat{I}_N^{\text{CV},\beta}=\frac{1}{N}\sum_{i=1}^N[\varphi(Z_i)-\beta(S_T(Z_i)-S_0e^{rT})]$.

It is unbiased for every β, because we subtract a zero-mean random variable: $\mathbb{E}[\varphi(Z) - \beta Y] = \mathbb{E}[\varphi(Z)] = I$.

The parameter β is chosen to reduce the variance. Indeed, $\mathrm{Var}(\varphi(Z) - \beta Y) = \mathrm{Var}(\varphi(Z)) + \beta^2 \mathrm{Var}(Y) - 2\beta \mathrm{Cov}(\varphi(Z), Y)$.

The optimal coefficient is $\beta^* = \mathrm{Cov}(\varphi(Z), Y)/\mathrm{Var}(Y)$.

Since $Y = S_T - S_0 e^{rT}$, this is equivalent to $\beta^* = \mathrm{Cov}(\varphi(Z), S_T)/\mathrm{Var}(S_T)$.

With this choice, $\mathrm{Var}(\varphi(Z) - \beta^* Y) = \mathrm{Var}(\varphi(Z))(1 - \rho^2_{\varphi,Y})$.

Thus the stronger the correlation between the payoff and the control variate, the larger the variance reduction.

In this example the control variate is useful because the call payoff is strongly related to S_T. When S_T is large, the payoff is large; when S_T is small, the payoff is small.

The corresponding estimator of the option price is $\widehat{C}^{\mathrm{CV},\beta}_{0,N} = e^{-rT} \widehat{I}^{\mathrm{CV},\beta}_N$.

In practice, β^* is usually estimated from the simulated sample or from a preliminary pilot run.

Stratified sampling. The same expectation can also be estimated by stratifying the uniforms used to generate the Gaussian random variables.

Since $Z = \Phi^{-1}(U)$, with $U \sim \mathrm{Unif}([0,1])$, we can write $I = \mathbb{E}[\varphi(\Phi^{-1}(U))]$.

Instead of drawing iid uniforms, we divide $[0,1]$ into N intervals and force one observation inside each interval: $U_i \sim \mathrm{Unif}([(i-1)/N, i/N])$, for $i = 1, \dots, N$.

Then we set $Z_i = \Phi^{-1}(U_i)$.

The stratified estimator of I is $\widehat{I}^{\mathrm{STRAT}}_N = \frac{1}{N} \sum_{i=1}^N \varphi(Z_i)$.

The corresponding estimator of the option price is $\widehat{C}^{\mathrm{STRAT}}_{0,N} = e^{-rT} \widehat{I}^{\mathrm{STRAT}}_N$.

The estimator is unbiased, because each stratum is sampled according to the correct conditional uniform distribution and each stratum has probability $1/N$.

The variance reduction comes from the fact that stratification avoids random clustering. In standard Monte Carlo, many uniforms may fall in the same region of $[0,1]$, while other regions may be poorly represented. In stratified sampling, every region is represented by construction.

Equivalently, stratification removes part of the randomness due to the random allocation of samples across the probability space. This is why it is particularly useful when different strata have different conditional means.

Thus the normal transformation is unchanged, but the uniforms are forced to cover the probability mass more regularly than in the standard iid Monte Carlo method.

REMARK – Remark

Pros and cons, and summary of the methods.

All three methods estimate the same quantity $I = \mathbb{E}[\varphi(Z)]$, but reduce variance in different ways:

- antithetic variables create negative dependence between paired payoffs;
- control variates subtract a zero-mean random variable strongly correlated with the payoff;
- stratified sampling forces the simulation to cover the probability space more regularly.

After estimating I, the option price is obtained by multiplying by the deterministic discount factor e^{-rT}.

Chapter 8

Importance Sampling

PROBLEM – Problem

In vanilla Monte Carlo, samples are drawn directly from the target distribution π. This works well when sampling from π is easy and the relevant regions are visited often enough, but it becomes inefficient for rare events.

For example, if $X \sim \mathcal{N}(0,1)$, then $\mathbb{P}(X > 4) \simeq 3.16 \cdot 10^{-5}$. With a moderate number of simulations, naive Monte Carlo may produce no observations larger than 4, giving an estimator equal to zero.

Another issue arises when $\pi(x) = \frac{\gamma(x)}{Z}$, $\qquad Z = \int \gamma(x)\,dx$, where γ is known but Z is unknown or hard to compute.

SOLUTION – Solution

Importance sampling samples from a proposal density q and corrects the change of distribution using weights.

Choosing q to visit important regions more often, such as tails, can reduce variance and avoid zero estimates in rare-event problems. It also works when π is known only up to a normalising constant, using weights proportional to $\gamma(x)/q(x)$.

8.1 Ordinary Importance Sampling

DEFINITION – Definition

Importance-sampling identity.

Let π be a normalised target density and let φ be a test function. If q is a proposal density such that $\mathrm{supp}(\pi) \subseteq \mathrm{supp}(q)$, then $\pi(\varphi) := \int \varphi(x)\pi(x)\,dx = \int \varphi(x)\frac{\pi(x)}{q(x)}q(x)\,dx = \int \varphi(x)w(x)q(x)\,dx$, where $w(x) := \frac{\pi(x)}{q(x)}$.

The density q is called the proposal distribution, and w is the importance weight function.

The support condition means $q(x) > 0$ whenever $\pi(x) > 0$. Unlike rejection sampling, importance sampling does not require q to dominate π pointwise. However, the proposal must not miss regions where the target has positive mass.

Given samples $X_1, \ldots, X_N \sim q$, the importance-sampling estimator is $\widehat{\pi(\varphi)}^{IS} = \frac{1}{N}\sum_{i=1}^{N} W_i\varphi(X_i)$, where $W_i = w(X_i) = \frac{\pi(X_i)}{q(X_i)}$.

REMARK – Remark

To make the estimator useful:

- q must be easy to sample from;
- the target π must be evaluable with its normalising constant;
- the weights must be numerically stable.

Algorithm: Ordinary Importance Sampling

Algorithm

1. Input: number of samples N, proposal density q, weight function $w = \pi/q$.
2. Sample $X_i \sim q$, for $i = 1, \dots, N$.
3. Compute weights $W_i = w(X_i)$, for $i = 1, \dots, N$.
4. Evaluate the test function $\varphi(X_i)$, for $i = 1, \dots, N$.
5. Output $\widehat{\pi(\varphi)}^{IS} = \frac{1}{N}\sum_{i=1}^{N} W_i \varphi(X_i)$.

REMARK – Remark

The key idea is that, instead of throwing away samples as in rejection sampling, we keep all samples drawn from q and reweight them according to their importance.

EXAMPLE – Example

Gaussian rare event.

Consider the problem of estimating $\mathbb{P}(X > 4)$, where $X \sim \mathcal{N}(0,1)$. The exact value is $\mathbb{P}(X > 4) = 3.16 \cdot 10^{-5}$.

Naive Monte Carlo samples $X_i \sim \mathcal{N}(0,1)$ and estimates the probability with $\widehat{p}_{\mathrm{MC}} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}_{\{X_i>4\}}$. Since the event $\{X > 4\}$ is very rare, with $N = 10000$ it may happen that no simulated point satisfies $X_i > 4$, giving estimate 0.

Importance sampling solves this by sampling instead from a proposal distribution with more mass in the tail, for example $Y_i \sim q = \mathcal{N}(6,1)$. Since the samples are drawn from q and not from the target density $\pi = \mathcal{N}(0,1)$, each observation must be corrected using the weight $w(y) = \frac{\pi(y)}{q(y)}$.

In this case, $w(y) = \frac{\mathcal{N}(y;0,1)}{\mathcal{N}(y;6,1)} = \exp(18 - 6y)$. Therefore, the importance sampling estimator is $\widehat{p}_{\mathrm{IS}} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}_{\{Y_i>4\}} w(Y_i)$, with $Y_i \sim \mathcal{N}(6,1)$.

A standard run with $N = 10000$ gives approximately $3.18 \cdot 10^{-5}$, which is much closer to the true value than the zero returned by naive Monte Carlo in an unlucky run.

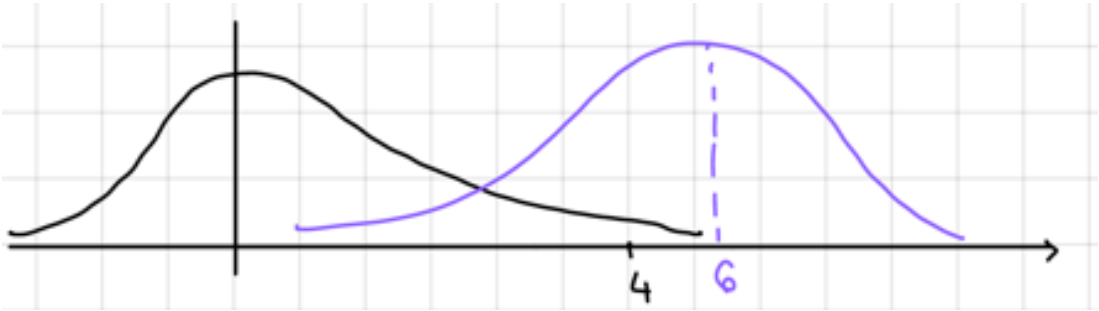


Figure 8.1: Importance sampling proposal shifted towards the rare-event region.

In the figure, the original target is centred at zero, while the proposal is centred near 6. The threshold 4 marks the rare event. Sampling from q gives many more observations in the important tail, and the weights $w = \pi/q$ correct the change of distribution.

Without the weights, $\frac{1}{N}\sum_{i=1}^{N} \varphi(X_i)$, with $X_i \sim q$, estimates $q(\varphi)$, not $\pi(\varphi)$.

With the weights, $\frac{1}{N}\sum_{i=1}^{N} w(X_i)\varphi(X_i)$ estimates $q(w\varphi) = \pi(\varphi)$.

8.2 Weighted empirical measure interpretation

Vanilla Monte Carlo uses particles sampled directly from the target: $X_i \sim \pi$, and $\pi_N(dx) = \frac{1}{N}\sum_{i=1}^{N} \delta_{X_i}(dx)$. All particles have equal mass $1/N$.

Importance sampling instead uses particles sampled from the proposal: $X_i \sim q$, and attaches the weight $w(X_i)$ to each particle:

$\pi_N^{IS}(dx) = \frac{1}{N}\sum_{i=1}^N w(X_i)\delta_{X_i}(dx)$.

This corresponds to a weighted empirical probability measure.



Figure 8.2: Vanilla Monte Carlo gives equal empirical mass to sampled points; importance sampling uses weighted particles sampled from the proposal q.

REMARK – Remark

This is why importance sampling is useful both for variance reduction and when direct sampling from π is difficult, provided that the weights can be evaluated.

8.3 Properties of ordinary IS

PROPOSITION – Proposition

Assume $X_1, \dots, X_N \sim q$ independently and $\text{supp}(\pi) \subseteq \text{supp}(q)$. Then:

1. the IS estimator is unbiased;
2. if $\mathbb{E}_q[(w(X)\varphi(X))^2] < +\infty$, its variance is finite and equals $Var(\widehat{\pi(\varphi)}^{IS}) = \frac{1}{N}\left[\int \frac{\pi(x)^2}{q(x)}\varphi(x)^2\,dx - \left(\int \pi(x)\varphi(x)\,dx\right)^2\right]$;
3. the estimator is consistent;
4. a central limit theorem holds under the same finite-variance condition.

Proof

Set $W_i = w(X_i) = \frac{\pi(X_i)}{q(X_i)}$. The support condition $\text{supp}(\pi) \subseteq \text{supp}(q)$ ensures that the ratio $\frac{\pi(x)}{q(x)}$ is well-defined on the region where the target is relevant.

Unbiasedness. Since $X_1, \dots, X_N \sim q$ independently, $\mathbb{E}_q\left[\widehat{\pi(\varphi)}^{IS}\right] = \mathbb{E}_q\left[\frac{1}{N}\sum_{i=1}^N W_i\varphi(X_i)\right] = \frac{1}{N}\sum_{i=1}^N \mathbb{E}_q\left[\frac{\pi(X_i)}{q(X_i)}\varphi(X_i)\right]$.

Moreover, $\mathbb{E}_q\left[\frac{\pi(X_i)}{q(X_i)}\varphi(X_i)\right] = \int \frac{\pi(x)}{q(x)}\varphi(x)q(x)\,dx = \int \pi(x)\varphi(x)\,dx$. Therefore, $\mathbb{E}_q\left[\widehat{\pi(\varphi)}^{IS}\right] = \frac{1}{N}\sum_{i=1}^N \int \pi(x)\varphi(x)\,dx = \int \pi(x)\varphi(x)\,dx = \pi(\varphi)$. Hence the IS estimator is unbiased.

In particular, taking $\varphi(x) = 1$, we also get $\mathbb{E}_q[W_i] = \int \frac{\pi(x)}{q(x)}q(x)\,dx = \int \pi(x)\,dx = 1$.

Variance. By independence, $Var\left(\widehat{\pi(\varphi)}^{IS}\right) = Var\left(\frac{1}{N}\sum_{i=1}^N W_i\varphi(X_i)\right) = \frac{1}{N^2}\sum_{i=1}^N Var_q\left(W_i\varphi(X_i)\right) = \frac{1}{N}Var_q\left(W\varphi(X)\right)$.

Moreover, $\mathbb{E}_q\left[(W\varphi(X))^2\right] = \int \left(\frac{\pi(x)}{q(x)}\right)^2 \varphi(x)^2 q(x)\,dx = \int \frac{\pi(x)^2}{q(x)}\varphi(x)^2\,dx$. Since $\mathbb{E}_q[W\varphi(X)] = \pi(\varphi)$, we obtain $Var_q\left(W\varphi(X)\right) = \mathbb{E}_q\left[(W\varphi(X))^2\right] - (\mathbb{E}_q[W\varphi(X)])^2 = \int \frac{\pi(x)^2}{q(x)}\varphi(x)^2\,dx - \left(\int \pi(x)\varphi(x)\,dx\right)^2$. Therefore, $Var\left(\widehat{\pi(\varphi)}^{IS}\right) = \frac{1}{N}\left[\int \frac{\pi(x)^2}{q(x)}\varphi(x)^2\,dx - \left(\int \pi(x)\varphi(x)\,dx\right)^2\right]$. This variance is finite whenever $\mathbb{E}_q\left[(W\varphi(X))^2\right] = \int \frac{\pi(x)^2}{q(x)}\varphi(x)^2\,dx < +\infty$.

Consistency. Consistency follows from the law of large numbers applied to the iid random variables $W_i\varphi(X_i)$. Indeed,

$\frac{1}{N}\sum_{i=1}^{N} W_i\varphi(X_i) \to \mathbb{E}_q[W\varphi(X)] = \pi(\varphi)$ as $N \to +\infty$.

Central limit theorem. Under the same finite-variance condition, the central limit theorem gives $\sqrt{N}\left(\widehat{\pi(\varphi)}^{IS} - \pi(\varphi)\right) \xrightarrow{d} \mathcal{N}(0, \sigma_{IS}^2)$, where $\sigma_{IS}^2 = Var_q(W\varphi(X)) = \int \frac{\pi(x)^2}{q(x)}\varphi(x)^2\,dx - \left(\int \pi(x)\varphi(x)\,dx\right)^2$.

Finite variance condition. A sufficient practical condition for finite variance is that the weights are bounded, namely $w(x) = \frac{\pi(x)}{q(x)} \le M < +\infty$, together with the required integrability of φ. More generally, the finite second moment condition is $\mathbb{E}_q[w(X)^2\varphi(X)^2] = \int \frac{\pi(x)^2}{q(x)}\varphi(x)^2\,dx < +\infty$. Thus q should not be too small in the regions where $\pi(x)|\varphi(x)|$ is relevant. Proposals with tails thinner than the target often create unstable or explosive weights. ■

EXAMPLE – Example

Infinite-variance warning.

Let $\pi(x) = \frac{1}{\pi}\frac{1}{1+x^2}$ be the Cauchy density, and choose $q(x) = \mathcal{N}(x; 0, 1) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$.

Then $w(x) = \frac{\pi(x)}{q(x)}$ explodes in the tails, because the Gaussian proposal has much thinner tails than the Cauchy target. This may give unstable estimates, for example when estimating $\mathbb{P}(2 < X < 6)$ for a Cauchy random variable.

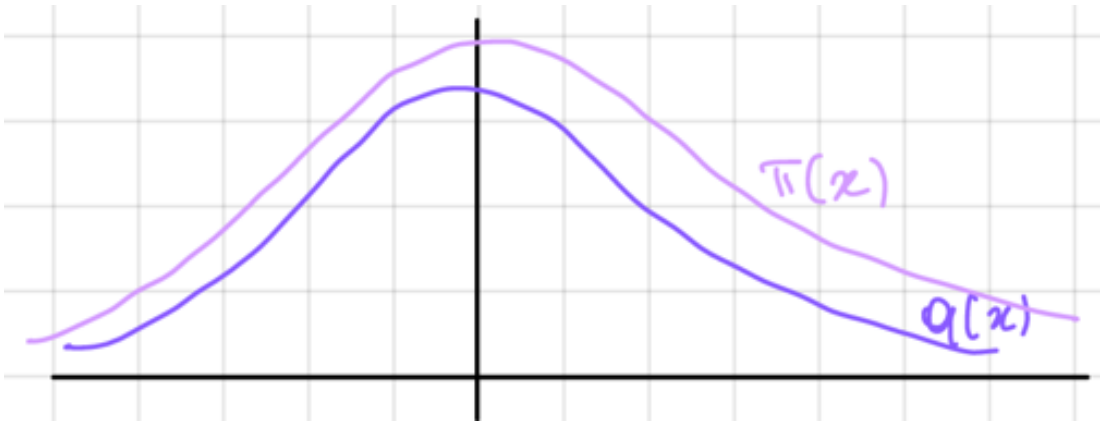


Figure 8.3: Tail comparison between target and proposal.

REMARK – Remark

The proposal must be positive wherever the target is positive, and it should not have thinner tails than the target. If $q(x)$ becomes too small where $\pi(x)$ is still relevant, then the ratio $w(x) = \pi(x)/q(x)$ can explode.

8.4 Self-Normalised Importance Sampling

PROBLEM – Problem

Ordinary IS requires access to the normalised density π. If $\pi(x) = \frac{\gamma(x)}{Z}$, with $Z = \int \gamma(x)\,dx$, the ordinary weight $w(x) = \pi(x)/q(x)$ cannot be computed because Z is unknown.

SOLUTION – Solution

Use the unnormalised weight $\widetilde{w}(x) = \frac{\gamma(x)}{q(x)}$. The unknown normalising constant cancels in a ratio.

Indeed, $\pi(\varphi) = \int \varphi(x)\pi(x)\,dx = \dfrac{\int \varphi(x)\gamma(x)\,dx}{\int \gamma(x)\,dx} = \dfrac{\int \varphi(x)\widetilde{w}(x)q(x)\,dx}{\int \widetilde{w}(x)q(x)\,dx}$.

Approximating numerator and denominator by Monte Carlo under q gives $\widehat{\pi(\varphi)}^{SNIS} = \dfrac{\sum_{i=1}^N \widetilde{W}_i\varphi(X_i)}{\sum_{i=1}^N \widetilde{W}_i} = \sum_{i=1}^N W_i\varphi(X_i)$, where $X_i \sim q$, $\widetilde{W}_i = \widetilde{w}(X_i)$, and $W_i = \dfrac{\widetilde{W}_i}{\sum_{j=1}^N \widetilde{W}_j}$.

The normalised weights satisfy $0 \le W_i \le 1$ and $\sum_{i=1}^N W_i = 1$. The support condition remains $\mathrm{supp}(\pi) = \mathrm{supp}(\gamma) \subseteq \mathrm{supp}(q)$.

Equivalently, we can write numerator and denominator separately: $NUM \approx \frac{1}{N}\sum_{i=1}^N \varphi(X_i)\widetilde{w}(X_i)$, and $DEN \approx \frac{1}{N}\sum_{i=1}^N \widetilde{w}(X_i)$. Then we take NUM/DEN. The common factor $1/N$ cancels.

Algorithm: Self-Normalised Importance Sampling

Algorithm

1. Input: number of samples N, proposal density q, unnormalised weight function $\widetilde{w} = \gamma/q$.
2. Sample $X_i \sim q$, for $i = 1, \dots, N$.
3. Compute unnormalised weights $\widetilde{W}_i = \widetilde{w}(X_i)$.
4. Compute normalised weights $W_i = \widetilde{W}_i / \sum_{j=1}^N \widetilde{W}_j$.
5. Output $\widehat{\pi(\varphi)}^{SNIS} = \sum_{i=1}^N W_i\varphi(X_i)$.

REMARK – Remark

Multiplying all unnormalised weights by a constant does not change the estimator, because the constant cancels in the normalisation.

If $q = \pi$, then $\widetilde{w}(X_i) = Z$, $W_i = 1/N$, and SNIS reduces to vanilla Monte Carlo.

In measure notation, $\pi_N^{SNIS}(dx) = \sum_{i=1}^N W_i\delta_{X_i}(dx)$, which is a probability measure because $W_i \ge 0$ and $\sum_i W_i = 1$.

PROPOSITION – Proposition

Basic properties of SNIS.

Under the usual integrability assumptions:

1. $\widehat{\pi(\varphi)}^{SNIS}$ is biased for finite N;
2. the bias is of order $O(N^{-1})$;
3. if $\mathbb{E}_q[(\widetilde{w}(X)\varphi(X))^2] < +\infty$, then the variance is finite;
4. the variance and mean squared error are of order $O(N^{-1})$;
5. the estimator is consistent;
6. a central limit theorem holds under the standard finite-variance conditions for the weighted numerator and denominator.

Proof

For consistency, write $\widehat{\pi(\varphi)}^{SNIS} = \frac{N^{-1}\sum_{i=1}^N \widetilde{w}(X_i)\varphi(X_i)}{N^{-1}\sum_{i=1}^N \widetilde{w}(X_i)}$.

By the law of large numbers, $\frac{1}{N}\sum_{i=1}^N \widetilde{w}(X_i)\varphi(X_i) \to \int q(x)\widetilde{w}(x)\varphi(x)\,dx = \int \gamma(x)\varphi(x)\,dx$, and $\frac{1}{N}\sum_{i=1}^N \widetilde{w}(X_i) \to \int q(x)\widetilde{w}(x)\,dx = \int \gamma(x)\,dx = Z$.

Since $Z > 0$, the ratio converges to $\frac{\int \gamma(x)\varphi(x)\,dx}{\int \gamma(x)\,dx} = \int \pi(x)\varphi(x)\,dx = \pi(\varphi)$. ■

REMARK – Remark

The reason SNIS is biased is that it is a ratio of two unbiased estimators. In general, $\mathbb{E}\left[\frac{\widehat{A}}{\widehat{B}}\right] \neq \frac{\mathbb{E}[\widehat{A}]}{\mathbb{E}[\widehat{B}]}$.

Since SNIS is biased, performance should be evaluated using the mean squared error: $MSE(\widehat{\pi(\varphi)}^{SNIS}) = \mathbb{E}[(\widehat{\pi(\varphi)}^{SNIS} - \pi(\varphi))^2] = bias^2 + Var(\widehat{\pi(\varphi)}^{SNIS})$.

The rates are $Var(\widehat{\pi(\varphi)}^{SNIS}) \simeq \frac{C}{N}$, and $MSE(\widehat{\pi(\varphi)}^{SNIS}) \simeq \frac{C'}{N}$, for constants C, C' depending on π, q, φ.

8.5 Normalising constant estimation

SNIS also suggests a simple estimator of the normalising constant: $Z = \int \gamma(x)\,dx = \int q(x)\frac{\gamma(x)}{q(x)}\,dx = \mathbb{E}_q[\widetilde{w}(X)]$.

Thus $\widehat{Z}_N = \frac{1}{N}\sum_{i=1}^N \widetilde{w}(X_i)$, with $X_i \sim q$.

PROPOSITION – Proposition

The estimator $\widehat{Z}_N$ is unbiased and consistent.

Proof

$\mathbb{E}[\widehat{Z}_N] = \frac{1}{N}\sum_{i=1}^N \mathbb{E}_q[\widetilde{w}(X_i)] = \int \gamma(x)\,dx = Z$. Consistency follows from the law of large numbers. ■

REMARK – Remark

This is one of the practical advantages of SNIS over rejection sampling in the unnormalised setting: the same weighted sample can estimate both $\pi(\varphi)$ and the unknown normalising constant Z.

EXAMPLE – Example

Marginal likelihood.

Let $x \sim \rho(x)$ and $y \mid x \sim p(y \mid x)$. The marginal likelihood is $\ell(y) = \int \rho(x)p(y \mid x)\,dx$.

Vanilla Monte Carlo samples $X_i \sim \rho$ and estimates $\widehat{\ell}_N(y) = \frac{1}{N}\sum_{i=1}^N p(y \mid X_i)$, with $Var(\widehat{\ell}_N) = \frac{Var_\rho(p(y|X))}{N}$.

The zero-variance ordinary IS proposal is $q^\star(x) = \frac{\rho(x)p(y|x)}{\int \rho(x')p(y|x')\,dx'} = p(x \mid y)$, the posterior distribution of x given y.

Since the denominator is the unknown marginal likelihood, one instead tries to build a proposal close to this posterior and uses SNIS.

For the Gaussian observation model $Y = X + \varepsilon$, with $\varepsilon \sim \mathcal{N}(0, \sigma^2)$, one has $p(y \mid x) = \mathcal{N}(y; x, \sigma^2)$.

The practical recipe is: understand the shape of $\rho(x)p(y \mid x)$, choose q close to it, and then use SNIS to obtain a low-variance estimate of the marginal likelihood.

Chapter 9

Choice of Proposal

PROBLEM – Problem

Many Monte Carlo methods use a proposal distribution q. The practical question is how to choose q so that rejection sampling, ordinary importance sampling, or self-normalised importance sampling is stable and efficient.

SOLUTION – Solution

Choose a parametric family q_θ whose support contains the support of the target, and then optimise a criterion with respect to θ.

9.1 General strategy

Rejection sampling and importance sampling both use a proposal q, but the correction mechanism is different.

For rejection sampling one needs an envelope: $\pi(x) \leq Mq(x)$ for some $M \geq 1$ when the target is normalised, or $\gamma(x) \leq Mq(x)$ when the target is only known up to a normalising constant.

For ordinary importance sampling the basic support condition is $\mathrm{supp}(\pi) \subseteq \mathrm{supp}(q)$. A stronger useful condition is that the weights do not explode: $w(x) = \frac{\pi(x)}{q(x)} < +\infty$ in the relevant region.

For SNIS the same idea applies to the unnormalised weights: $\widetilde{w}(x) = \frac{\gamma(x)}{q(x)}$.

REMARK – Remark

A practical method is:

1. choose a parametric family q_θ whose support contains the support of π;
2. optimise a criterion with respect to θ.

The criterion depends on the algorithm:

- acceptance probability for rejection sampling;
- variance or second moment for importance sampling;
- asymptotic variance or effective sample size for SNIS;
- sometimes a divergence such as KL for general approximation.

9.2 Proposal choice for rejection sampling

Let q_θ be the proposal. For a fixed θ, the best envelope constant is $M_\theta = \sup_x \frac{\pi(x)}{q_\theta(x)}$ for a normalised target, or $M_\theta = \sup_x \frac{\gamma(x)}{q_\theta(x)}$ for an unnormalised target.

The acceptance rate is $1/M_\theta$ if π is normalised, or Z/M_θ when only the unnormalised target γ is available.

Hence the proposal parameter is selected by solving $\theta^\star = \arg\max_\theta \frac{1}{M_\theta}$, equivalently $\theta^\star = \arg\min_\theta M_\theta$.

REMARK – Remark

Equivalently, one often minimises $\log M_\theta$, because ratios of densities are easier and more stable in log-space.

EXAMPLE – Example

Gamma target with exponential proposal.

Let $\pi(x) = \frac{x^{\alpha-1}e^{-x}}{\Gamma(\alpha)}\mathbf{1}_{\{x>0\}}$, with $\alpha > 1$, and use the exponential proposal $q_\lambda(x) = \lambda e^{-\lambda x}\mathbf{1}_{\{x\geq 0\}}$.

The ratio is $R_\lambda(x) = \frac{\pi(x)}{q_\lambda(x)} = \frac{x^{\alpha-1}e^{-(1-\lambda)x}}{\lambda\Gamma(\alpha)}$.

For this ratio to be bounded, one needs $0 < \lambda < 1$. Indeed, if $\lambda \geq 1$, the exponential term does not dominate the polynomial term in the right tail.

Differentiate the log-ratio: $\frac{d}{dx}\log R_\lambda(x) = \frac{\alpha-1}{x} - (1-\lambda)$.

The maximiser in x is $x^\star_\lambda = \frac{\alpha-1}{1-\lambda}$.

Therefore $M_\lambda = R_\lambda(x^\star_\lambda) = \frac{\left(\frac{\alpha-1}{1-\lambda}\right)^{\alpha-1}e^{-(\alpha-1)}}{\lambda\Gamma(\alpha)}$.

The optimal λ minimises M_λ. Differentiating $-\log M_\lambda$ gives $\frac{1}{\lambda} + \frac{1-\alpha}{1-\lambda} = 0$, and hence $\lambda^\star = \frac{1}{\alpha}$.

At the optimum, $M_{\lambda^\star} = \frac{\alpha^\alpha e^{-(\alpha-1)}}{\Gamma(\alpha)}$, and acceptance rate $= \frac{1}{M_{\lambda^\star}} = \frac{\Gamma(\alpha)e^{\alpha-1}}{\alpha^\alpha}$.

The optimised rejection sampler samples $X' \sim \text{Exp}(1/\alpha)$, $U \sim \text{Unif}(0,1)$, and accepts X' when $U \leq \frac{\pi(X')}{M_{\lambda^\star}q_{\lambda^\star}(X')} = \left(\frac{X'}{\alpha}\right)^{\alpha-1}\exp\left[-\frac{\alpha-1}{\alpha}X' + \alpha - 1\right]$.

9.3 Proposal choice for ordinary IS

For ordinary IS, the target is to minimise the variance. With $X \sim q_\theta$, $Var_{q_\theta}(\widehat{\pi(\varphi)}^{IS}) = \frac{1}{N}\left(\mathbb{E}_{q_\theta}[w_\theta(X)^2\varphi(X)^2] - \pi(\varphi)^2\right)$, where $w_\theta(x) = \frac{\pi(x)}{q_\theta(x)}$.

The factor $1/N$ and the term $\pi(\varphi)^2$ do not change the minimiser. Therefore, within the family q_θ, one chooses $\theta^\star = \arg\min_\theta \mathbb{E}_{q_\theta}[w_\theta(X)^2\varphi(X)^2]$.

PROPOSITION – Proposition

The variance of the ordinary IS estimator is minimised by $q^\star(x) = \frac{\pi(x)|\varphi(x)|}{\int \pi(x')|\varphi(x')|\,dx'}$.

Proof

For one sample, the variance term is $Var_q(W\varphi(X)) = \mathbb{E}_q[(W\varphi(X))^2] - \left(\int \pi(x)\varphi(x)\,dx\right)^2$.

Using $W = \pi/q$, $\mathbb{E}_q[(W\varphi(X))^2] = \int \frac{\pi(x)^2\varphi(x)^2}{q(x)}\,dx$.

By Jensen's inequality, $\mathbb{E}_q\left[\left(\frac{\pi(X)\varphi(X)}{q(X)}\right)^2\right] \geq \left(\mathbb{E}_q\left[\left|\frac{\pi(X)\varphi(X)}{q(X)}\right|\right]\right)^2 = \left(\int \pi(x)|\varphi(x)|\,dx\right)^2$.

The proposal $q^\star$ attains this lower bound because $\frac{\pi(x)|\varphi(x)|}{q^\star(x)} = \int \pi(x')|\varphi(x')|\,dx'$ is constant. Therefore it minimises the variance. ■

With the optimal proposal, $Var(\widehat{\pi(\varphi)}^{IS}) = \frac{1}{N}\left[\left(\int \pi(x)|\varphi(x)|\,dx\right)^2 - \left(\int \pi(x)\varphi(x)\,dx\right)^2\right]$.

By Jensen's inequality, $\left(\int \pi(x)|\varphi(x)|\,dx\right)^2 \leq \int \pi(x)\varphi(x)^2\,dx$, so IS with the optimal proposal gives a variance no larger than the vanilla Monte Carlo variance.

REMARK – Remark

Zero variance for nonnegative test functions.

If $\varphi \geq 0$, then $|\varphi| = \varphi$ and the optimal proposal gives $Var_q(W\varphi(X)) = 0$.

In practice this construction is usually unavailable because it requires $\int \pi(x)|\varphi(x)|\,dx$, which is precisely the type of integral we are trying to estimate.

REMARK – Remark

Without the optimal proposal, the variance comparison gives the heuristic:

- where $\varphi(x)^2\pi(x)$ is large, choose $q(x) > \pi(x)$;
- where $\varphi(x)^2\pi(x)$ is small, choose $q(x) < \pi(x)$.

This is why IS can be more difficult to apply than control variates or antithetic sampling: we need some knowledge of the important regions of the integrand.

EXAMPLE – Example

Gaussian target with Gaussian proposal.

Let $\pi(x) = \mathcal{N}(x; 0, 1)$, and $q(x) = \mathcal{N}(x; 0, \sigma^2)$.

Then $w(x) = \frac{\pi(x)}{q(x)} = \sigma \exp\left\{\frac{x^2}{2}\left(\frac{1}{\sigma^2} - 1\right)\right\}$.

If $\sigma = 1$, then $q = \pi$ and $w = 1$. As $\sigma \to 0$, the weights explode. As $\sigma \to +\infty$, the weights are bounded above, but the sampler may become inefficient because many samples fall in regions where the target has almost no mass.

9.4 Lecture 8.05: optimal proposal for an exponential target

EXAMPLE – Example

One page of the lecture notes works out an optimal proposal for an importance-sampling exercise. The target is an exponential density on $[0, +\infty)$: $\pi(x) = \lambda e^{-\lambda x}\mathbf{1}_{\{x\geq 0\}}$, and the test function is approximately $\varphi(x) = \exp\left[-\frac{(x-10)^2}{2}\right]$.

The zero-variance ordinary IS proposal is proportional to $|\varphi(x)|\pi(x)$: $q^\star(x) \propto \exp\left[-\frac{(x-10)^2}{2} - \lambda x\right]\mathbf{1}_{\{x\geq 0\}}$.

Completing the square, $(x-10)^2 + 2\lambda x = x^2 + (2\lambda - 20)x + 100 = (x - (10-\lambda))^2 + 100 - (10-\lambda)^2$.

Therefore $q^\star(x) \propto \exp\left[-\frac{1}{2}(x - (10-\lambda))^2\right]\mathbf{1}_{\{x\geq 0\}}$.

Thus $q^\star$ is a normal distribution with mean $10 - \lambda$ and variance 1, truncated to $[0, +\infty)$.

A simple simulation recipe is:

1. sample $Y \sim \mathcal{N}(10 - \lambda, 1)$;
2. accept Y if $Y > 0$, otherwise resample.

9.5 Minimum-variance exponential proposal

EXAMPLE – Example

General exponential rare event.

Let $X \sim \text{Exp}(\lambda)$ and estimate $\mathbb{P}(X > K) = \int_K^\infty \lambda e^{-\lambda x}\,dx$.

Use an exponential proposal $q_\mu(x) = \mu e^{-\mu x}\mathbf{1}_{\{x\geq 0\}}$.

The second moment term is $g(\mu) = \mathbb{E}_{q_\mu}[w(X)^2\mathbf{1}_{\{X>K\}}] = \frac{\lambda^2}{\mu(2\lambda-\mu)}e^{-(2\lambda-\mu)K}$, with $0 < \mu < 2\lambda$.

Ignoring constants independent of μ, $\log g(\mu) = -\log\mu - \log(2\lambda - \mu) + \mu K$.

The first-order condition is $-\frac{1}{\mu} + \frac{1}{2\lambda-\mu} + K = 0$, or $K\mu^2 - 2K\lambda\mu - 2\mu + 2\lambda = 0$.

The admissible root is $\mu^\star = \frac{K\lambda+1-\sqrt{K^2\lambda^2+1}}{K}$.

For large K, this rate is smaller than λ, hence the proposal has a heavier right tail and samples the rare-event region more often.

EXAMPLE – Example

Lecture 11.05 special case: $X \sim \text{Exp}(2)$ and $K = 6$.

Let $X \sim \text{Exp}(2)$, and estimate $\mathbb{P}(X > K) = \mathbb{P}(X > 6)$.

Use proposals $q_\theta(x) = \theta e^{-\theta x}\mathbf{1}_{\{x\geq 0\}}$.

The weight is $w_\theta(x) = \frac{2e^{-2x}}{\theta e^{-\theta x}} = \frac{2}{\theta}e^{-(2-\theta)x}$.

For the indicator $\varphi(x) = \mathbf{1}_{\{x>K\}}$, $\mathbb{E}_{q_\theta}[w_\theta(X)^2\varphi(X)^2] = \frac{4}{\theta(4-\theta)}e^{-K(4-\theta)}$, with $0 < \theta < 4$.

Differentiating its logarithm gives $-\frac{1}{\theta} + \frac{1}{4-\theta} + K = 0$, or $K\theta^2 - 2(2K+1)\theta + 4 = 0$.

The admissible minimiser is $\theta^\star = \frac{2K+1-\sqrt{4K^2+1}}{K}$.

For $K = 6$, $\theta^\star = \frac{13-\sqrt{145}}{6} \simeq 0.1597$.

The probability is $\mathbb{P}(X > 6) = e^{-12} \simeq 6.14 \cdot 10^{-6}$.

This explains why plain Monte Carlo often returns zero at moderate sample sizes, while the optimised IS proposal samples much more often in the relevant tail.

9.6 Lecture 8.05: shifted exponential proposal for a Gaussian tail

The notes revisit an importance-sampling exercise for a truncated Gaussian tail. Define $\pi_c(x) = \frac{\varphi_0(x)\mathbf{1}_{\{x\geq c\}}}{\overline{\Phi}(c)}$, where $\varphi_0(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$, $\overline{\Phi}(c) = \mathbb{P}(Z \geq c)$, and $Z \sim \mathcal{N}(0,1)$.

Use the shifted exponential proposal $g_c(x) = e^{-(x-c)}\mathbf{1}_{\{x\geq c\}}$.

The weight is $w(x) = \frac{\pi_c(x)}{g_c(x)} = \frac{1}{\sqrt{2\pi}\,\overline{\Phi}(c)}\exp\left(-\frac{x^2}{2} + x - c\right)$.

It satisfies $\mathbb{E}_g[w(X)] = 1$.

For the second moment, $\mathbb{E}_g[w(X)^2] = \int_c^\infty \frac{\pi_c(x)^2}{g_c(x)}\,dx = \frac{1}{2\pi\,\overline{\Phi}(c)^2}\int_c^\infty e^{-x^2+x-c}\,dx$.

Completing the square, $x^2 - x + c = (x - 1/2)^2 + c - \frac{1}{4}$, so $\mathbb{E}_g[w(X)^2] = \frac{e^{-c+1/4}}{2\sqrt{\pi}\,\overline{\Phi}(c)^2}\overline{\Phi}(\sqrt{2}(c - 1/2))$.

Therefore $Var_g(w) = \mathbb{E}_g[w^2] - 1$.

REMARK – Remark

As $c \to +\infty$, this variance grows. This explains why rare-event proposals still need to be chosen carefully.

9.7 Optimal proposal for SNIS

PROPOSITION – Proposition

The asymptotic variance of the SNIS estimator is minimised by a proposal proportional to $q^\star(x) \propto \pi(x)|\varphi(x) - \pi(\varphi)|$.

REMARK – Remark

This proposal is usually not directly usable because it depends on the unknown target expectation $\pi(\varphi)$. It is nevertheless useful as a guide: a good proposal should resemble the regions where $\pi(x)|\varphi(x) - \pi(\varphi)|$ is large and where the weighted contribution to the asymptotic variance is important.

EXAMPLE – Example

Optimal SNIS proposal for an indicator.

Let A be a measurable set and let $\varphi(x) = \mathbf{1}_A(x)$. Then $\pi(\varphi) = \int \mathbf{1}_A(x)\pi(x)\,dx = \pi(A)$.

For SNIS, the asymptotically optimal proposal is $q^\star(x) \propto \pi(x)\,|\mathbf{1}_A(x) - \pi(A)|$.

To understand why the final expression has this form, set $p = \pi(A)$. Then, if $x \in A$, we have $\mathbf{1}_A(x) = 1$. Therefore $|\mathbf{1}_A(x) - \pi(A)| = |1 - p| = 1 - p$. So, before normalisation, $q^\star(x) \propto \pi(x)(1-p), \qquad x \in A$.

If instead $x \notin A$, then $\mathbf{1}_A(x) = 0$. Therefore $|\mathbf{1}_A(x) - \pi(A)| = |0 - p| = p$.

So, before normalisation, $q^\star(x) \propto \pi(x)p, \qquad x \notin A$.

The normalising constant is $C = \int \pi(x)\,|\mathbf{1}_A(x) - p|\,dx$. Splitting the integral over A and A^c, we obtain

$$C = \int_A \pi(x)(1-p)\,dx + \int_{A^c} \pi(x)p\,dx.$$

Since $\int_A \pi(x)\,dx = p, \qquad \int_{A^c} \pi(x)\,dx = 1 - p$, we get $C = (1-p)p + p(1-p) = 2p(1-p)$.

Hence, after normalisation, $q^\star(x) = \begin{cases} \dfrac{\pi(x)(1-p)}{2p(1-p)}, & x \in A, \\ \dfrac{\pi(x)p}{2p(1-p)}, & x \notin A. \end{cases}$

Therefore $q^\star(x) = \begin{cases} \dfrac{\pi(x)}{2\pi(A)}, & x \in A, \\ \dfrac{\pi(x)}{2(1-\pi(A))}, & x \notin A. \end{cases}$

Notice the important point: inside A, the factor $1 - \pi(A)$ appears before normalisation, but it cancels with the same factor in the normalising constant. Outside A, the factor $\pi(A)$ appears before normalisation, but it also cancels with the same factor in the normalising constant.

Finally, $\int_A q^\star(x)\,dx = \int_A \dfrac{\pi(x)}{2\pi(A)}\,dx = \dfrac{1}{2\pi(A)} \int_A \pi(x)\,dx = \dfrac{1}{2}$,

and $\int_{A^c} q^\star(x)\,dx = \int_{A^c} \dfrac{\pi(x)}{2(1-\pi(A))}\,dx = \dfrac{1}{2(1-\pi(A))} \int_{A^c} \pi(x)\,dx = \dfrac{1}{2}$.

Thus the optimal SNIS proposal allocates one half of its mass inside A and one half outside A. This is especially useful when A is a rare event, because the proposal forces the sampler to explore both the event A and its complement A^c.

9.8 Effective Sample Size

DEFINITION – Definition

Effective sample size.

The effective sample size of the normalised weights is $ESS = \dfrac{1}{\sum_{i=1}^N W_i^2} = \dfrac{\left(\sum_{i=1}^N \widetilde{w}(X_i)\right)^2}{\sum_{i=1}^N \widetilde{w}(X_i)^2}$.

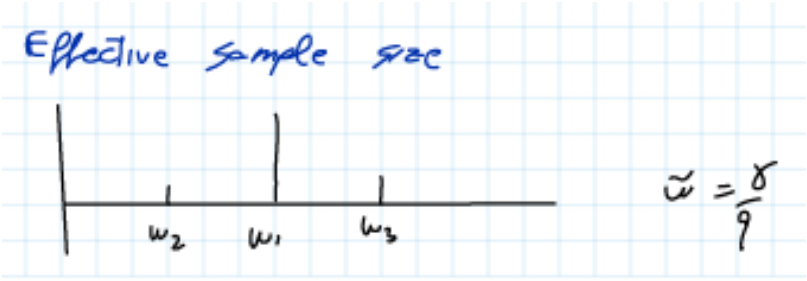


Figure 9.1: Effective sample size measures how many equally weighted samples the weighted sample behaves like.

PROPOSITION – Proposition

The effective sample size satisfies $1 \leq ESS \leq N$. Moreover, $ESS = N$ if and only if $W_i = 1/N$ for all i.

Proof

Since $0 \leq W_i \leq 1$ and $\sum_i W_i = 1$, $0 \leq \sum_{i=1}^N W_i^2 \leq \sum_{i=1}^N W_i = 1$, so $ESS \geq 1$.

For the upper bound, Cauchy–Schwarz gives $\left(\sum_{i=1}^N \widetilde{w}(X_i)\right)^2 = \left(\sum_{i=1}^N 1 \cdot \widetilde{w}(X_i)\right)^2 \leq N \sum_{i=1}^N \widetilde{w}(X_i)^2$.

Therefore $ESS \leq N$. Equality holds when the weights are constant. ■

REMARK – Remark

The closer the ESS is to N, the better. A practical warning rule is: $ESS < N/2 \Rightarrow$ the proposal is not working well. In that case too much mass is carried by too few weighted samples.

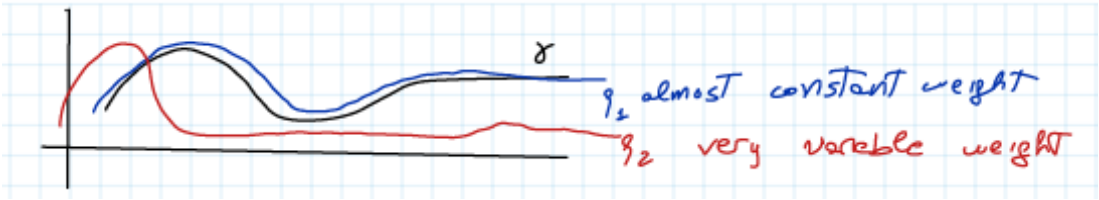


Figure 9.2: A proposal q_1 close to the target gives nearly constant weights; a poor proposal q_2 gives highly variable weights.

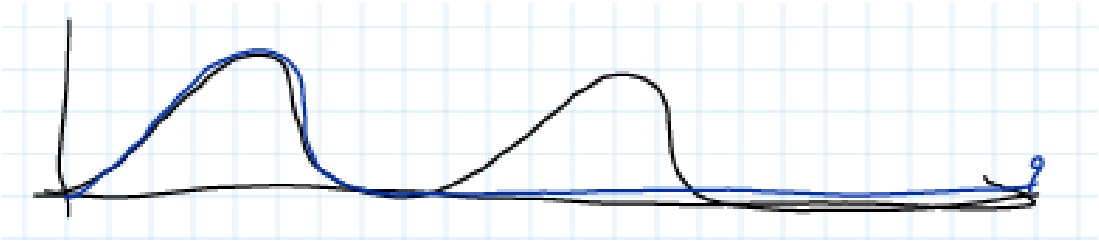


Figure 9.3: When the proposal misses an important region of the target, only a small number of weighted particles drive the estimator.

REMARK – Remark

Checklist for a usable proposal:

1. $\mathrm{supp}(\pi) \subseteq \mathrm{supp}(q)$;
2. the unnormalised weights $\widetilde{w}(x) = \gamma(x)/q(x)$ should not explode;
3. the effective sample size should not be too small relative to N.

EXAMPLE – Example

ESS with five weights.

For $N = 5$, take $W = \left(\frac{7}{10}, \frac{1}{10}, \frac{2}{10}, 0, 0\right)$.

Then $ESS = \left(\frac{49}{100} + \frac{1}{100} + \frac{4}{100}\right)^{-1} = \frac{100}{54} \simeq 1.85$.

Since $N/2 = 2.5$, this is a poor set of weights.

EXAMPLE – Example

Gaussian proposal scale for SNIS.

Let the unnormalised target be $\gamma(x) = \exp(-x^2/2)$, and let $q(x) = \mathcal{N}(x; 0, \sigma^2)$.

Then $\widetilde{w}(x) = \frac{\gamma(x)}{q(x)} = \sqrt{2\pi}\,\sigma \exp\left\{\frac{x^2}{2}\left(\frac{1}{\sigma^2} - 1\right)\right\}$.

If $\sigma = 1$, then $q = \pi$ and the weights are constant.

If $\sigma > 1$, the exponent is negative and the weights are bounded, although very large σ^2 can still give low ESS because many samples fall in regions with negligible target mass.

If $\sigma < 1$, the exponent is positive and the weights explode in the tails, which is dangerous for SNIS.

Thus σ^2 close to 1 gives high ESS, moderate over-dispersion can still be stable, and severe over-dispersion lowers ESS because many samples receive almost zero weight.

9.9 A simpler robust criterion

If $|\varphi(x)| \le \overline{\varphi}$, then $\mathbb{E}_q[w(X)^2\varphi(X)^2] \le \overline{\varphi}^2\mathbb{E}_q[w(X)^2]$.

REMARK – Remark

Thus one can sometimes choose q by minimising the second moment of the weights, or equivalently by making q close to π. This loses the integrand-specific optimality, but can be easier to tune.

9.10 Kullback–Leibler criterion and variational approximation

Another proposal-selection criterion is the Kullback–Leibler divergence. In the continuous case, $KL(q_\theta\|\pi) = \int q_\theta(x) \log \frac{q_\theta(x)}{\pi(x)}\,dx$.

In the discrete case, $KL(q_\theta\|\pi) = \sum_{k=1}^{K} q_\theta(x_k) \log \frac{q_\theta(x_k)}{\pi(x_k)}$.

The variational approximation is $q_{\theta^\star} = \arg\min_{q_\theta} KL(q_\theta\|\pi)$.

REMARK – Remark

The direction $KL(q\|\pi)$ is the classical variational direction. It tends to penalise putting mass where π is small. This can lead to approximations that focus on one high-density region when the target is multimodal.

EXAMPLE – Example

Gaussian variational approximation.

Let $Z \sim \mathcal{N}(0,1)$, and $Y \mid Z = z \sim \mathcal{N}(z,1)$.

Then $p(z \mid y) \propto p(z)p(y \mid z)$.

Completing the square gives $p(z \mid y) = \mathcal{N}\left(z; \frac{y}{2}, \frac{1}{2}\right)$.

Consider the approximating family $q_\phi(z) = \mathcal{N}\left(z; ay+b, \frac{2}{3}\right)$, with $\phi = (a,b)$.

For univariate Gaussians, $KL\left(\mathcal{N}(m_1, s_1^2)\middle\|\mathcal{N}(m_2, s_2^2)\right) = \frac{1}{2}\left[\log \frac{s_2^2}{s_1^2} + \frac{s_1^2+(m_1-m_2)^2}{s_2^2} - 1\right]$.

Only the mean term depends on a, b, so one minimises $\left(ay+b-\frac{y}{2}\right)^2$.

The optimum is $a^\star = \frac{1}{2}$, and $b^\star = 0$.

Thus the best member of the chosen variational family is $q_{\phi^\star}(z) = \mathcal{N}\left(z; \frac{y}{2}, \frac{2}{3}\right)$, whereas the exact posterior has variance $1/2$.

Parte II

Appunti unificati e practical

Edizione complementare conservata integralmente. Include esercitazioni pratiche con testo completo.

Numerical and Statistical Methods for Finance

Unofficial Unified Notes and Exercises

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Chapter 1

1_EDCFandPlugIn — ECDF & Plug-In Principle

In this first part of this module we focus on statistical inference, and develop computational methods which allow inferences to be based on less strict assumptions than the likelihood-based methods introduced in the first semester.[1]

1.1 Statistical inference about distributions

This chapter studies methods for making inferences about probability distributions. Given a sample of data, we may wish to:

- check whether the distribution has a particular parametric form;
- estimate its cumulative distribution function;
- estimate its probability density function without making a specific distributional assumption;
- estimate quantities of interest, such as the mean, the variance, or other functionals of the distribution.

Suppose

$$y_1, \ldots, y_n$$

are iid observations from a random variable Y with cumulative distribution function F.

Remark. *When we only assume that the observations are iid, we are naturally led to a nonparametric approach. By contrast, likelihood-based procedures such as maximum likelihood estimation are typically associated with parametric approaches, where a specific family for the distribution is assumed in advance.*

1.2 The empirical cumulative distribution function

A useful tool for learning about the unknown distribution of a random variable Y is the empirical cumulative distribution function.

Definition. *For univariate data $y_1, \ldots, y_n$, the empirical cumulative distribution function (ecdf) is defined for every $y \in \mathbb{R}$ by*

$$\hat{F}_n(y) = \frac{\#\{i : y_i \leq y\}}{n},$$

where $\#A$ denotes the number of elements in the set A.

Equivalently,

$$\hat{F}_n(y) = \frac{\text{n}^\circ \text{ of data points smaller than or equal to } y}{\text{total n}^\circ \text{ of data points}}.$$

Thus $\hat{F}_n(y)$ is the proportion of observations smaller than or equal to y.

Example. *Consider the observations*

$$2.0,\ 2.1,\ 2.8,\ 3.6,\ 4.9,\ 6.1,\ 7.2,\ 8.3.$$

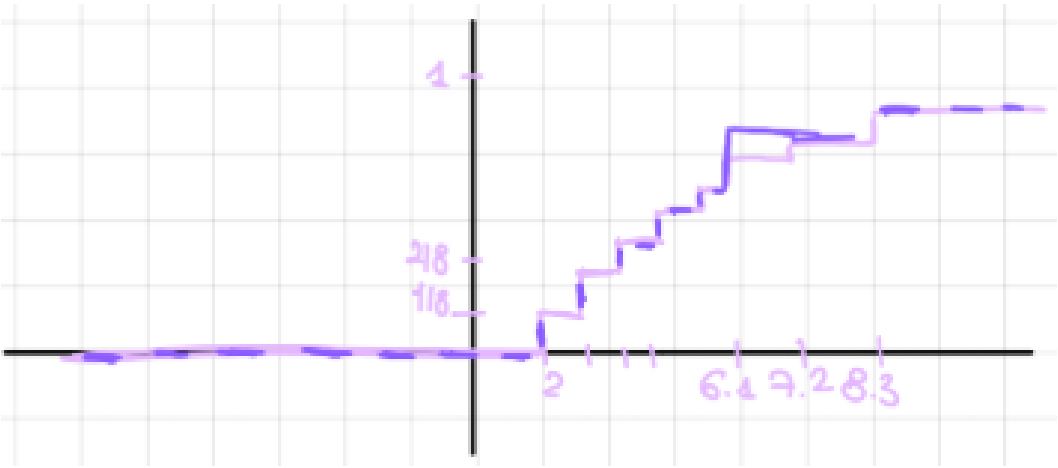
[1] These notes are based on lecture notes developed by (in alphabetic order): Ö. Deniz Akyildiz, Nicolas Chopin, Heiko Großmann, Adam M. Johansen, Amir Khorrami Chokami, Nikolas Nüsken, Art Owen. Many thanks to them for making their material available.

Then

$$\hat{F}_n(y) = \begin{cases} 0, & y < 2.0, \\ 1/8, & 2.0 \le y < 2.1, \\ 2/8, & 2.1 \le y < 2.8, \\ 3/8, & 2.8 \le y < 3.6, \\ 4/8, & 3.6 \le y < 4.9, \\ 5/8, & 4.9 \le y < 6.1, \\ 6/8, & 6.1 \le y < 7.2, \\ 7/8, & 7.2 \le y < 8.3, \\ 1, & y \ge 8.3. \end{cases}$$

So the ecdf is a step function, with jumps of size $1/8$ *at the observed sample points.*

If some observations are repeated, the jump at that value is larger. For instance, if the sample contains two equal observations at 7.2, then the ecdf jumps by $2/n$ at 7.2.



Proposition. *The ECDF satisfies the following properties:*

1. $\hat{F}_n(y) \ge 0$ *for every* $y \in \mathbb{R}$*;*
2. $\hat{F}_n(y) \le 1$ *for every* $y \in \mathbb{R}$*;*
3. $\hat{F}_n$ *is increasing;*
4. $\hat{F}_n$ *is right-continuous;*
5. $\hat{F}_n(y) = 0$ *for values below the smallest sample point;*
6. $\hat{F}_n(y) = 1$ *for all* $y \ge y_{(n)}$*, where* $y_{(n)}$ *is the largest sample point.*

For a generic cdf F, we also recall the boundary conditions

$$\lim_{y \to -\infty} F(y) = 0, \qquad \lim_{y \to +\infty} F(y) = 1.$$

Remark. *An equivalent expression for the ECDF is*

$$\hat{F}_n(y) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}_{(-\infty, y]}(y_i),$$

where

$$\mathbf{1}_{(-\infty,y]}(t) = \begin{cases} 1, & \text{if } t \in (-\infty, y], \\ 0, & \text{otherwise.} \end{cases}$$

This representation is very useful because it allows us to study the ecdf using Bernoulli and Binomial random variables.

Fix $y \in \mathbb{R}$. If Y_i is a random variable, then

$$Z_i := \mathbf{1}_{(-\infty, y]}(Y_i)$$

is a Bernoulli random variable with parameter

$$p = \mathbb{P}(Z_i = 1) = \mathbb{P}(Y_i \le y) = F(y).$$

Therefore,

$$Z_i \sim \text{Bernoulli}(F(y)).$$

Since the Y_i are iid, the variables $Z_1, \ldots, Z_n$ are iid Bernoulli with parameter $F(y)$. Hence

$$\sum_{i=1}^{n} \mathbf{1}_{(-\infty,y]}(Y_i) \sim \text{Binomial}(n, F(y)).$$

This gives immediately the expectation of the ecdf:

$$\mathbb{E}[\hat{F}_n(y)] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} \mathbf{1}_{(-\infty,y]}(Y_i)\right] = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}[\mathbf{1}_{(-\infty,y]}(Y_i)] = \frac{1}{n}\sum_{i=1}^{n} F(y) = F(y).$$

Proposition. *For every fixed $y \in \mathbb{R}$, the ecdf $\hat{F}_n(y)$ is an unbiased estimator of $F(y)$.*

Its variance is

$$\text{Var}(\hat{F}_n(y)) = \text{Var}\left[\frac{1}{n}\sum_{i=1}^{n} \mathbf{1}_{(-\infty,y]}(Y_i)\right] = \frac{1}{n^2}\sum_{i=1}^{n} \text{Var}(\mathbf{1}_{(-\infty,y]}(Y_i)) = \frac{1}{n^2} \cdot n \cdot F(y)(1 - F(y)).$$

Hence

$$\text{Var}(\hat{F}_n(y)) = \frac{F(y)(1 - F(y))}{n}.$$

Proposition. *For every fixed $y \in \mathbb{R}$,*

$$\text{Var}(\hat{F}_n(y)) = \frac{1}{n}F(y)(1 - F(y)).$$

Since $\hat{F}_n(y)$ is unbiased,

$$\text{MSE}(\hat{F}_n(y)) = \text{Var}(\hat{F}_n(y)).$$

Moreover, for every fixed y,

$$\text{Var}(\hat{F}_n(y)) \to 0 \qquad \text{as } n \to \infty,$$

so $\hat{F}_n(y)$ is consistent.

1.3 The empirical probability density function

The ecdf is the exact cdf of a discrete random variable $\tilde{Y}$ which only takes values among the observed data points. This suggests a corresponding empirical probability density function.

Definition. *Given univariate data $y_1, \ldots, y_n$, the empirical probability density function is the function*

$$\hat{f}_n : \mathbb{R} \to [0, 1]$$

defined by

$$\hat{f}_n(y) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}_{\{y_i = y\}} = \frac{\#\{i : y_i = y\}}{n}.$$

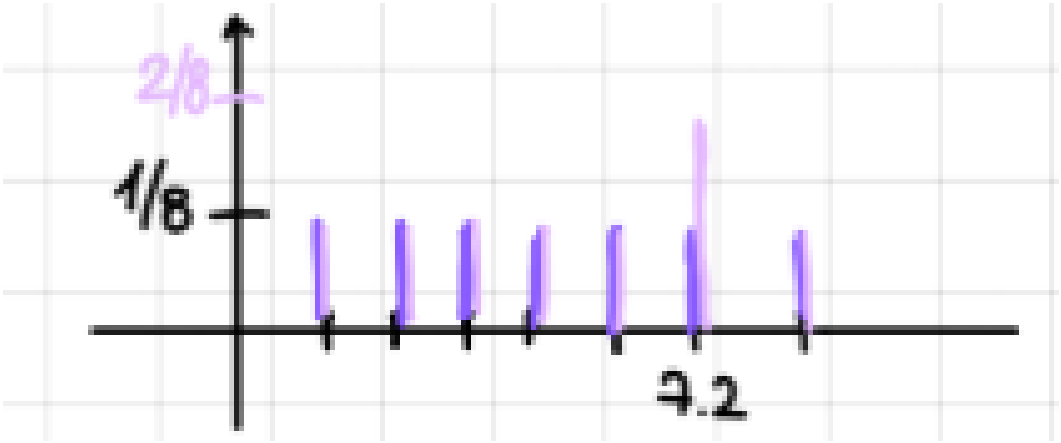
Thus, for every distinct sample value y, the quantity $\hat{f}_n(y)$ is the proportion of observations equal to y.
If all data points are distinct, then

$$\hat{f}_n(y_i) = \frac{1}{n} \qquad \text{for each } i,$$

and

$$\hat{f}_n(y) = 0 \qquad \text{for all other } y.$$

Repeated observations are reflected through multiplicities.



The empirical pdf has some undesirable features if the underlying distribution is continuous:

1. it is not continuous;
2. it is zero everywhere except at the data points;
3. it does not behave like a continuous density in the sense that

$$\int_{-\infty}^{+\infty} \hat{f}_n(y)\, dy \neq 1.$$

Remark. *Even if the empirical pdf is not a good estimator of a continuous density, it is still very useful for estimating expectations, moments, and other distributional quantities through the plug-in principle.*

1.4 Estimating expectations and integrals

Let ψ be a test function. Then

$$\mathbb{E}[\psi(Y)] = \int_{-\infty}^{+\infty} f(y)\psi(y)\, dy.$$

A natural empirical approximation is obtained by replacing f with $\hat{f}_n$:

$$\hat{\mathbb{E}}_n[\psi(Y)] = \int_{-\infty}^{+\infty} \hat{f}_n(y)\psi(y)\, dy.$$

If the data points are distinct, since the empirical pdf is nonzero only at the sample points, this becomes

$$\hat{\mathbb{E}}_n[\psi(Y)] = \sum_{i=1}^{n} \hat{f}_n(y_i)\psi(y_i) = \frac{1}{n}\sum_{i=1}^{n} \psi(y_i).$$

This is the key idea behind the plug-in principle: replace the unknown distribution by its empirical version.

Example (Empirical mean)**.** *Take $\psi(y) = y$. Then*

$$\hat{\mathbb{E}}_n[Y] = \frac{1}{n}\sum_{i=1}^{n} y_i = \bar{y}.$$

So the sample mean is the plug-in estimator of the population mean.

Example (Empirical variance)**.** *Since*

$$\mathrm{Var}(Y) = \mathbb{E}[(Y - \mathbb{E}[Y])^2],$$

its empirical analogue is

$$\begin{aligned}
\widehat{\mathrm{Var}}_n(Y) &= \hat{\mathbb{E}}_n\left[(Y - \hat{\mathbb{E}}_n[Y])^2\right] \\
&= \hat{\mathbb{E}}_n\left[(Y - \bar{y})^2\right] \\
&= \sum_{i=1}^{n} \hat{f}_n(y_i)(y_i - \bar{y})^2 \\
&= \frac{1}{n}\sum_{i=1}^{n} (y_i - \bar{y})^2.
\end{aligned}$$

This differs from the usual unbiased sample variance, which uses the factor $1/(n-1)$ instead of $1/n$.

1.5 Histogram estimator

Because the empirical pdf is made of point masses, it is not satisfactory as an estimator of a continuous pdf. A more useful alternative is the histogram estimator.

Suppose that we observe an empirical sample

$$Y_1, \dots, Y_n$$

from a distribution with unknown continuous density f. Here:

- f is the true density, unknown;
- n is the sample size;
- $Y_1, \dots, Y_n$ are the observed data.

Assume that the density f has support on a finite interval $[a, b]$. Partition this interval into m non-overlapping bins

$$T_k = (t_k, t_{k+1}], \qquad k = 1, \dots, m,$$

where

$$a = t_1 < t_2 < \cdots < t_{m+1} = b.$$

Thus:

- m is the number of bins;
- $t_1, \dots, t_{m+1}$ are the endpoints of the bins;
- for each $k = 1, \dots, m$, T_k is the kth bin.

Let

$$h_k = t_{k+1} - t_k, \qquad k = 1, \dots, m,$$

be the width of the kth bin, and let

$$n_k = \#\{i : Y_i \in T_k\}, \qquad k = 1, \dots, m,$$

be the number of data points in the bin T_k.

So:

- h_k is the horizontal length of the interval T_k;
- n_k is a count, namely the number of observations falling in T_k;
- n_k is not the height of the histogram bar.

Definition. *The histogram estimator of f is*

$$\hat{f}_H(y) = \sum_{k=1}^{m} \frac{n_k}{n\, h_k} \mathbf{1}_{T_k}(y).$$

Here $\mathbf{1}_{T_k}(y)$ is the indicator function:

$$\mathbf{1}_{T_k}(y) = \begin{cases} 1, & y \in T_k, \\ 0, & y \notin T_k. \end{cases}$$

Hence, if y belongs to one specific bin, say

$$y \in T_\ell \quad \text{for some } \ell \in \{1, \dots, m\},$$

then all the terms in the sum are equal to zero except the one corresponding to $k = \ell$, and therefore

$$\hat{f}_H(y) = \frac{n_\ell}{n\, h_\ell}.$$

This notation should be read carefully:

- k is a generic index running over all bins $1, \dots, m$;
- ℓ is the specific index of the bin containing the point y;
- therefore $\ell \in \{1, \dots, m\}$.

It is important to interpret correctly the quantity

$$\frac{n_k}{n\, h_k}.$$

For the bar corresponding to the bin T_k:

- the base is h_k;

- the height is $\dfrac{n_k}{n\,h_k}$;
- the area is

$$h_k \cdot \frac{n_k}{n\,h_k} = \frac{n_k}{n}.$$

Thus the area of the kth bar is the empirical relative frequency of observations in T_k. This is why we divide by both n and h_k: the factor $1/n$ turns counts into relative frequencies, and the factor $1/h_k$ turns relative frequencies into density heights.

So $\hat{f}_H$ is again a step function, but unlike the empirical pdf it is constant on intervals rather than concentrated only at individual points. Also, once the empirical sample $Y_1, \ldots, Y_n$ has been observed, the counts $n_1, \ldots, n_m$ are fixed and the histogram becomes a concrete deterministic function. Before observing the sample, however, the counts $n_1, \ldots, n_m$ and hence $\hat{f}_H$ are random.

Proposition. *The histogram estimator satisfies:*

(a) $\hat{f}_H(y) \geq 0$ *for every* y*;*
(b) the total area under the graph of $\hat{f}_H$ *is equal to* 1.

Indeed, for each $k = 1, \ldots, m$,

$$\frac{n_k}{n\,h_k} \geq 0,$$

since $n_k \geq 0$, $n > 0$, and $h_k > 0$. Moreover, the area of the kth bar is

$$h_k \cdot \frac{n_k}{n\,h_k} = \frac{n_k}{n}.$$

Summing over all bins gives

$$\sum_{k=1}^{m} \frac{n_k}{n} = \frac{1}{n} \sum_{k=1}^{m} n_k = \frac{n}{n} = 1,$$

because every observation belongs to exactly one of the bins $T_1, \ldots, T_m$.

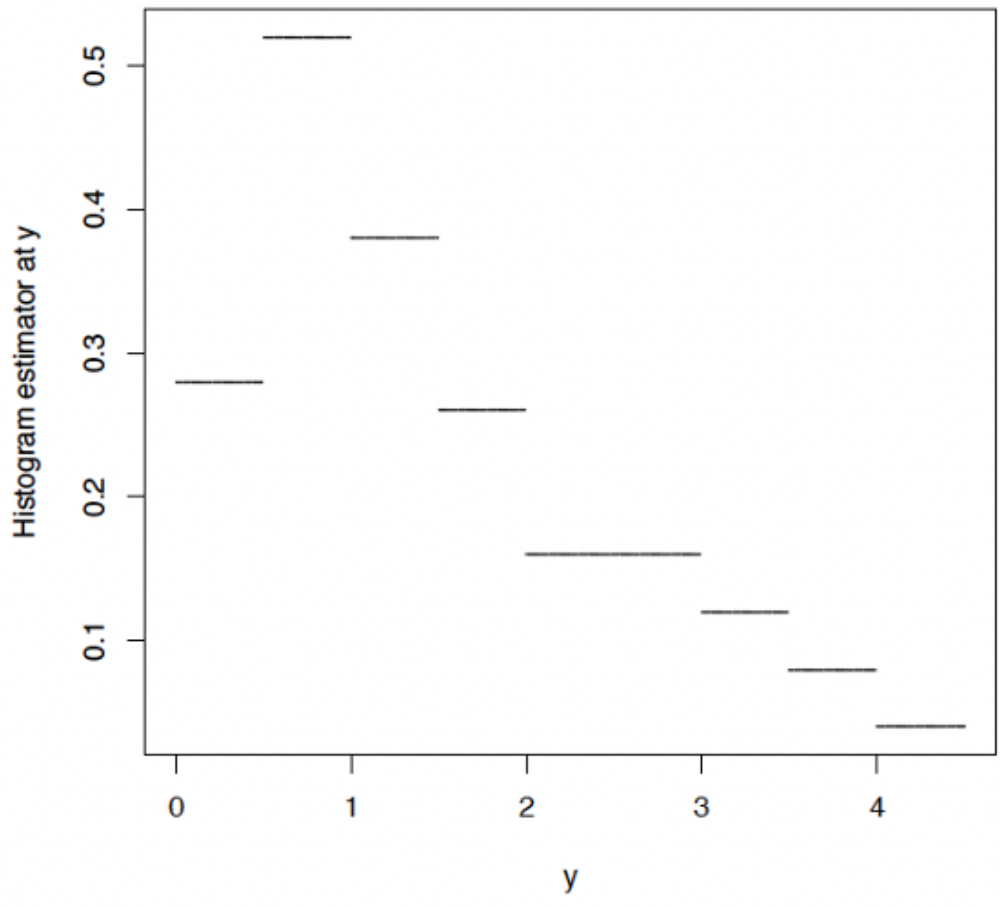


Figure 1.1: Histogram estimator.

Example. *Take* $[a, b] = (0, 4.5]$, $n = 100$, *and equally spaced bins with width*

$$h_k = 0.5, \qquad k = 1, \ldots, 9.$$

Suppose the frequencies are given by

Class interval	*Frequency*
$T_1 = (0.00, 0.50]$	*14*
$T_2 = (0.50, 1.00]$	*26*
$T_3 = (1.00, 1.50]$	*19*
$T_4 = (1.50, 2.00]$	*13*
$T_5 = (2.00, 2.50]$	*8*
$T_6 = (2.50, 3.00]$	*8*
$T_7 = (3.00, 3.50]$	*6*
$T_8 = (3.50, 4.00]$	*4*
$T_9 = (4.00, 4.50]$	*2*

Then, for example,

$$\frac{n_1}{nh_1} = \frac{14}{100 \cdot 0.5} = 0.28.$$

So in the first bin T_1:

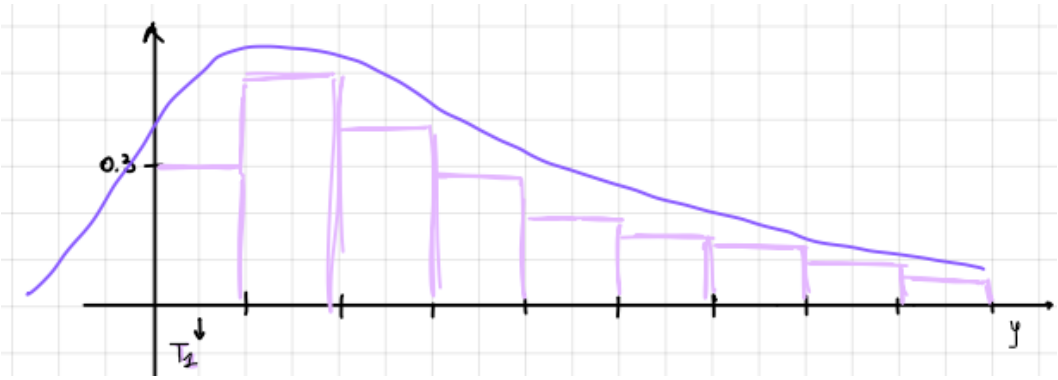
- *the base is $h_1 = 0.5$;*
- *the height is* 0.28*;*
- *the area is*

$$0.5 \cdot 0.28 = 0.14 = \frac{14}{100},$$

which is exactly the empirical relative frequency of the first bin.

The histogram estimator may be written as

$$\hat{f}_H(y) = \frac{1}{50}\left[14 \cdot 1_{(0.00,0.50]}(y) + 26 \cdot 1_{(0.50,1.00]}(y) + \cdots + 2 \cdot 1_{(4.00,4.50]}(y)\right].$$



A natural question is whether the histogram estimator is unbiased for the true density f.

Fix a point y, and suppose that

$$y \in T_\ell \quad \text{for some } \ell \in \{1, \ldots, m\}.$$

Then

$$\hat{f}_H(y) = \frac{n_\ell}{nh_\ell}.$$

Taking expectations,

$$\begin{aligned}
\mathbb{E}[\hat{f}_H(y)] &= \mathbb{E}\left[\frac{n_\ell}{nh_\ell}\right] \\
&= \frac{1}{nh_\ell}\mathbb{E}[n_\ell] \\
&= \frac{1}{nh_\ell}\mathbb{E}\left[\sum_{i=1}^{n}\mathbf{1}_{T_\ell}(Y_i)\right] \\
&= \frac{1}{nh_\ell}\sum_{i=1}^{n}\mathbb{E}[\mathbf{1}_{T_\ell}(Y_i)] \\
&= \frac{1}{nh_\ell}\sum_{i=1}^{n}\mathbb{P}(Y_i \in T_\ell) \\
&= \frac{1}{nh_\ell}\sum_{i=1}^{n}\big(F(t_{\ell+1}) - F(t_\ell)\big) \\
&= \frac{F(t_{\ell+1}) - F(t_\ell)}{h_\ell}.
\end{aligned}$$

Now,

$$F(t_{\ell+1}) - F(t_\ell) = \int_{t_\ell}^{t_{\ell+1}} f(u)\,du,$$

hence

$$\mathbb{E}[\hat{f}_H(y)] = \frac{1}{h_\ell}\int_{t_\ell}^{t_{\ell+1}} f(u)\,du.$$

Therefore $\mathbb{E}[\hat{f}_H(y)]$ is the average value of the true density over the whole bin T_ℓ, not the pointwise value $f(y)$.
In general,

$$\frac{F(t_{\ell+1}) - F(t_\ell)}{h_\ell} \neq f(y),$$

so the histogram estimator is biased.
Its bias is

$$\frac{F(t_{\ell+1}) - F(t_\ell)}{h_\ell} - f(y).$$

We now derive the variance formula carefully. Since

$$\hat{f}_H(y) = \frac{n_\ell}{nh_\ell},$$

and n and h_ℓ are constants, we have

$$\begin{aligned}
\mathrm{Var}(\hat{f}_H(y)) &= \mathrm{Var}\left(\frac{n_\ell}{nh_\ell}\right) \\
&= \frac{1}{n^2h_\ell^2}\,\mathrm{Var}(n_\ell).
\end{aligned}$$

Next, recall that

$$n_\ell = \sum_{i=1}^{n}\mathbf{1}_{T_\ell}(Y_i),$$

that is, n_ℓ counts how many observations fall in the bin T_ℓ.
Set

$$I_i = \mathbf{1}_{T_\ell}(Y_i), \qquad i = 1, \ldots, n.$$

Then

$$n_\ell = \sum_{i=1}^{n} I_i.$$

Each I_i is a Bernoulli random variable with parameter

$$p_\ell = \mathbb{P}(Y_i \in T_\ell) = F(t_{\ell+1}) - F(t_\ell).$$

Therefore,

$$I_i \sim \text{Bernoulli}(p_\ell), \qquad i = 1, \ldots, n,$$

and, since the observations $Y_1, \ldots, Y_n$ are independent, also $I_1, \ldots, I_n$ are independent.

Hence

$$\begin{aligned}\operatorname{Var}(n_\ell) &= \operatorname{Var}\left(\sum_{i=1}^{n} I_i\right) \\ &= \sum_{i=1}^{n} \operatorname{Var}(I_i) \qquad \text{(by independence)} \\ &= \sum_{i=1}^{n} p_\ell(1-p_\ell) \\ &= n\, p_\ell(1-p_\ell).\end{aligned}$$

Substituting this into the variance of $\hat{f}_H(y)$ gives

$$\begin{aligned}\operatorname{Var}(\hat{f}_H(y)) &= \frac{1}{n^2 h_\ell^2} \operatorname{Var}(n_\ell) \\ &= \frac{1}{n^2 h_\ell^2}\, n\, p_\ell(1-p_\ell) \\ &= \frac{p_\ell(1-p_\ell)}{n h_\ell^2}.\end{aligned}$$

Finally, replacing

$$p_\ell = F(t_{\ell+1}) - F(t_\ell),$$

we obtain

$$\operatorname{Var}(\hat{f}_H(y)) = \frac{\big[F(t_{\ell+1}) - F(t_\ell)\big]\big[1 - (F(t_{\ell+1}) - F(t_\ell))\big]}{n h_\ell^2}.$$

So the variance formula comes from two facts:

- $\hat{f}_H(y)$ is a constant multiple of n_ℓ;
- n_ℓ is a binomial count, namely

$$n_\ell \sim \operatorname{Bin}(n, p_\ell), \qquad p_\ell = F(t_{\ell+1}) - F(t_\ell).$$

Remark. *The histogram estimator illustrates the classical bias–variance trade-off. Smaller bin widths reduce the bias, because the average density over a small interval is closer to the pointwise density, but they increase the variance because fewer observations fall into each bin. Larger bin widths do the opposite: they reduce variability but increase smoothing and hence bias. This motivates more refined density estimators, such as kernel density estimators.*

A general result of Rosenblatt (1956) shows that any estimator $\hat{f}$ satisfying $\hat{f}(y) \geq 0$ for all y and total area equal to 1 must be biased for some density f.

1.6 Empirical quantiles

For a continuous random variable Y with cdf F, any value y_α satisfying

$$F(y_\alpha) = \alpha$$

is called an α-quantile.

For a general random variable, the following definition guarantees existence.

Definition. *Let Y be a random variable and let $\alpha \in (0,1)$. Every value y_α such that*

$$\mathbb{P}(Y \leq y_\alpha) \geq \alpha \qquad \text{and} \qquad \mathbb{P}(Y < y_\alpha) \leq 1-\alpha$$

is called an α-quantile of the distribution of Y.

Given a sample $y_1, \ldots, y_n$, let

$$y_{(1)} \leq y_{(2)} \leq \cdots \leq y_{(n)}$$

be the ordered sample.

Definition. *For $i = 1, \ldots, n$, the $(i-0.5)/n$-quantile is defined as $y_{(i)}$ and is called the ith sample quantile.*

Thus the ordered observations $y_{(1)}, \ldots, y_{(n)}$ are interpreted as empirical quantiles of the sample distribution.

1.7 Quantile–Quantile plots

Given iid data $y_1, \ldots, y_n$, we may wish to assess whether the sample comes from a specific continuous distribution.

Histograms are useful for detecting broad features such as symmetry, skewness, outliers, or kurtosis, but they depend strongly on the bin choice and are less effective for identifying subtle departures from a reference model.

A more informative graphical tool is the *quantile–quantile plot* or *Q–Q plot*.

The point is that, in practice, we only observe the sample and do not know the true density. The histogram gives a rough empirical visualization of the distribution shape, whereas the Q–Q plot is designed to compare the empirical sample with a theoretical distribution chosen in advance.

Let the reference distribution have cdf F, strictly increasing and continuous on the set where $0 < F(y) < 1$. The Q–Q plot consists of the points

$$\left(F^{-1}\left(\frac{i-0.5}{n}\right), y_{(i)} \right), \qquad i = 1, \ldots, n.$$

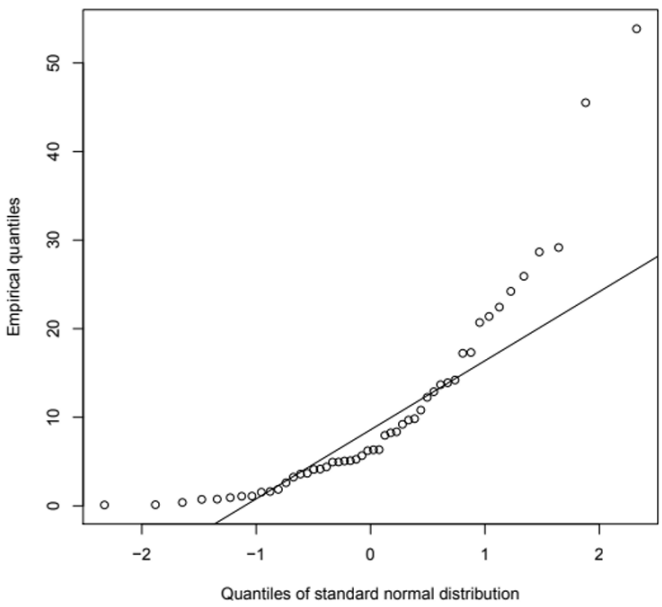
Thus we compare the empirical quantiles $y_{(i)}$ with the theoretical quantiles

$$F^{-1}\left(\frac{i-0.5}{n}\right).$$

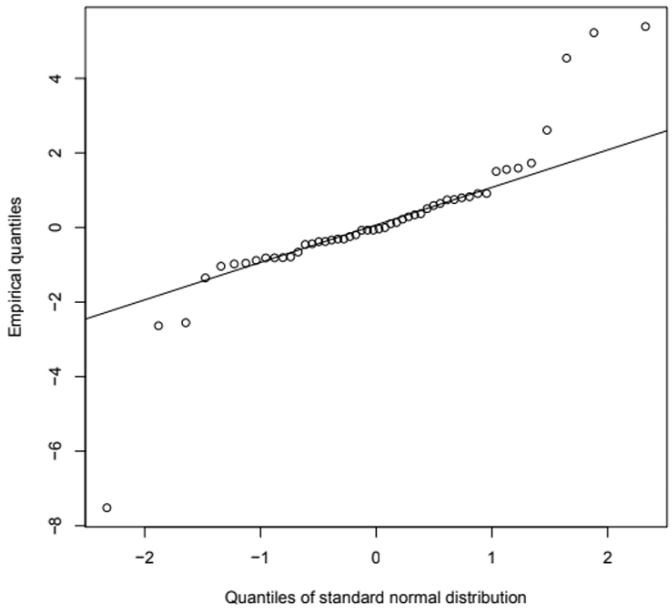
So:

- on the horizontal axis we place the theoretical quantiles;
- on the vertical axis we place the empirical quantiles.

If the chosen reference distribution describes the data well, then the empirical quantiles should be close to the theoretical quantiles, and the plotted points should lie approximately on a straight line.



If the dots align approximately on the straight line, the distribution is approximately normal.



This pattern is still reasonably close to normality, but the deviation in the upper tail may indicate a heavy-tailed distribution.

Figure 1.2: Q–Q plots for a positively skewed distribution and for a distribution with heavier tails than the normal distribution.

If the reference distribution is correct, the points should lie close to a straight line with intercept 0 and slope 1.

Typical deviations have an interpretation:

- positively skewed data often produce a convex shape, as happens for chi-square or exponential distributions;
- heavier tails than the normal distribution typically bend away from the line at the extremes, as in the case of a t-distribution.

Therefore, the main role of the Q–Q plot is not to estimate the true density, but to assess graphically whether the empirical sample is compatible with a chosen reference distribution.

1.8 The plug-in principle

Many quantities of interest can be viewed as functions of the underlying distribution F.

For example, if Y has cdf F and density f, then:

- its mean is
$$\mu = \mathbb{E}(Y) = \int_{-\infty}^{+\infty} y f(y)\, dy;$$
- its variance is
$$\sigma^2 = \mathrm{Var}(Y) = \int_{-\infty}^{+\infty} (y-\mu)^2 f(y)\, dy;$$
- for any test function ψ,
$$\mathbb{E}[\psi(Y)] = \int_{-\infty}^{+\infty} \psi(y) f(y)\, dy.$$

More generally, if a parameter of interest can be written as

$$\theta = t(F),$$

then the *plug-in principle* says that we estimate θ by replacing the unknown cdf F with the empirical cdf $\hat{F}_n$:

$$\hat{\theta} = t(\hat{F}_n).$$

Remark. *To emphasise the dependence on the underlying distribution, one may write θ_F for the population quantity and $\hat{\theta}_{\hat{F}_n}$ for the empirical plug-in estimator.*

Example. *Three standard plug-in estimators are:*

1. *If $\theta_F = \mathbb{E}(Y)$, then*
$$\hat{\theta}_{\hat{F}_n} = \mathbb{E}(\tilde{Y}) = \frac{1}{n}\sum_{i=1}^{n} y_i = \bar{y}.$$
2. *If $\theta_F = \mathrm{Var}(Y)$, then*
$$\hat{\theta}_{\hat{F}_n} = \mathrm{Var}(\tilde{Y}) = \frac{1}{n}\sum_{i=1}^{n} (y_i - \bar{y})^2.$$
3. *If $\theta_F = F(c)$, then*
$$\hat{\theta}_{\hat{F}_n} = \hat{F}_n(c) = \frac{\#\{i : y_i \le c\}}{n}.$$

To assess the quality of a plug-in estimator, we study three key quantities:

$$\mathrm{bias}_F(\hat{\theta}_{\hat{F}_n}) = \mathbb{E}_F(\hat{\theta}_{\hat{F}_n}) - \theta_F,$$

$$\mathrm{se}_F(\hat{\theta}_{\hat{F}_n}) = \sqrt{\mathrm{Var}_F(\hat{\theta}_{\hat{F}_n})},$$

$$\mathrm{MSE}_F(\hat{\theta}_{\hat{F}_n}) = \mathbb{E}_F\big[(\hat{\theta}_{\hat{F}_n} - \theta_F)^2\big].$$

Corollary. *If $\hat{\theta}_{\hat{F}_n}$ is unbiased, then*

$$\mathrm{MSE}_F(\hat{\theta}_{\hat{F}_n}) = \mathrm{Var}_F(\hat{\theta}_{\hat{F}_n}).$$

If it is biased, then

$$\mathrm{MSE}_F(\hat{\theta}_{\hat{F}_n}) = \mathrm{Var}_F(\hat{\theta}_{\hat{F}_n}) + \big(\mathrm{bias}_F(\hat{\theta}_{\hat{F}_n})\big)^2.$$

If F is known, these quantities can in principle be computed analytically. When this is difficult, one may approximate them by simulation:

- generate many random samples of size n from a population with cdf F;
- compute the estimator for each simulated sample;
- use the sample mean and sample variance of the simulated values to approximate the expectation and the variance of the estimator.

If F is unknown, direct simulation from F is impossible. This is one of the motivations for resampling methods studied later.

1.9 Exercise: coefficient of skewness

Consider the coefficient of skewness

$$\gamma = \frac{\mathbb{E}[(Y-\mu)^3]}{\sigma^3},$$

where

$$\mu = \mathbb{E}[Y], \qquad \sigma = \sqrt{\mathrm{Var}(Y)} = \sqrt{\mathbb{E}[(Y-\mu)^2]}.$$

Using the sample

$$4.3,\ 6.7,\ 7.2,\ 3.2,\ 10,$$

we first estimate the mean by the plug-in estimator:

$$\mu \approx \hat{\mu} = \frac{4.3+6.7+7.2+3.2+10}{5} = \frac{31.4}{5} = 6.28.$$

Next we estimate the variance:

$$\begin{aligned}
\sigma^2 \approx \hat{\sigma}^2 &= \frac{1}{5}\sum_{i=1}^{5}(y_i-\hat{\mu})^2 \\
&= \frac{1}{5}\left[(4.3-6.28)^2+(6.7-6.28)^2+(7.2-6.28)^2+(3.2-6.28)^2+(10-6.28)^2\right] \\
&= 5.6536.
\end{aligned}$$

Then we estimate the third centred moment:

$$\begin{aligned}
\mathbb{E}[(Y-\mu)^3] &\approx \frac{1}{5}\left[(4.3-6.28)^3+(6.7-6.28)^3+(7.2-6.28)^3+(3.2-6.28)^3+(10-6.28)^3\right] \\
&= 3.07.
\end{aligned}$$

Therefore the plug-in estimator of the skewness is

$$\hat{\gamma} = \frac{3.07}{(5.6536)^{3/2}} = 0.2.$$

Remark. *This example shows the plug-in principle in action for a more complicated functional of the distribution. We first replace the unknown mean and variance with their empirical counterparts, then use those empirical quantities inside the formula for skewness.*

Chapter 2

2_JackknifeBootstrap — Bootstrap & Jackknife

Here we consider methods for assessing the quality of estimators and for making predictions based on resampling ideas.[1]

2.1 Why resampling methods are needed

Let $y_1, \ldots, y_n$ be iid observations from a random variable Y with cdf F. Suppose we build an estimator $\hat{\theta}$ for an unknown parameter θ.

When studying the quality of $\hat{\theta}$, we are usually interested not only in the point estimate itself, but also in:

- the bias:
$$\text{bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta,$$
- the variance and the standard error:
$$\text{Var}(\hat{\theta}), \qquad \text{se}(\hat{\theta}) = \sqrt{\text{Var}(\hat{\theta})},$$
- the mean squared error:
$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + \text{bias}(\hat{\theta})^2.$$

For simple estimators, these quantities can sometimes be computed analytically. For more complicated estimators, explicit formulae may be difficult or impossible to obtain. A way around this is to use *resampling methods*, which repeatedly reuse the observed data.

The two main resampling methods considered here are:

- the **jackknife**, based on systematically deleting one observation at a time;
- the **bootstrap**, based on repeatedly sampling with replacement from the observed data, or from a fitted parametric model.

2.2 The jackknife

Denote the original sample by
$$y = (y_1, \ldots, y_n).$$

The **jackknife sample** obtained by removing the ith observation is
$$y_{[-i]} = (y_1, \ldots, y_{i-1}, y_{i+1}, \ldots, y_n).$$

In the slides this was also denoted by $y_{[i]}$; both notations refer to the same leave-one-out sample.

There are n such jackknife samples,
$$y_{[-1]}, \ldots, y_{[-n]},$$
each of size $n - 1$.

Applying the estimator to each jackknife sample yields the **jackknife replications**
$$\hat{\theta}_{[-1]}, \ldots, \hat{\theta}_{[-n]}.$$

[1] These notes are based on lecture notes developed by (in alphabetic order): Ö. Deniz Akyildiz, Nicolas Chopin, Heiko Großmann, Adam M. Johansen, Amir Khorrami Chokami, Nikolas Nüsken, Art Owen. Many thanks to them for making their material available.

Original sample	Jackknife samples of size $n-1$	Jackknife replications of $\hat{\theta}$
$y = (y_1, \ldots, y_n)$	$y_{[-1]} = (y_2, y_3, \ldots, y_n)$	$\hat{\theta}_{[-1]}$
	$y_{[-2]} = (y_1, y_3, \ldots, y_n)$	$\hat{\theta}_{[-2]}$
$\vdots$	$\vdots$	$\vdots$
	$y_{[-n]} = (y_1, y_2, \ldots, y_{n-1})$	$\hat{\theta}_{[-n]}$

The method uses all possible subsamples obtained by deleting exactly one observation.

Definition. *The average of the jackknife replications is*

$$\hat{\theta}_{[\cdot]} = \frac{1}{n} \sum_{i=1}^{n} \hat{\theta}_{[-i]}.$$

Definition. *The jackknife estimate of bias is*

$$\widehat{\text{bias}}_{\text{jack}} = (n-1)(\hat{\theta}_{[\cdot]} - \hat{\theta}).$$

This quantity should be compared with the true bias

$$\text{bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta.$$

The jackknife bias estimate can be viewed as a plug-in approximation of this quantity, where the expectation is approximated through the leave-one-out replicates.

Definition. *The bias-corrected jackknife estimator is*

$$\begin{aligned} \hat{\theta}_{\text{jack}} &= \hat{\theta} - \widehat{\text{bias}}_{\text{jack}} \\ &= \hat{\theta} - (n-1)(\hat{\theta}_{[\cdot]} - \hat{\theta}) \\ &= n\hat{\theta} - (n-1)\hat{\theta}_{[\cdot]}. \end{aligned}$$

This estimator often has considerably smaller bias than $\hat{\theta}$.

Definition. *The jackknife variance estimator is*

$$\widehat{\text{Var}}_{\text{jack}} = \frac{n-1}{n} \sum_{i=1}^{n} (\hat{\theta}_{[-i]} - \hat{\theta}_{[\cdot]})^2.$$

The corresponding jackknife standard error is

$$\widehat{\text{se}}_{\text{jack}} = \sqrt{\widehat{\text{Var}}_{\text{jack}}}.$$

Finally, the jackknife estimate of mean squared error is

$$\widehat{\text{MSE}}_{\text{jack}} = \widehat{\text{Var}}_{\text{jack}} + (\widehat{\text{bias}}_{\text{jack}})^2.$$

Remark. *The true variance* $\text{Var}(\hat{\theta})$ *cannot be computed directly from a single observed sample. The jackknife variance estimator approximates it using the variability of the jackknife replicates.*

Remark. *If n is very small, and in particular if* $n = 1$*, then the jackknife is not informative, since there is not enough data to form meaningful subsamples.*

2.3 Example: jackknife for the sample mean

Let

$$\theta = \mathbb{E}(Y), \qquad \hat{\theta} = \bar{y} = \frac{1}{n} \sum_{i=1}^{n} y_i.$$

For the jackknife sample $y_{[-i]}$, the corresponding replicate is

$$\hat{\theta}_{[-i]} = \frac{1}{n-1} \sum_{j \neq i} y_j.$$

Using

$$\sum_{j \neq i} y_j = n\bar{y} - y_i,$$

we get

$$\hat{\theta}_{[-i]} = \frac{n\bar{y} - y_i}{n-1}.$$

Now compute the average of the jackknife replicates:

$$\hat{\theta}_{[\cdot]} = \frac{1}{n}\sum_{i=1}^{n} \hat{\theta}_{[-i]}.$$

Then

$$\begin{aligned}
\hat{\theta}_{[\cdot]} &= \frac{1}{n}\sum_{i=1}^{n} \frac{n\bar{y} - y_i}{n-1} \\
&= \frac{1}{n(n-1)}\left(n^2\bar{y} - \sum_{i=1}^{n} y_i\right) \\
&= \frac{1}{n(n-1)}(n^2\bar{y} - n\bar{y}) \\
&= \bar{y}.
\end{aligned}$$

Therefore,

$$\widehat{\text{bias}}_{\text{jack}} = (n-1)(\hat{\theta}_{[\cdot]} - \hat{\theta}) = (n-1)(\bar{y} - \bar{y}) = 0,$$

and thus

$$\hat{\theta}_{\text{jack}} = \hat{\theta}.$$

For the variance,

$$\widehat{\text{Var}}_{\text{jack}} = \frac{n-1}{n}\sum_{i=1}^{n}(\hat{\theta}_{[-i]} - \hat{\theta}_{[\cdot]})^2.$$

Since $\hat{\theta}_{[\cdot]} = \bar{y}$,

$$\hat{\theta}_{[-i]} - \bar{y} = \frac{n\bar{y} - y_i}{n-1} - \bar{y} = \frac{\bar{y} - y_i}{n-1}.$$

Hence

$$\begin{aligned}
\widehat{\text{Var}}_{\text{jack}} &= \frac{n-1}{n}\sum_{i=1}^{n}\left(\frac{\bar{y} - y_i}{n-1}\right)^2 \\
&= \frac{1}{n(n-1)}\sum_{i=1}^{n}(y_i - \bar{y})^2 \\
&= \frac{s^2}{n},
\end{aligned}$$

where

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2$$

is the sample variance.

Remark. *Thus the jackknife estimate of the variance of the sample mean coincides with the usual variance estimator of the sample mean.*

2.4 Example: sample correlation coefficient

Two techniques for measuring blood flow, DYE and EFP, were used on $n = 9$ dogs. The data are

DYE	1.15	1.70	1.42	1.38	2.80	4.70	4.80	1.41	3.90
EFP	1.38	1.72	1.59	1.47	1.66	3.45	3.87	1.31	3.75

The estimator of interest is the sample correlation coefficient, whose observed value is

$$\hat{\theta} = r = 0.9448.$$

Leaving out one pair at a time yields the jackknife replications:

i	$\hat{\theta}_{[-i]}$	i	$\hat{\theta}_{[-i]}$	i	$\hat{\theta}_{[-i]}$
1	0.9406	4	0.9408	7	0.9205
2	0.9434	5	0.9766	8	0.9396
3	0.9434	6	0.9396	9	0.9583

The average of the jackknife replications is

$$\hat{\theta}_{[\cdot]} = \frac{1}{9}\sum_{i=1}^{9}\hat{\theta}_{[-i]} = 0.9448,$$

so the jackknife estimate of bias is

$$\widehat{\text{bias}}_{\text{jack}} = 8(0.9448 - 0.9448) = 0.$$

The jackknife variance is

$$\widehat{\text{Var}}_{\text{jack}} = \frac{8}{9}\sum_{i=1}^{9}\left(\hat{\theta}_{[-i]} - \hat{\theta}_{[\cdot]}\right)^2 = \frac{8}{9} \times 0.00188 = 0.00167,$$

hence

$$\widehat{\text{se}}_{\text{jack}} = \sqrt{0.00167} = 0.04.$$

□

2.5 Example with 4 observations

Suppose the data are

$$3.50,\ 3.55,\ 1.13,\ 1.86, \qquad n = 4.$$

We want to estimate

$$\hat{\theta} = \frac{\mathbb{E}[Y]^2}{\mathbb{E}[Y^2] - \mathbb{E}[Y]^2} = \frac{\mathbb{E}[Y]^2}{\text{Var}(Y)}.$$

Using plug-in estimates:

$$\mathbb{E}[Y] \approx \frac{1}{n}\sum_{i=1}^{n} y_i = 2.51,$$

$$\mathbb{E}[Y^2] \approx \frac{1}{n}\sum_{i=1}^{n} y_i^2 = 7.4.$$

Thus

$$\hat{\theta} = \frac{(2.51)^2}{7.40 - (2.51)^2} = 5.74.$$

The jackknife table is:

Sample	$\bar{y}$	$\overline{y^2}$	$\hat{\theta}$
y	2.51	7.4	5.74
$y_{[-1]}$	2.18	5.78	5.63
$y_{[-2]}$	2.16	5.66	4.77
$y_{[-3]}$	2.87	8.49	14.31
$y_{[-4]}$	2.73	8.71	5.83

and

$$\hat{\theta}_{[\cdot]} = 7.38.$$

Usually we order the sample as

$$1.13,\ 1.86,\ 3.50,\ 3.55.$$

Then

$$\widehat{\text{bias}}_{\text{jack}} = (n-1)(\hat{\theta}_{[\cdot]} - \hat{\theta}) = 3(7.38 - 5.74) = 4.92,$$

$$\widehat{\text{Var}}_{\text{jack}} = \frac{n-1}{n}\sum_{i=1}^{n}\left(\hat{\theta}_{[-i]} - \hat{\theta}_{[\cdot]}\right)^2 = 48.62,$$

and therefore

$$\widehat{\text{se}}_{\text{jack}} = \sqrt{\widehat{\text{Var}}_{\text{jack}}} = 6.97.$$

2.6 The bootstrap

For this section, recall the plug-in principle from Chapter 1: when the true cdf F is unknown, we replace it with an estimate and then study the estimator under that estimated distribution.

The **bootstrap** follows exactly this logic. Instead of deriving analytically the sampling distribution of $\hat{\theta}$, we approximate it by repeated simulation.

2.7 Nonparametric bootstrap

The **nonparametric bootstrap**, also called the simple bootstrap, aims to estimate the bias and standard error of an estimator $\hat{\theta}$ when the cdf F of the data-generating distribution is unknown.

Given the original sample

$$y = (y_1, \ldots, y_n),$$

draw with replacement new samples

$$y^* = (y_1^*, \ldots, y_n^*)$$

of size n from the observed data. These are called **bootstrap samples**. For each bootstrap sample, compute a **bootstrap replication** $\hat{\theta}^*$.

Sampling bootstrap datasets in this way is equivalent to drawing from the empirical cdf $\hat{F}_n$.

The practical procedure is:

1. For N sufficiently large, draw N independent bootstrap samples $y_1^*, \ldots, y_N^*$, each of size n, from y with replacement.
2. For each bootstrap sample y_i^*, compute the bootstrap replication $\hat{\theta}_i^*$.

Definition. *The ideal bootstrap bias is*

$$\text{bias}_{\hat{F}_n}(\hat{\theta}^*) = \mathbb{E}_{\hat{F}_n}(\hat{\theta}^*) - \hat{\theta}_{\hat{F}_n},$$

and the ideal bootstrap standard error is

$$\text{se}_{\hat{F}_n}(\hat{\theta}^*) = \sqrt{\text{Var}_{\hat{F}_n}(\hat{\theta}^*)}.$$

In practice, these are approximated by

$$\widehat{\text{bias}}_N = \hat{\theta}_{(\cdot)}^* - \hat{\theta}_{\hat{F}_n}, \tag{4.3}$$

where

$$\hat{\theta}_{(\cdot)}^* = \frac{1}{N}\sum_{i=1}^{N} \hat{\theta}_i^*,$$

and by

$$\widehat{\text{se}}_N = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}\left(\hat{\theta}_i^* - \hat{\theta}_{(\cdot)}^*\right)^2}. \tag{4.4}$$

Thus, to estimate the true bias and true standard error of $\hat{\theta}$:

1. generate N bootstrap samples;
2. compute one bootstrap replication per sample;
3. use (4.3) and (4.4) to estimate bias and standard error.

2.8 Example: bioequivalence of patches

Eight subjects wore placebo, old, and new medical patches. Define

$$x_i = \text{old} - \text{placebo}, \qquad y_i = \text{new} - \text{old}.$$

The data are:

subject	placebo	old patch	new patch	$x_i = \text{old} - \text{placebo}$	$y_i = \text{new} - \text{old}$
1	9243	17649	16449	8406	-1200
2	9671	12013	14614	2342	2601
3	11792	19979	17274	8187	-2705
4	13357	21816	23798	8459	1982
5	9055	13850	12560	4795	-1290
6	6290	9806	10157	3516	351
7	12412	17208	16570	4796	-638
8	18806	29044	26325	10238	-2719
mean				6342	-452.3

The FDA criterion for bioequivalence is

$$\frac{|\mathbb{E}(\text{new}) - \mathbb{E}(\text{old})|}{|\mathbb{E}(\text{old}) - \mathbb{E}(\text{placebo})|} \leq 0.2.$$

Define

$$\theta = \frac{\mathbb{E}(\text{new}) - \mathbb{E}(\text{old})}{\mathbb{E}(\text{old}) - \mathbb{E}(\text{placebo})} = \frac{\mathbb{E}_F(Y)}{\mathbb{E}_F(X)}.$$

The plug-in estimate is

$$\hat{\theta} = \frac{\bar{y}}{\bar{x}} = \frac{-452.3}{6342} = -0.0713.$$

Its absolute value is less than 0.2.

Using the bootstrap with $N = 400$, the estimates are

$$\widehat{\text{bias}}_{400} = 0.01059, \qquad \widehat{\text{se}}_{400} = 0.10153.$$

Since

$$\frac{\widehat{\text{bias}}_{400}}{\widehat{\text{se}}_{400}} = 0.1043,$$

the bias is small relative to the variability.

The bias-corrected estimate is

$$\hat{\theta} - \widehat{\text{bias}}_{400} = -0.0713 - 0.01059 = -0.0819,$$

whose absolute value is still below 0.2. Therefore, the new patch is considered equivalent to the old one.

2.9 Parametric bootstrap

If the unknown cdf F is assumed to belong to a parametric family

$$\{F_\psi : \psi \in \Psi\},$$

this extra information can be used to improve the bootstrap procedure.

Example. *Typical examples are:*

- *Normal family:*
$$F \in \{\Phi(\mu, \sigma^2) : \mu \in \mathbb{R}, \sigma^2 > 0\}, \qquad \psi = (\mu, \sigma^2).$$
- *Exponential family:*
$$F \in \{F_\lambda : \lambda > 0\}, \qquad \psi = \lambda.$$

The idea is the following:

1. estimate the parameter $\hat{\psi}$ from the original sample;
2. set
$$\hat{F} = F_{\hat{\psi}};$$
3. generate bootstrap samples from $\hat{F}$ instead of from the empirical cdf $\hat{F}_n$;
4. compute the bootstrap replications and estimate bias and standard error using (4.3) and (4.4).

So the full procedure is:

1. estimate $\hat{\psi}$ from the original sample;
2. generate N bootstrap samples of size n from $F_{\hat{\psi}}$;
3. compute $\hat{\theta}_i^*$ for each sample;
4. estimate bias and standard error using (4.3) and (4.4).

2.10 Example: exponential lifetime model

Suppose the lifetime Y of an electronic component is exponential with parameter λ, so that

$$\mathbb{E}(Y) = \frac{1}{\lambda}.$$

We want to estimate

$$\theta = F_\lambda(24),$$

that is, the fraction of components that fail within 24 hours.

From $n = 100$ observations, the sample mean is $\bar{y} = 50$ hours and 35 components failed before 24 hours. The plug-in estimate based on the ecdf is

$$\hat{\theta} = \hat{F}_n(24) = \frac{35}{100} = 0.35.$$

For the parametric bootstrap, estimate

$$\hat{\lambda} = \frac{1}{\bar{y}} = \frac{1}{50} = 0.02,$$

and simulate from

$$\hat{F} = F_{0.02}.$$

Using $N = 200$ bootstrap samples gives

$$\hat{\theta}^*_{(\cdot)} = 0.3889.$$

Hence the bootstrap estimate of bias is

$$\widehat{\text{bias}}_{200} = \hat{\theta}^*_{(\cdot)} - \hat{\theta} = 0.3889 - 0.35 = 0.0389.$$

The bootstrap estimate of standard error is

$$\widehat{\text{se}}_{200} = \sqrt{\frac{1}{199}\sum_{i=1}^{200}\left(\hat{\theta}^*_i - \hat{\theta}^*_{(\cdot)}\right)^2} = 0.0452.$$

Since the ecdf estimator $\hat{F}_n(24)$ is unbiased for $F(24)$, this nonzero estimated bias is due to the fact that the bootstrap samples are generated from the fitted exponential distribution $F_{0.02}$, for which

$$F_{0.02}(24) = 0.3812.$$

Also,

$$\sqrt{\frac{1}{n}F_{0.02}(24)\big(1 - F_{0.02}(24)\big)} = \sqrt{\frac{1}{100}(0.3812 \times 0.6188)} = 0.04857,$$

which is very close to the simulated value 0.0452.

2.11 Bootstrap confidence intervals

In addition to estimating bias and standard error, one often wants a confidence interval for θ.

For the nonparametric bootstrap, if

$$\hat{\theta}^*_1, \ldots, \hat{\theta}^*_N$$

are the bootstrap replications, let $\hat{\theta}^{*(\alpha)}$ be the 100αth percentile of the ordered bootstrap values.

Definition. *The percentile interval is*

$$\left(\hat{\theta}^{*(\alpha/2)}, \hat{\theta}^{*(1-\alpha/2)}\right).$$

This interval uses the empirical quantiles of the bootstrap distribution directly. It is simple and often useful, but it does not explicitly correct for bias or for asymmetry in the sampling distribution of the estimator.

A more refined interval is the BCa interval, where BCa stands for *bias corrected and accelerated*. To define it, we first introduce the bias-correction term and the acceleration term.

Definition. *The **bias-correction term** is*

$$\hat{z}^* = \Phi^{-1}\left(\frac{\#\{i : \hat{\theta}^*_i < \hat{\theta}\}}{N}\right).$$

The proportion

$$\frac{\#\{i : \hat{\theta}_i^* < \hat{\theta}\}}{N}$$

tells us where the original estimate $\hat{\theta}$ lies inside the bootstrap distribution. If this proportion is close to 0.5, then $\hat{\theta}$ is near the centre of the bootstrap distribution. If it is much larger or much smaller than 0.5, then the bootstrap distribution is not centred at $\hat{\theta}$, which suggests bias.

The function Φ^{-1} is the quantile function of the standard normal distribution. Its role here is not to assume that the data are normal. Rather, it converts the percentile position of $\hat{\theta}$ inside the bootstrap distribution into a standard normal score, or *z-score*. Thus $\hat{z}^*$ expresses the location of $\hat{\theta}$ inside the bootstrap distribution on the standard normal scale.

For example:

- if
$$\frac{\#\{i : \hat{\theta}_i^* < \hat{\theta}\}}{N} = 0.5,$$
then
$$\hat{z}^* = \Phi^{-1}(0.5) = 0,$$
so the original estimate $\hat{\theta}$ is at the centre of the bootstrap distribution;
- if the proportion is greater than 0.5, then $\hat{z}^* > 0$, so the original estimate $\hat{\theta}$ lies to the right of the centre of the bootstrap distribution;
- if the proportion is smaller than 0.5, then $\hat{z}^* < 0$, so the original estimate $\hat{\theta}$ lies to the left of the centre of the bootstrap distribution;
- therefore, if the proportion differs substantially from 0.5, the bootstrap distribution of the bootstrap estimator $\hat{\theta}^*$ is not centred at the original estimate $\hat{\theta}$, suggesting that the original estimator may be biased.

Thus $\hat{z}^*$ shifts the percentile levels in order to correct for lack of centring of the bootstrap distribution around $\hat{\theta}$.

Definition. *The **acceleration term** is*

$$\hat{a} = \frac{\sum_{i=1}^{n} (\hat{\theta}_{[\cdot]} - \hat{\theta}_{[-i]})^3}{6 \left[\sum_{i=1}^{n} (\hat{\theta}_{[\cdot]} - \hat{\theta}_{[-i]})^2\right]^{3/2}},$$

where $\hat{\theta}_{[-i]}$ are the jackknife replications and

$$\hat{\theta}_{[\cdot]} = \frac{1}{n} \sum_{i=1}^{n} \hat{\theta}_{[-i]}.$$

The role of $\hat{a}$ is to measure the degree of asymmetry or skewness in the behaviour of the estimator.

More precisely, the acceleration term is computed from the jackknife replications, so it reflects how the estimator changes when one observation is deleted at a time:

- if the estimator reacts in a roughly symmetric way to these deletions, then $\hat{a}$ is close to 0;
- if the estimator reacts in an asymmetric or nonlinear way, then $\hat{a}$ is different from 0.

Therefore $\hat{a}$ adjusts for the shape of the sampling distribution, not for its location. In particular, it modifies the lower and upper percentile levels differently when the estimator has a skewed distribution.

Definition. *The **BCa interval** is*

$$\left(\hat{\theta}^{*(\alpha_1)}, \hat{\theta}^{*(\alpha_2)}\right),$$

where

$$\alpha_1 = \Phi\left(\hat{z}^* + \frac{\hat{z}^* + z_{\alpha/2}}{1 - \hat{a}(\hat{z}^* + z_{\alpha/2})}\right),$$

$$\alpha_2 = \Phi\left(\hat{z}^* + \frac{\hat{z}^* + z_{1-\alpha/2}}{1 - \hat{a}(\hat{z}^* + z_{1-\alpha/2})}\right).$$

So, in summary:

- $\hat{z}^*$ corrects for **bias**, that is, for lack of centring of the bootstrap distribution around $\hat{\theta}$;
- $\hat{a}$ corrects for **acceleration**, that is, for skewness, asymmetry, or nonlinear variation in the estimator.

Remark. *If both $\hat{z}^* = 0$ and $\hat{a} = 0$, then there is no bias correction and no acceleration correction. In this case the BCa interval reduces to the percentile interval.*

Remark. *A BCa interval is only approximately exact, but its accuracy improves with larger sample size. Its main advantage over the percentile interval is that it adjusts both for bias and for asymmetry in the bootstrap distribution.*

Remark. *If the sample size is n, then the number of distinct bootstrap samples is*

$$\binom{2n-1}{n-1} = \frac{(2n-1)!}{(n-1)!n!}.$$

Thus it is sensible to choose

$$N \leq \binom{2n-1}{n-1}.$$

In practice, $N = 400$ is often sufficient for estimating standard errors, while for bootstrap confidence intervals one usually takes at least $N = 2000$.

Chapter 3

3_PRNGTransformationMethods — PRNG & Transformation Methods

The second part of this module is about simulating random variables, or put differently sampling from probability distributions. This task arises in many real-world scenarios and critical applications. We also look at generating samples from intractable distributions and at variance reduction techniques.[1]

3.1 Why Monte Carlo methods?

A typical motivation comes from stochastic differential equations. For instance, a simple discrete-time dynamics may be written as

$$X_t = X_{t-1} + \bar{Z}_t, \qquad \bar{Z}_t \sim N(0,1).$$

If we want to simulate a trajectory (X_t), then we need a way to simulate the innovations $\bar{Z}_t$ and then propagate the recursion forward.

This leads naturally to the following simulation programme:

1. generate uniform random variables on $[0,1]$;
2. transform uniforms into known random variables;
3. generate samples from a chosen target distribution.

A generic target density can often be written as

$$\pi(x) = \frac{\gamma(x)}{Z}, \qquad \gamma(x) > 0, \qquad Z = \int_{-\infty}^{+\infty} \gamma(x)\, dx.$$

Here γ is an unnormalised density and Z is the normalising constant. In many applications γ is known but Z is difficult or impossible to compute explicitly.

3.2 Generating uniform random variates

One of the central pillars of sampling algorithms is the ability to sample from the uniform distribution. Since exact real-valued uniform random numbers are difficult to generate, much of the literature focuses on *pseudo*-uniform random numbers.

Definition. *A sequence of pseudo-random numbers $u_1, u_2, \ldots$ is a deterministic sequence whose statistical properties mimic those of random numbers from a desired distribution.*

A useful generator should be:

- fast;
- reliable;
- and its output should look random.

[1] These notes are based on lecture notes developed by (in alphabetic order): Ö. Deniz Akyildiz, Nicolas Chopin, Heiko Großmann, Adam M. Johansen, Amir Khorrami Chokami, Nikolas Nüsken, Art Owen. Many thanks to them for making their material available.

3.3 Linear Congruential Generator

The most popular classical method is the **Linear Congruential Generator (LCG)**:

$$x_{n+1} = (ax_n + b) \bmod m,$$

where x_0 is the seed, m is the modulus, a the multiplier, and b the shift. If $b = 0$, the LCG is *multiplicative*; if $b \neq 0$, it is *mixed*. Choose

$$m \in \mathbb{N}, \qquad x_0, a, b \in \{0, \dots, m-1\}.$$

Then

$$x_n \in \{0, 1, \dots, m-1\}$$

and the pseudo-uniforms are

$$u_n = \frac{x_n}{m}, \qquad u_n \in [0, 1) \quad \forall n.$$

The sequence $(u_n)_{n\geq 0}$ is periodic with period $T \leq m$. Ideally one chooses parameters giving full period $T = m$. The choice of m is linked to precision, for example $m \approx 2^{32}$ for single precision.

Recall the modular arithmetic notation:

$$a \equiv b \mod m \iff a - b = km \quad \text{for some } k \in \mathbb{N}.$$

For example,

$$3 \mod 2 = 1, \qquad 24 \mod 10 = 4.$$

LCGs have known shortcomings, especially in high dimensions. Even when the histogram looks uniform, plots of (u_k, u_{k+1}) may reveal a clear lattice structure.

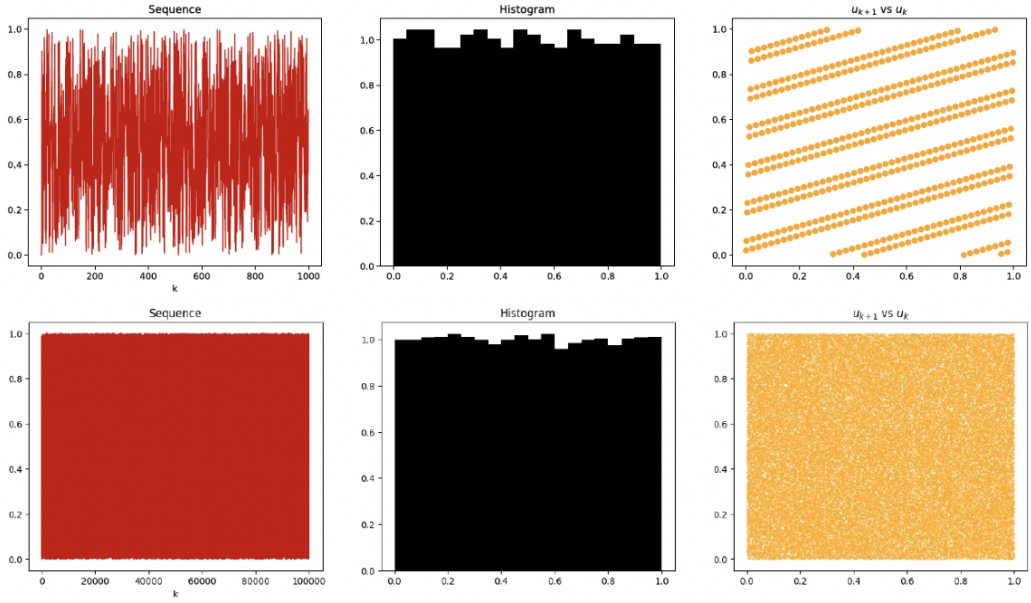


Figure 3.1: Top Left: Samples generated by an LCG with parameters $m = 2048$, $a = 43$, $b = 0$, $x_0 = 1$. Top Middle: The histogram of the samples showing that it conforms the uniform distribution. Top Right: However, plotting (u_k, u_{k+1}) pairs shows that the samples are not random enough. Bottom Left: Samples generated by an LCG with parameters $m = 2^{32}$, $a = 1664525$, $b = 1013904223$. Bottom Middle: The histogram of the samples showing that it conforms the uniform distribution. Bottom Right: Plotting (u_k, u_{k+1}) pairs shows that the samples do not look nonrandom.

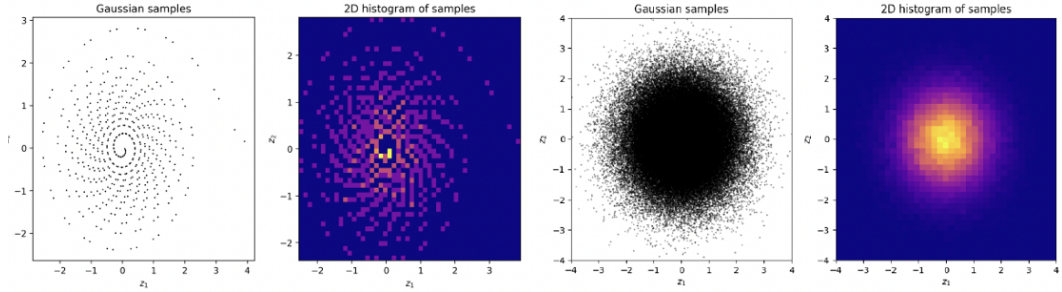


Figure 3.2: Left two figures: Gaussian samples generated using the samples in the top row of Fig. 3.1. Right two figures: Gaussian samples generated using the samples in the bottom row of Fig. 3.1 with a better parameterisation: $m = 2^{32}$, $a = 1664525$, $b = 1013904223$.

Credit to Ö. Deniz Akyildiz.

Modern implementations typically use much more sophisticated generators, such as the Mersenne Twister or newer default generators. In the rest of the course we use properly initialised generators such as `rng.uniform(0,1)`.

```
rng = np.random.default_rng(10)
u = rng.uniform(0, 1)
v = np.random.uniform(0, 1)
```

3.4 Transformation methods

Given a pseudo-uniform generator, we can describe exact sampling methods based on deterministic transformations

$$Y = g(U), \qquad U \sim \text{Unif}(0,1).$$

The basic idea is that if we can write a target random variable as a transformation of a variable that is easy to simulate, then simulation becomes immediate.

The inverse-cdf method is a special case:

$$X = F^{-1}(U).$$

3.5 Inverse transform method

This method uses the cdf F of a target distribution.

Theorem. ***Probability integral transform.*** *Let X be a random variable with cdf F_X and let*

$$U \sim \text{Unif}(0,1).$$

Then

$$F_X^{-1}(U)$$

has the same distribution as X.

Proof. Let $Y = F_X^{-1}(U)$. Then

$$\mathbb{P}(Y \le y) = \mathbb{P}(F_X^{-1}(U) \le y) = \mathbb{P}(U \le F_X(y)).$$

Since U is uniform on $[0,1]$,

$$\mathbb{P}(U \le F_X(y)) = F_X(y).$$

Thus the cdf of Y is exactly F_X, hence

$$F_X^{-1}(U) \overset{d}{=} X.$$

□

Algorithm 1 Inverse transform sampling (continuous random variable)

1: **Input:** The number of samples n and cdf F_X
2: **for** $i = 1, \dots, n$ **do**
3: Generate $U_i \sim \text{Unif}(0,1)$
4: Set $X_i = F_X^{-1}(U_i)$
5: **end for**
6: **Output:** $X_1, \dots, X_n$ with cdf F_X

Example. ***Exponential distribution.*** *Let*

$$X \sim \text{Exp}(\lambda), \qquad f_X(x) = \lambda e^{-\lambda x} \mathbf{1}_{[0,+\infty)}(x).$$

Then

$$F_X(x) = \int_0^x \lambda e^{-\lambda x'} \, dx' = 1 - e^{-\lambda x}.$$

Solving

$$U = 1 - e^{-\lambda X}$$

gives

$$X = -\frac{1}{\lambda} \log(1 - U).$$

Hence

$$F_X^{-1}(U) = -\lambda^{-1}\log(1-U).$$

Since

$$1 - U \overset{d}{=} U,$$

one may equivalently use

$$X = -\frac{\log U}{\lambda}.$$

Indeed,

$$\mathbb{P}(1-U \le u) = \mathbb{P}(U \ge 1-u) = 1 - \mathbb{P}(U < 1-u) = 1-(1-u) = u.$$

Thus the algorithm is:

- *generate* $U_i \sim \mathrm{Unif}(0,1)$*;*
- *set*

$$X_i = -\lambda^{-1}\log(1-U_i),$$

or, equivalently,

$$X_i = -\lambda^{-1}\log(U_i).$$

Example. ***Cauchy distribution.*** *Suppose*

$$\pi(x) = \frac{1}{\pi(1+x^2)}.$$

Its cdf is

$$F_X(x) = \frac{1}{2} + \frac{1}{\pi}\tan^{-1}(x),$$

and therefore

$$F_X^{-1}(u) = \tan\left[\pi\left(u - \frac{1}{2}\right)\right].$$

So the sampler is:

- *generate* $U_i \sim \mathrm{Unif}(0,1)$*;*
- *set*

$$X_i = \tan\left[\pi\left(U_i - \frac{1}{2}\right)\right].$$

3.6 Discrete random variables and inverse transform

Let π be a discrete distribution on

$$S = \{s_1, \dots, s_K\}, \qquad \mathbb{P}(X = s_k) = \pi_k, \qquad \sum_{k=1}^{K} \pi_k = 1.$$

Then the inverse-cdf procedure becomes:

- sample $U_i \sim \mathrm{Unif}(0,1)$;
- find

$$k_i = \min\{k : F_X(s_k) \ge U_i\};$$

- set $X_i = s_{k_i}$.

Equivalently,

$$F^{-1}(u) = s_{\min\{i:F(s_i)\ge u\}}.$$

Example. ***Bernoulli distribution.*** *For*

$$\mathbb{P}(X = k) = p^k(1-p)^{1-k}, \qquad k = 0, 1,$$

we have

$$\mathbb{P}(X = 1) = p, \qquad \mathbb{P}(X = 0) = 1 - p.$$

Therefore, with the convention above, the inverse-cdf method is:

- *sample* $U \sim \mathrm{Unif}(0,1)$*;*
- *if* $U < 1 - p$ *return* 0*, otherwise return* 1*.*

If instead one denotes by p *the probability of returning* 0*, then the threshold becomes* $U \le p$ *for the value* 0*.*

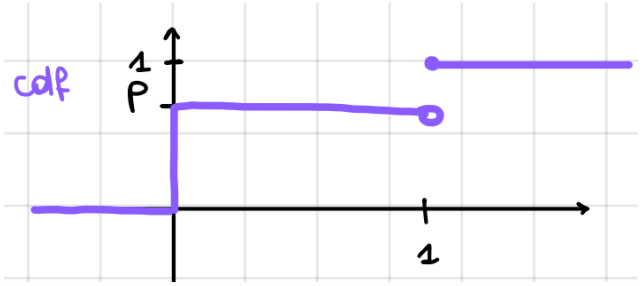


Figure 3.3: Cdf sketch for a Bernoulli-type discrete distribution.

A practical version for discrete distributions is:

Algorithm 2 Inverse transform sampling (discrete random variable)

Input: The number of samples n, the pmf $\pi_1, \dots, \pi_K$, and sorted uniforms $U^{(1)} \leq \cdots \leq U^{(n)}$
Set $m = 1$, $q = \pi_1$
for $i = 1, \dots, n$ **do**
    while $q < U^{(i)}$ **do**
        Set $m = m + 1$ and $q = q + \pi_m$
    end while
    Set $X_i = s_m$
end for

Example. ***Poisson distribution.*** *Let*

$$\mathbb{P}(X = k) = \text{Pois}(k; \lambda) = \frac{\lambda^k e^{-\lambda}}{k!}, \qquad k = 0, 1, 2, \dots.$$

To generate one sample:

- *sample* $U \sim \text{Unif}(0, 1)$;
- *set* $m = 0$, $q = e^{-\lambda}$;
- *while* $q < U$, *update*

$$m = m + 1, \qquad q = q + \frac{\lambda^m e^{-\lambda}}{m!};$$

- *return* m.

Then $m \in \mathbb{N}$ *is the sample.*

Remark. *A bad example for direct inversion is*

$$X \sim N(0, 1),$$

whose cdf

$$\Phi(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-s^2/2}\, ds$$

does not admit a simple closed-form inverse. For such distributions one often uses other transformations, approximations, or accept–reject methods.

3.7 General transformation method

If we can write

$$X = g(Y),$$

where Y is a random variable we can simulate from, then we can simulate from X via the transformation method. The inverse-cdf method is the special case where

$$Y = U \sim \text{Unif}(0, 1), \qquad g = F^{-1}.$$

Algorithm 3 Pseudocode for transformation method

Input: The number of samples n
for $i = 1, \dots, n$ **do**
    Generate Y_i from a distribution we can simulate
    Set $X_i = g(Y_i)$
end for

Remark. *If $X \sim \pi(x)$ and $Y = g(X)$, then in one dimension*

$$p_Y(y) = \pi(g^{-1}(y)) \left| \frac{dg^{-1}(y)}{dy} \right|. \tag{1}$$

For an n-dimensional transformation,

$$p_Y(y) = \pi(g^{-1}(y)) \, |\det J_{g^{-1}}(y)|, \tag{2}$$

where $J_{g^{-1}}$ is the Jacobian matrix of the inverse map.

For $d = 2$,

$$J_{g^{-1}} = \begin{pmatrix} \partial g_1^{-1}/\partial y_1 & \partial g_1^{-1}/\partial y_2 \\ \partial g_2^{-1}/\partial y_1 & \partial g_2^{-1}/\partial y_2 \end{pmatrix}.$$

3.8 Examples of transformation methods

Example. *Let*

$$X_1 \sim \text{Exp}\left(\frac{1}{2}\right), \qquad X_2 \sim \text{Unif}(-\pi, \pi),$$

independent, and define

$$Y_1 = \sqrt{X_1}\cos X_2, \qquad Y_2 = \sqrt{X_1}\sin X_2.$$

Then Y_1 and Y_2 are independent standard normal random variables.
Indeed,

$$(y_1, y_2) = g(x_1, x_2) = (\sqrt{x_1}\cos x_2, \sqrt{x_1}\sin x_2),$$

with inverse

$$x_1 = y_1^2 + y_2^2, \qquad x_2 = \arctan(y_2/y_1).$$

Using the change-of-variables formula and $|\det J_{g^{-1}}| = 2$, one gets

$$p_{y_1,y_2}(y_1, y_2) = \frac{1}{2\pi} e^{-\frac{1}{2}(y_1^2+y_2^2)} = N(y_1; 0, 1)\, N(y_2; 0, 1).$$

Example. ***Sampling uniformly on the unit disk.*** *Let*

$$r \sim \text{Unif}(0, 1), \qquad \theta \sim \text{Unif}(-\pi, \pi),$$

and define

$$X_1 = \sqrt{r}\cos\theta, \qquad X_2 = \sqrt{r}\sin\theta.$$

The inverse transformation is

$$g^{-1}(x_1, x_2) = \left(x_1^2 + x_2^2, \arctan\left(\frac{x_2}{x_1}\right)\right),$$

with Jacobian matrix

$$J_{g^{-1}}(x_1, x_2) = \begin{pmatrix} 2x_1 & 2x_2 \\ \dfrac{1}{1+(x_2/x_1)^2}\left(-\dfrac{x_2}{x_1^2}\right) & \dfrac{1}{1+(x_2/x_1)^2}\dfrac{1}{x_1} \end{pmatrix},$$

and

$$\det J_{g^{-1}}(x_1, x_2) = 2.$$

Therefore

$$p_{x_1,x_2}(x_1, x_2) = \begin{cases} \dfrac{1}{\pi}, & x_1^2 + x_2^2 < 1, \\ 0, & \textit{otherwise}. \end{cases}$$

Thus the pair is uniformly distributed on the unit disk.

Example. ***Affine transformation of a Gaussian.*** *If*

$$Y \sim N(0, 1), \qquad X = \sigma Y + \mu,$$

then

$$g(y) = \sigma y + \mu, \qquad g^{-1}(x) = \frac{x-\mu}{\sigma}, \qquad \frac{dg^{-1}(x)}{dx} = \frac{1}{\sigma}.$$

By (1),

$$f_X(x) = \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right] \frac{1}{\sigma} = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right].$$

Hence $X \sim N(\mu, \sigma^2)$.

Example. ***From*** $U \sim \text{Unif}([0,1])$ ***to*** $\text{Unif}([a,b])$. *Set*

$$X = (b-a)U + a.$$

Then

$$g(u) = (b-a)u + a, \qquad g^{-1}(x) = \frac{x-a}{b-a}, \qquad \frac{\partial g^{-1}(x)}{\partial x} = \frac{1}{b-a}.$$

Thus

$$f_X(x) = \mathbf{1}_{[0,1]}\left(\frac{x-a}{b-a}\right)\frac{1}{b-a} = \mathbf{1}_{[a,b]}(x)\frac{1}{b-a}.$$

3.9 Box–Müller transform

If

$$U_1, U_2 \sim \text{Unif}(0,1)$$

are independent, then

$$Z_1 = \sqrt{-2\log U_1}\cos(2\pi U_2), \qquad Z_2 = \sqrt{-2\log U_1}\sin(2\pi U_2)$$

are independent standard normal random variables. This is the **Box–Müller transform**.

3.10 Exercise: logistic distribution

Sample from $X \sim \text{Logistic}(0,1)$, with cdf

$$F(x) = \frac{1}{1+e^{-x}}.$$

Set

$$u = \frac{1}{1+e^{-x}}.$$

Then

$$1 + e^{-x} = \frac{1}{u}, \qquad e^{-x} = \frac{1-u}{u}, \qquad x = -\log(1-u) + \log u.$$

Thus, if $U_i \sim \text{Unif}([0,1])$,

$$X_i = -\log(1-U_i) + \log(U_i).$$

Remark. *Even if* $1 - U_i \overset{d}{=} U_i$*, we cannot simplify*

$$-\log(1-U_i) + \log(U_i)$$

to 0*, because the two terms depend on the same random variable.*

3.11 Exercise: $Z = X + U$ with $X \sim \text{Exp}(1)$ and $U \sim \text{Unif}(0,1)$

Let

$$Z = X + U, \qquad X \sim \text{Exp}(1), \qquad U \sim \text{Unif}(0,1),$$

with X and U independent. Then

$$f_Z(z) = \int_{-\infty}^{+\infty} f_X(x) f_U(z-x)\,dx = \int_0^{+\infty} e^{-x}\mathbf{1}_{[0,1]}(z-x)\,dx.$$

It follows that

$$f_Z(z) = (1-e^{-z})\mathbf{1}_{[0,1]}(z) + (e-1)e^{-z}\mathbf{1}_{[1,+\infty)}(z).$$

Indeed:

- if $0 \le z \le 1$, $\int_0^z e^{-x}\,dx = 1 - e^{-z}$;
- if $z \ge 1$, $\int_{z-1}^z e^{-x}\,dx = (e-1)e^{-z}$;
- if $z \le 0$, the density is 0.

To simulate from Z, simply simulate

$$U \sim \text{Unif}([0,1]), \qquad \widetilde{X} \sim \text{Exp}(1),$$

and set

$$Z = U + \widetilde{X}.$$

Chapter 4

4_RejectionSampling — Rejection Sampling

Here we consider methods which can be applied when π does not admit an inverse cdf and cannot be written as a transformation of another pdf.[1]

4.1 Introduction to rejection sampling

Let

$$\pi(x) = \frac{\gamma(x)}{z}, \qquad z = \int_{-\infty}^{+\infty} \gamma(x)\, dx,$$

where γ is known, while the normalising constant z may be unknown.

If one can sample uniformly under the graph of a density or unnormalised density, then the x-marginal of those samples has the desired density. This is the geometric basis of rejection sampling.

Theorem. ***Fundamental Theorem of Simulation.*** *Assume $\gamma(x) \geq 0$ and define*

$$A = \{(x, u) \in \mathbb{R}^2 : \ 0 \leq u \leq \gamma(x)\}.$$

If (X, U) is uniformly distributed on A, then the marginal density of X is

$$\pi(x) = \frac{\gamma(x)}{z}.$$

Proof. The area of A is

$$|A| = \int_{-\infty}^{+\infty} \gamma(x)\, dx = z.$$

Since (X, U) is uniform on A, its joint density is

$$q(x, u) = \frac{1}{|A|} = \frac{1}{z}, \qquad (x, u) \in A.$$

For fixed x, the vertical section has length $\gamma(x)$, hence

$$q_X(x) = \int_0^{\gamma(x)} \frac{1}{z}\, du = \frac{\gamma(x)}{z} = \pi(x).$$

Therefore the marginal law of X is exactly π. □

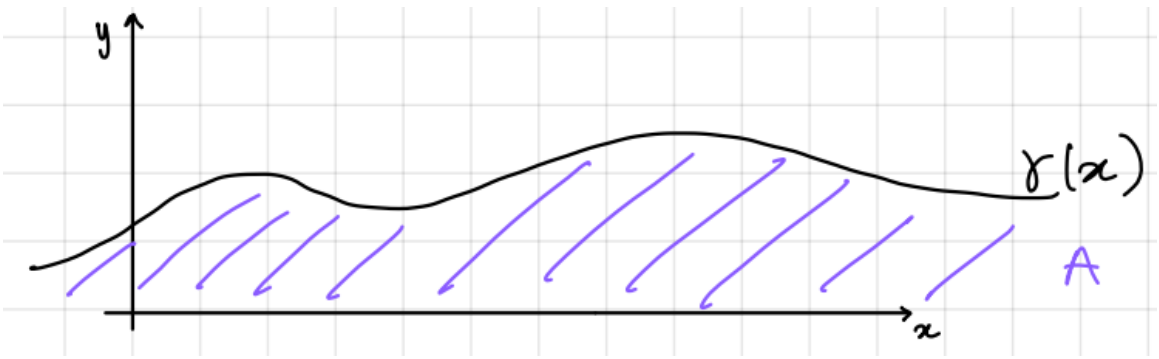


Figure 4.1: Geometric interpretation of sampling uniformly under the curve.

[1] These notes are based on lecture notes developed by (in alphabetic order): Ö. Deniz Akyildiz, Nicolas Chopin, Heiko Großmann, Adam M. Johansen, Amir Khorrami Chokami, Nikolas Nüsken, Art Owen. Many thanks to them for making their material available.

4.2 The rectangle method

A first simple version of rejection sampling is the rectangle method. Suppose the support of π is contained in $[a, b]$ and that

$$\pi(x) \leq p^* \qquad \text{for all } x \in [a, b].$$

If

$$(X', U') \sim \text{Unif}([a, b] \times [0, p^*]),$$

and we accept X' whenever

$$U' \leq \pi(X'),$$

then the accepted values have density π.

Remark. *This method works well when the support is bounded and the density can be enclosed in a convenient rectangle. If the support is unbounded, a finite rectangle is not available, so this construction is not directly applicable.*

Remark. *Any bounding region containing the graph of π is valid. However, the larger the bounding region, the more rejections we get, so efficiency decreases.*

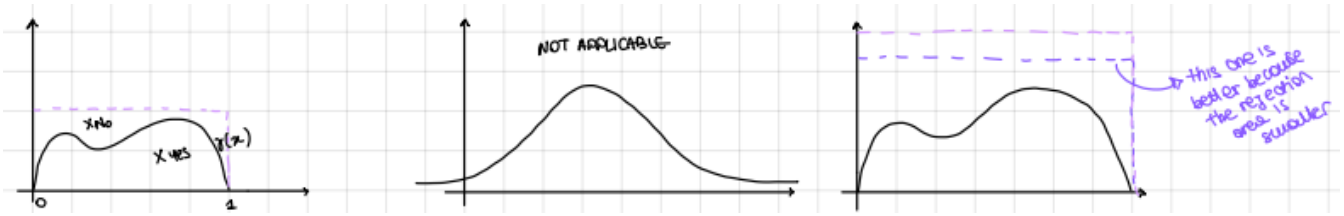


Figure 4.2: The rectangle method can be inefficient when the box leaves a large amount of empty area above the target.

Example. ***Beta density under a box.*** *Consider*

$$\pi(x) = \text{Beta}(x; 2, 2).$$

Its support is $[0, 1]$ *and its maximum is*

$$p_{\max} = \max_x \text{Beta}(x; 2, 2) = 1.5.$$

Thus one may sample uniformly on the box

$$[0, 1] \times [0, p_{\max}],$$

say

$$(X', U') \sim \text{Unif}([0, 1] \times [0, p_{\max}]),$$

and accept if

$$U' \leq \pi(X').$$

4.3 The real rejection sampler

The rectangle method is often too crude. A better idea is to use a proposal density q whose shape is closer to the target.

Definition. *Let q be a proposal density. Assume there exists a constant $M > 0$ such that*

$$\gamma(x) \leq Mq(x) \qquad \textit{for all } x.$$

Then Mq is called an envelope for γ.

Remark. *Using Mq is often much better than using a rectangle, because the envelope can adapt to the shape of the target density.*

The conceptual algorithm is:

- sample $X' \sim q(x)$;
- sample $U \sim \text{Unif}(0, 1)$;
- accept X' if

$$U \leq a(X') = \frac{\pi(X')}{Mq(X')} \leq 1 \tag{3}$$

  for a normalised target π.

For an unnormalised target γ, the acceptance probability is

$$\alpha(X') = \frac{\gamma(X')}{Mq(X')}.$$

Equivalently, one may generate

$$U' \sim \text{Unif}(0, Mq(X'))$$

and accept whenever

$$U' \leq \gamma(X').$$

Algorithm 1 Rejection Sampling (Continuous random variable)

1: **Input:** The number of iterations n and scaling factor M.
2: **for** $i = 1, \ldots, n$ **do**
3: Generate $X_i' \sim q(x')$.
4: $U \sim \text{Unif}(0,1)$.
5: **if** $U \leq \dfrac{\pi(X_i')}{Mq(X_i')}$ **then**
6: Accept X_i'.
7: **end if**
8: **end for**
9: Return accepted samples

If we want N accepted samples, we repeat the procedure until N acceptances are obtained.

```
i = 0
while i < N
    draw X from q
    draw U from Unif(0,1)
    if U <= gamma(X)/(M q(X))
        i = i + 1
        output X
    end
end
```

4.4 Distribution of the accepted values

Proposition. *The distribution of the accepted values in the rejection sampling algorithm is the target distribution π.*

Proof. Let A denote the event that X' is accepted. Then

$$\mathbb{P}(X' \leq x \mid A) = \frac{\mathbb{P}(X' \leq x,\ A)}{\mathbb{P}(A)}.$$

For an unnormalised density γ, the denominator is

$$\mathbb{P}(A) = \mathbb{P}\left(U \leq \frac{\gamma(X')}{Mq(X')}\right) = \int q(t)\frac{\gamma(t)}{Mq(t)}\,dt = \frac{z}{M}.$$

For the numerator,

$$\begin{aligned}\mathbb{P}(X' \leq x,\ A) &= \int_{-\infty}^{x} q(t)\frac{\gamma(t)}{Mq(t)}\,dt \\ &= \frac{1}{M}\int_{-\infty}^{x}\gamma(t)\,dt.\end{aligned}$$

Therefore,

$$\mathbb{P}(X' \leq x \mid A) = \frac{\frac{1}{M}\int_{-\infty}^{x}\gamma(t)\,dt}{z/M} = \frac{\int_{-\infty}^{x}\gamma(t)\,dt}{z} = \int_{-\infty}^{x}\pi(t)\,dt.$$

Hence the accepted value has density π. □

For a normalised target π, the same proof gives

$$\mathbb{P}(X \leq x \mid X \text{ accepted}) = \int_{-\infty}^{x}\pi(x')\,dx'.$$

4.5 Discrete random variables

Given a target pmf $\boldsymbol{\pi} = (\pi_1, \ldots, \pi_K)$ and proposal pmf $\mathbf{q} = (q_1, \ldots, q_K)$, if

$$\pi_i \leq M q_i, \qquad i = 1, \ldots, K,$$

then the discrete rejection sampler is:

Algorithm 2 Rejection Sampling (Discrete random variable)

1: **Input:** The number of iterations n and scaling factor M.
2: **for** $i = 1, \ldots, n$ **do**
3: Sample s_i with probability $\mathbf{q} = (q_1, \ldots, q_K)$.
4: $U \sim \text{Unif}(0, 1)$.
5: **if** $U \leq \dfrac{\pi_i}{M q_i}$ **then**
6: Accept s_i.
7: **end if**
8: **end for**
9: Return accepted samples

4.6 Acceptance rate

An important measure of efficiency is the acceptance rate: accepted samples divided by total proposals.

Proposition. *When the target density $\pi(x)$ is normalised and M is fixed, the acceptance rate is*

$$\hat{a} = \frac{1}{M}.$$

Proof.

$$\hat{a} = P(\text{accept}) = P\left(U \leq \frac{\pi(X')}{M q(X')}\right).$$

Conditioning on $X' = x'$,

$$P(\text{accept} \mid X' = x') = \frac{\pi(x')}{M q(x')}.$$

Therefore

$$P(\text{accept}) = \int \frac{\pi(x')}{M q(x')} q(x')\, dx' = \frac{1}{M} \int \pi(x')\, dx' = \frac{1}{M}.$$

□

Proposition. *For an unnormalised target density $\gamma(x)$ with normalising constant*

$$z = \int \gamma(x)\, dx,$$

the acceptance rate is

$$\hat{a} = \frac{z}{M}.$$

Proof.

$$\hat{a} = \int \frac{\gamma(x')}{M q(x')} q(x')\, dx' = \frac{1}{M} \int \gamma(x')\, dx' = \frac{z}{M}.$$

□

Proposition. *The number of proposals needed until the first acceptance is geometrically distributed with success probability $1/M$ in the normalised case, and with success probability z/M in the unnormalised case.*

Remark. *There is a difference between*

$$a(x) = \frac{\pi(x)}{M q(x)}$$

and $\hat{a}$. The first is the acceptance ratio *or pointwise acceptance probability at the point x, while the second is its average, i.e. the* acceptance rate.

4.7 Rejection sampling without normalising constants

Suppose the target density is only known up to a normalising constant:

$$\pi(x) = \frac{\gamma(x)}{Z}, \qquad Z = \int \gamma(x)\, dx.$$

This is common in computational statistics.

Rejection sampling still works, provided

$$\gamma(x) \le Mq(x) \qquad \text{for all } x.$$

Algorithm 3 Rejection sampling without normalising constants

Input: The number of iterations n and scaling factor M.
for $i = 1, \dots, n$ **do**
  Generate $X_i' \sim q(x')$.
  $U \sim \text{Unif}(0,1)$.
  if $U \le \dfrac{\gamma(X_i')}{Mq(X_i')}$ **then**
    Accept X_i'.
  end if
end for
Return accepted samples

Example. ***Truncated Gaussian.*** *Consider*

$$\gamma(x) = \mathcal{N}(x; 0, 1)\, 1_{|x| \le a}(x).$$

The normalised target is

$$\pi(x) = \frac{\gamma(x)}{Z} = \frac{\mathcal{N}(x; 0, 1)\, 1_{|x| \le a}(x)}{\int_{-a}^{a} \mathcal{N}(y; 0, 1)\, dy}. \tag{4}$$

Take

$$q(x) = \mathcal{N}(x; 0, 1),$$

so that $\gamma(x) \le q(x)$ and therefore $M = 1$. In this case one simply samples from the Gaussian and accepts if the draw lies in $[-a, a]$. The acceptance rate is

$$\hat{a} = Z = \int_{-a}^{a} \mathcal{N}(y; 0, 1)\, dy.$$

4.8 Designing the optimal rejection sampler

For a given target π or unnormalised target γ and proposal q, the optimal scaling constant is the smallest valid envelope constant. For a normalised target,

$$M_{\max} = \sup_x \frac{\pi(x)}{q(x)}.$$

For an unnormalised target,

$$M^* = \sup_x \frac{\gamma(x)}{q(x)}.$$

This is the smallest valid M and therefore gives the highest possible acceptance rate.

Remark. *To maximise the acceptance rate, M should be chosen as small as possible, subject to*

$$\gamma(x) \le Mq(x) \qquad \forall x.$$

The optimal choice is therefore

$$M^* = \sup_x \frac{\gamma(x)}{q(x)}.$$

4.9 Example: Gaussian target with Cauchy proposal

Example. *Consider*

$$\gamma(x) = e^{-x^2/2},$$

so that the target density is proportional to a standard Gaussian density. Take as proposal the Cauchy density

$$q(x) = \frac{1}{\pi(1+x^2)}.$$

We can sample from q, so rejection sampling is feasible.

We need to find M such that

$$\gamma(x) \le Mq(x) \qquad \forall x.$$

The smallest such constant is

$$M^* = \sup_x \frac{\gamma(x)}{q(x)}.$$

Now

$$\frac{\gamma(x)}{q(x)} = \pi(1+x^2)e^{-x^2/2}.$$

Define

$$R(x) = \pi(1+x^2)e^{-x^2/2}.$$

To maximise R, it is enough to maximise $\log R$:

$$\log R(x) = \log \pi + \log(1+x^2) - \frac{x^2}{2}.$$

Hence

$$\frac{d}{dx}\log R(x) = \frac{2x}{1+x^2} - x.$$

Setting this equal to zero gives

$$\frac{2x}{1+x^2} - x = 0 \iff x\left(\frac{2}{1+x^2} - 1\right) = 0 \iff x(x^2-1) = 0.$$

Thus the critical points are

$$x = 0, \pm 1.$$

Comparing values, the maximum is attained at

$$x = \pm 1,$$

while $x = 0$ is a minimum. Therefore

$$M^* = R(\pm 1) = \pi(1+1)e^{-1/2} = 2\pi e^{-1/2}.$$

So the rejection sampler becomes:

- draw $X \sim q$ with
$$q(x) = \frac{1}{\pi(1+x^2)};$$
- draw
$$U \sim \text{Unif}(0,1);$$
- accept X if
$$U \le \frac{\gamma(X)}{M^* q(X)} = \frac{e^{-X^2/2}}{2\pi e^{-1/2}\, q(X)}.$$

Since

$$z = \int_{-\infty}^{+\infty} e^{-x^2/2}\, dx = \sqrt{2\pi},$$

the corresponding acceptance rate is

$$\frac{z}{M^*} = \frac{\sqrt{2\pi}}{2\pi e^{-1/2}} = \frac{e^{1/2}}{\sqrt{2\pi}}.$$

Example. *For the Beta*$(2,2)$ *density, the box method used* $M_{box} = 1.5$*, so*

$$\hat{a}_{box} = \frac{1}{1.5} = 0.67.$$

If instead one takes

$$q(x) = \mathcal{N}(x; 0.5, 0.25)$$

with $M_{Gauss} = 1.3$*, then*

$$\hat{a}_{Gauss} = \frac{1}{1.3} = 0.77.$$

Hence the Gaussian proposal is more efficient.

Chapter 5

5_MultivariateDistributions — Composition & Multivariate Simulation

We can use the ideas seen previously to sample from multivariate and conditional distributions.[1]

5.1 Composition

So far, the main ideas have been:

1. sample uniforms;
2. use transformations to sample from "basic" random variables;
3. use composition ideas to sample from less basic random variables.

When a density can be written as a composition of simpler operations, we can often sample from it straightforwardly. We begin with mixture densities.

5.2 Sampling from discrete mixtures of pdfs

Definition. *A random variable has a* discrete mixture *distribution if its density or pmf can be written as*

$$\pi(x) = w_1 q_1(x) + w_2 q_2(x) + \cdots + w_K q_K(x) = \sum_{i=1}^{K} w_i q_i(x),$$

where each q_i is a distribution and the weights satisfy

$$w_i \geq 0, \qquad \sum_{i=1}^{K} w_i = 1.$$

To sample from such a mixture:

- first sample an index J from the pmf $\mathbb{P}(J = j) = w_j$;
- then sample $X \sim q_J(x)$.

Theorem. *If*

$$\mathbb{P}(J = j) = w_j, \qquad j = 1, \ldots, K,$$

and conditional on $J = j$ we generate

$$X \sim q_j,$$

then the unconditional distribution of X is

$$\pi(x) = \sum_{j=1}^{K} w_j q_j(x).$$

Proof. For any x,

$$\mathbb{P}(X \in dx) = \sum_{j=1}^{K} \mathbb{P}(X \in dx \mid J = j)\mathbb{P}(J = j) = \sum_{j=1}^{K} q_j(x) w_j \, dx.$$

[1] These notes are based on lecture notes developed by (in alphabetical order): Ö. Deniz Akyildiz, Nicolas Chopin, Heiko Großmann, Adam M. Johansen, Amir Khorrami Chokami, Nikolas Nüsken, Art Owen. Many thanks to them for making their material available.

Hence the density/pmf of X is exactly

$$\pi(x) = \sum_{j=1}^{K} w_j q_j(x).$$

□

Algorithm 1 Sampling discrete mixtures

1: Input: The number of samples N.
2: for $i = 1, \dots, N$ do
3: Generate $J \sim \pi(j)$ with $\mathbb{P}(J = j) = w_j$.
4: Generate $X_i \sim q_J(x)$.
5: end for

Remark. *If you see a density or pmf written as a sum of distributions with positive coefficients summing to one, this is often a sign that you are dealing with a discrete mixture.*

Example. *Suppose*

$$\mathbb{P}(X = k) = \frac{1}{2}\frac{4^k e^{-4}}{k!} + \frac{1}{4}\frac{10^k e^{-10}}{k!} + \frac{1}{4}\frac{2^k e^{-2}}{k!}, \qquad k = 0, 1, 2, \dots.$$

Then X is a mixture of Poisson distributions:

$$X \sim \frac{1}{2}\,\mathrm{Poi}(4) + \frac{1}{4}\,\mathrm{Poi}(10) + \frac{1}{4}\,\mathrm{Poi}(2).$$

So we can sample from X as follows:

1. *generate J with*
$$\mathbb{P}(J = 1) = \frac{1}{2}, \qquad \mathbb{P}(J = 2) = \frac{1}{4}, \qquad \mathbb{P}(J = 3) = \frac{1}{4};$$
2. *if $J = 1$, sample $X \sim \mathrm{Poi}(4)$;*
3. *if $J = 2$, sample $X \sim \mathrm{Poi}(10)$;*
4. *if $J = 3$, sample $X \sim \mathrm{Poi}(2)$.*

Example. *Consider*

$$\pi(x) = \frac{1}{2}\frac{1}{\sqrt{2\pi}}e^{-x^2/2} + \frac{1}{2}\frac{1}{\sqrt{2\pi}}e^{-(x-1)^2/2}.$$

Then

$$\pi(x) = \frac{1}{2}\mathcal{N}(0,1) + \frac{1}{2}\mathcal{N}(1,1)$$

in mixture form.

Hence one may sample $X \sim \pi$ by:

1. *generating $J \sim \mathrm{Bernoulli}(1/2)$;*
2. *if $J = 0$, sampling $X \sim \mathcal{N}(0,1)$;*
3. *if $J = 1$, sampling $X \sim \mathcal{N}(1,1)$.*

Remark. *If*

$$X \sim \frac{1}{2}\mathcal{N}(0,1) + \frac{1}{2}\mathcal{N}(1,1),$$

this is not *the same as*

$$Z_1 + Z_2, \qquad Z_1 \sim \mathcal{N}(0,1), \quad Z_2 \sim \mathcal{N}(1,1).$$

Summing random variables is not the same as summing densities.

Example. *Let*

$$X_1 \sim \mathcal{N}(0,1), \qquad X_2 \sim \mathcal{N}(6,4),$$

independent. The random variable

$$S = \frac{1}{2}X_1 + \frac{1}{2}X_2$$

is Gaussian:

$$S \sim \mathcal{N}(3, 5/4).$$

By contrast, if Y has cdf

$$F_Y(x) = \frac{1}{2}F_{X_1}(x) + \frac{1}{2}F_{X_2}(x),$$

then

$$f_Y(x) = \frac{1}{2}f_{X_1}(x) + \frac{1}{2}f_{X_2}(x),$$

which is a Gaussian mixture, not the distribution of S.

Example. *A mixture of Gamma distributions:*

$$F_X(x) = \sum_{k=1}^{5} w_k F_{X_k}(x), \qquad X_k \sim \text{Gamma}(3, 1/k), \qquad w_k = \frac{k}{15}.$$

To generate one sample:

- *sample* $k \in \{1, 2, 3, 4, 5\}$ *with pmf* w_k*;*
- *sample from* $\text{Gamma}(3, 1/k)$*.*

5.3 Sampling from discrete mixtures of pmfs

In the discrete case, if $X_k \sim F_{X_k}$ and weights w_k satisfy $\sum_{k=1}^{K} w_k = 1$, then X is a discrete mixture if

$$F_X(x) = \sum_{k=1}^{K} w_k F_{X_k}(x).$$

Simulation proceeds exactly as above: first sample the component index I, then sample from F_{X_I}.

5.4 Sampling from conditional densities

If

$$\pi(y \mid x) = \mathcal{N}(y; x, 1),$$

then for fixed x, sampling from $\pi(y \mid x)$ is just ordinary Gaussian sampling with mean x and variance 1.
Let X have marginal distribution $\pi(x)$, and let the conditional distribution of Y given $X = x$ be

$$Y \mid X = x \sim \pi(y \mid x).$$

Theorem. *If we generate*

$$X \sim \pi(x), \qquad Y \mid X \sim \pi(y \mid X),$$

then the pair (X, Y) *has joint distribution*

$$\pi(x, y) = \pi(x)\pi(y \mid x).$$

Remark. *Algorithmically, to generate N samples:*

1. *for* $i = 1, \ldots, N$*, generate*
$$X_i \sim \pi(x);$$
2. *then generate*
$$Y_i \sim \pi(y \mid X_i).$$

If needed, one may keep only the Y_i*'s.*

5.5 Sampling from joint distributions

A joint density can be factorised as

$$\pi(x, y) = \pi(y \mid x)\pi(x).$$

Hence to sample $(X, Y) \sim \pi(x, y)$ one may:

$$X \sim \pi(x), \qquad Y \mid X = x \sim \pi(y \mid x).$$

More generally,

$$\pi(x_1, \ldots, x_d) = \pi(x_1)\pi(x_2 \mid x_1)\pi(x_3 \mid x_1, x_2) \cdots \pi(x_d \mid x_1, \ldots, x_{d-1}).$$

Equivalently, one may also factorize in reverse order:

$$\pi(x_1, \ldots, x_d) = \pi(x_d)\pi(x_{d-1} \mid x_d)\pi(x_{d-2} \mid x_{d-1}, x_d) \cdots \pi(x_1 \mid x_2, \ldots, x_d).$$

So one may sample sequentially:

$$\begin{aligned} X_1 &\sim \pi(x_1), \\ X_2 &\sim \pi(x_2 \mid x_1), \\ X_3 &\sim \pi(x_3 \mid x_1, x_2), \\ &\vdots \\ X_d &\sim \pi(x_d \mid x_1, \ldots, x_{d-1}). \end{aligned}$$

Remark. *This factorisation idea is fundamental in probabilistic modelling: complex joint distributions are built from simpler conditionals and marginals, and simulation follows the same order.*

Example. *Consider the joint density*

$$\pi(x,y,z) = \frac{e^{-x}}{(1+e^{-x})^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-(y-x)^2/2} \cdot \frac{1}{\sqrt{2\pi}} e^{-(z-(x+y))^2/2}.$$

This can be read as

$$X \sim \mathit{Logistic}(0,1), \qquad Y \mid X = x \sim \mathcal{N}(x,1), \qquad Z \mid X = x, Y = y \sim \mathcal{N}(x+y,1).$$

Therefore:

1. *sample*
$$X_i \sim \mathit{Logistic}(0,1)$$
for example using the inverse cdf method;
2. *sample*
$$\varepsilon_i \sim \mathcal{N}(0,1), \qquad Y_i = X_i + \varepsilon_i;$$
3. *sample*
$$\varepsilon_i' \sim \mathcal{N}(0,1), \qquad Z_i = X_i + Y_i + \varepsilon_i'.$$

Then

$$(X_i, Y_i, Z_i) \sim \pi.$$

5.6 Sampling from continuous mixtures or marginalisation

Since

$$\pi(y) = \int \pi(x,y)\,dx = \int \pi(y \mid x)\pi(x)\,dx,$$

one can sample from $\pi(y)$ by sampling from the joint distribution and retaining only the Y-coordinates.

Theorem. *If*

$$(X,Y) \sim \pi(x,y),$$

and we discard X, then the remaining variable Y has density

$$\pi(y) = \int \pi(x,y)\,dx.$$

Similarly, discarding Y leaves $X \sim \pi(x) = \int \pi(x,y)\,dy$.

Remark. *So one way to sample from a marginal distribution is:*

1. *sample from the joint density $\pi(x,y)$;*
2. *discard the unwanted coordinate.*

Remark. *In some cases it is convenient to factorize*

$$\pi(x,y) = \pi(y)\pi(x \mid y),$$

sample Y first, and then sample $X \mid Y$.

If instead only

$$\pi(x,y) = \pi(x)\pi(y \mid x)$$

is available and $\pi(x)$ is difficult to sample from, this factorization may be less useful for direct simulation of the marginal.

Example. *If*

$$\pi(x) = \mathcal{N}(x;\mu,\sigma^2), \qquad \pi(y \mid x) = \mathcal{N}(y;x,1),$$

then

$$\pi(y) = \mathcal{N}(y;\mu,\sigma^2+1).$$

Numerically one may verify this by sampling

$$X \sim \pi(x), \qquad Y \mid X = x \sim \pi(y \mid x),$$

and plotting the resulting Y values.

Example. *Suppose*

$$\pi(x, y) = \phi(x)\phi(y),$$

where ϕ denotes the standard normal density. Then

$$(X, Y) \sim \pi$$

means that X and Y are independent standard normal variables.
If we have available samples

$$(x_1, y_1), \ldots, (x_n, y_n) \sim \pi,$$

then simply discarding the first coordinate gives

$$y_1, \ldots, y_n \sim \pi(y).$$

Example. ***Linear model.*** *To simulate data with a linear relationship, consider*

$$y = ax + b + n, \qquad n \sim \mathcal{N}(0, \sigma^2).$$

Take, for example,

$$\pi(x) = \mathrm{Unif}(x; -10, 10), \qquad \pi(y \mid x) = \mathcal{N}(y; ax + b, \sigma^2).$$

Then generate

$$X_i \sim \pi(x), \qquad Y_i \mid X_i = x_i \sim \pi(y \mid x_i).$$

5.7 Sampling multivariate densities

A multivariate density is simply a joint density in d dimensions:

$$\pi(x_1, \ldots, x_d).$$

The previous factorisation ideas still apply when the conditionals are known.

5.8 Sampling a multivariate Gaussian

Let $x \in \mathbb{R}^d$ and

$$\pi(x) = (2\pi)^{-d/2} |\det \Sigma|^{-1/2} \exp\left(-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu)\right),$$

where $\mu \in \mathbb{R}^d$ and $\Sigma \in \mathbb{R}^{d \times d}$ is symmetric positive definite.
More generally, let

$$X = (X_1, \ldots, X_d) \sim \mathcal{N}(\mu, \Sigma),$$

where

$$\mu \in \mathbb{R}^d, \qquad \Sigma \in \mathbb{R}^{d \times d}$$

is symmetric positive semidefinite.
A useful special case is obtained from a standard Gaussian vector

$$Z = (Z_1, \ldots, Z_d) \sim \mathcal{N}(0, I_d),$$

that is,

$$Z_i \sim \mathcal{N}(0, 1) \qquad \text{for } i = 1, \ldots, d,$$

independently.

Theorem. *If $\Sigma^{1/2}$ is a matrix square root of Σ, meaning that*

$$\Sigma^{1/2}\Sigma^{1/2} = \Sigma,$$

then

$$X = \mu + \Sigma^{1/2} Z$$

has multivariate normal distribution

$$X \sim \mathcal{N}(\mu, \Sigma).$$

In practice one often computes $\Sigma^{1/2}$ via the Cholesky decomposition

$$\Sigma = LL^\top.$$

Then

$$X = \mu + LZ$$

has distribution $\mathcal{N}(\mu, \Sigma)$.

Algorithm 2 Sampling Multivariate Gaussian

1: Input: The number of samples N.
2: Compute L such that $\Sigma = LL^\top$ using the Cholesky decomposition.
3: for $i = 1, \ldots, N$ do
4: Draw d independent univariate normals $v_k \sim \mathcal{N}(0, 1)$ and form $v = [v_1, \ldots, v_d]^\top$.
5: Generate $X_i = \mu + Lv$.
6: end for

Chapter 6

6_MonteCarloIntegration — Monte Carlo Integration

We are now ready to introduce Monte Carlo integration. This is a powerful technique used to approximate integrals and expectations which cannot be computed analytically.[1]

6.1 Introduction to Monte Carlo integration

Given a probability distribution π, we are often interested in expectations of the form

$$\pi(\varphi) = \int \pi(x)\varphi(x)\,dx,$$

where φ is a **test function**. For instance, $\varphi(x) = x$ yields the mean, $\varphi(x) = x^2$ the second moment, and $\varphi(x) = \log x$ the entropy.

If

$$X_1, \ldots, X_N \sim \pi$$

are iid, then

$$\mathbb{E}_\pi[X] = \int x\pi(x)\,dx \approx \frac{1}{N}\sum_{i=1}^{N} X_i.$$

This corresponds to approximating π by its empirical measure

$$\pi^N(x) = \frac{1}{N}\sum_{i=1}^{N} \delta_{X_i}(x) = \frac{\#\{i : X_i = x\}}{N},$$

which coincides with the empirical pdf introduced in Chapter 1.

The Dirac measure δ_y may be viewed as a point mass at y, informally

$$\delta_y(x) = \begin{cases} 1, & x = y, \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$\pi(\varphi) = \int \varphi(x)\pi(x)\,dx \approx \int \varphi(x)\pi^N(x)\,dx$$

$$= \int \varphi(x)\frac{1}{N}\sum_{i=1}^{N}\delta_{X_i}(x)\,dx = \frac{1}{N}\sum_{i=1}^{N}\varphi(X_i) = \pi^N(\varphi).$$

This is the **vanilla Monte Carlo estimator**.

Remark. *Monte Carlo estimators can also approximate ordinary integrals. If*

$$I = \int f(x)\,dx,$$

[1] These notes are based on lecture notes developed by (in alphabetic order): Ö. Deniz Akyildiz, Nicolas Chopin, Heiko Großmann, Adam M. Johansen, Amir Khorrami Chokami, Nikolas Nüsken, Art Owen. Many thanks to them for making their material available.

and π is a density with support covering the domain of integration, then

$$I = \int \underbrace{\frac{f(x)}{\pi(x)}}_{\varphi(x)} \pi(x)\,dx \approx \frac{1}{N}\sum_{i=1}^{N} \frac{f(X_i)}{\pi(X_i)}, \qquad X_i \sim \pi.$$

For integrals over $[0,1]$, one often chooses π uniform, in which case the ratio simplifies.

6.2 Unbiasedness and variance

Proposition. *If $X_1, \dots, X_N$ are iid from π, then the Monte Carlo estimator*

$$\pi^N(\varphi) = \frac{1}{N}\sum_{i=1}^{N} \varphi(X_i)$$

is unbiased:

$$\mathbb{E}[\pi^N(\varphi)] = \pi(\varphi).$$

Proof.

$$\mathbb{E}[\pi^N(\varphi)] = \mathbb{E}\left[\frac{1}{N}\sum_{i=1}^{N}\varphi(X_i)\right] = \frac{1}{N}\sum_{i=1}^{N}\mathbb{E}[\varphi(X_i)] = \int \varphi(x)\pi(x)\frac{1}{N}\sum_{i=1}^{N} 1\,dx = \pi(\varphi).$$

□

Proposition. *If $X_1, \dots, X_N$ are iid from π, then*

$$\mathrm{Var}[\pi^N(\varphi)] = \frac{1}{N}\mathrm{Var}[\varphi(X)],$$

where

$$\mathrm{Var}[\varphi(X)] = \int (\varphi(x) - \pi(\varphi))^2 \pi(x)\,dx.$$

Proof.

$$\mathrm{Var}[\pi^N(\varphi)] = \mathrm{Var}\left(\frac{1}{N}\sum_{i=1}^{N}\varphi(X_i)\right) = \frac{1}{N^2}\sum_{i=1}^{N}\mathrm{Var}[\varphi(X_i)] = \frac{1}{N}\mathrm{Var}[\varphi(X)].$$

□

Provided $\mathrm{Var}[\varphi(X)] < \infty$, the estimator is consistent as $N \to \infty$.

Remark. *In practice the exact variance*

$$\mathrm{Var}[\pi^N(\varphi)] = \frac{1}{N}\mathrm{Var}[\varphi(X)]$$

depends on the unknown quantity $\pi(\varphi)$. Replacing it with $\pi^N(\varphi)$ yields the empirical estimator

$$\sigma^2_{\varphi,N} = \frac{1}{N^2}\sum_{i=1}^{N}\Big(\varphi(X_i) - \pi^N(\varphi)\Big)^2.$$

This leads to the asymptotic normal approximation

$$\frac{\pi^N(\varphi) - \pi(\varphi)}{\sigma_{\varphi,N}} \xrightarrow{d} \mathcal{N}(0,1) \qquad \text{as } N \to \infty.$$

Also,

$$\mathrm{Std}[\pi^N(\varphi)] = \frac{\sigma_\varphi}{\sqrt{N}},$$

which shows the typical Monte Carlo convergence rate $O(N^{-1/2})$.

Remark. *Probabilities are a special case of expectations. For an event A,*

$$P(X \in A) = \int_A \pi(x)\,dx = \int 1_A(x)\pi(x)\,dx.$$

Thus with $\varphi(x) = 1_A(x)$,

$$P(X \in A) \approx \frac{1}{N}\sum_{i=1}^{N} 1_A(X_i).$$

So one simply counts the simulated samples falling in A and divides by N.

6.3 Examples

Example. *Consider*

$$I = \int_0^1 [\cos(50x) + \sin(20x)]^2 \, dx.$$

The exact value is 0.965. *Choose*

$$\pi(x) = \text{Unif}(0,1), \qquad \varphi(x) = [\cos(50x) + \sin(20x)]^2,$$

and apply the standard Monte Carlo estimator.

Example. *Let* $X \sim \mathcal{N}(0,1)$. *To estimate*

$$P(X > 2),$$

take

$$\pi(x) = \mathcal{N}(0,1), \qquad \varphi(x) = 1_{\{x>2\}}(x).$$

Then

$$P(X > 2) \approx \frac{1}{N} \sum_{i=1}^{N} 1_{\{x>2\}}(X_i), \qquad X_i \sim \mathcal{N}(0,1).$$

Example. *Let*

$$\pi(x) = \frac{1}{\pi(1+x^2)}.$$

We want to estimate

$$P(X > 2) = \int_2^\infty \pi(x) \, dx.$$

Take

$$\varphi(x) = 1_{\{x>2\}}(x).$$

The exact value is

$$I = \int_2^\infty \pi(x) \, dx = \frac{1}{2} - \frac{1}{\pi} \tan^{-1}(2) = 0.1476.$$

Now

$$\text{Var}(\varphi) = 0.1476 - 0.1476^2 = 0.125,$$

so for $N = 10$,

$$\text{Var}(\pi^N(\varphi)) = \frac{0.125}{10} = 0.0125.$$

A better estimator uses symmetry:

$$P(X > 2) = P(X < -2).$$

Hence

$$P(X > 2) = \frac{1}{2} P(|X| > 2) = \int \frac{1}{2} 1_{\{|x|>2\}}(x) \pi(x) \, dx.$$

Now take

$$\varphi(x) = \frac{1}{2} 1_{\{|x|>2\}}(x).$$

Then

$$\text{Var}(\varphi) = \frac{1}{4} \cdot 2 \cdot 0.1476 - \frac{1}{4}(2 \cdot 0.1476)^2 = 0.052,$$

and for $N = 10$,

$$\text{Var}(\pi^N(\varphi)) = \frac{0.052}{10} = 0.0052.$$

This is an improvement.

Example. *When one integration limit is* $+\infty$,

$$\int_0^{+\infty} f(x) \, dx,$$

one may use the change of variable

$$y = \frac{1}{x+1}$$

to obtain an integral over $[0,1]$*:*

$$\int_0^{+\infty} f(x) \, dx = \int_0^1 y^{-2} f(y^{-1} - 1) \, dy.$$

On $(-\infty, 0)$ *this transformation is not suitable, and one should instead use*

$$y = \frac{1}{1-x}.$$

These transformations are useful, but if possible it is usually better to find a convenient density π *with sufficiently large support and use that instead.*

6.4 Multivariate integrals

Monte Carlo methods are especially useful in high dimension. If

$$I = \int_{[0,1]^d} f(x_1, \dots, x_d)\, dx_1 \cdots dx_d,$$

then

$$I = \mathbb{E}[f(U_1, \dots, U_d)], \qquad U_k \sim \text{Unif}(0,1),$$

and may be approximated by

$$\hat{I} = \frac{1}{N} \sum_{i=1}^{N} f(U_1^i, \dots, U_d^i).$$

Example. *To estimate π geometrically, consider the square $[-1,1] \times [-1,1]$ with uniform density and the unit disk*

$$A = \{(x,y) \in [-1,1] \times [-1,1] : x^2 + y^2 \le 1\}.$$

Then

$$P((X,Y) \in A) = \int 1_A(x,y)\pi(x,y)\, dx\, dy \approx \frac{1}{N} \sum_{i=1}^{N} 1_A(X_i, Y_i),$$

with (X_i, Y_i) uniform on the square. This converges to

$$\frac{\pi}{4}.$$

Chapter 7

Recap Exercises and Practicals

7.1 Bootstrap & Jackknife: Recap Exercises

Exercise 1

For a random sample $y_1, \ldots, y_n$ from the Gamma distribution, the method of moments estimator of the shape parameter θ is

$$\hat{\theta} = \frac{m_1^2}{m_2 - m_1^2},$$

where

$$m_1 = \frac{1}{n}\sum_{i=1}^{n} y_i, \qquad m_2 = \frac{1}{n}\sum_{i=1}^{n} y_i^2.$$

Assume that

$$3.50, \quad 3.55, \quad 1.13, \quad 1.86$$

are a random sample from a Gamma distribution. Find by hand the jackknife estimates of bias and standard error of $\hat{\theta}$.

Exercise 2

1. The coefficient of skewness of a random variable Y with mean $\mu = E(Y)$ and standard deviation $\sigma = \sqrt{\text{Var}(Y)}$ is

   $$\gamma = \frac{E[(Y-\mu)^3]}{\sigma^3}.$$

   For a random sample $y_1, \ldots, y_n$, find the plug-in estimate of γ.
2. For the data

   $$4.3 \qquad 6.7 \qquad 7.2 \qquad 3.2 \qquad 10$$

   compute the plug-in estimate.

Exercise 3

When estimating $\theta = \mathbb{E}(Y)$, the sample mean may be strongly affected by outliers. A robust alternative is the trimmed mean. Consider

$$1.2, \quad 3.5, \quad 4.7, \quad 7.3, \quad 8.6, \quad 12.4, \quad 13.8, \quad 18.1.$$

Let $\hat{\theta}$ be the trimmed mean obtained by deleting the two smallest and the two largest observations and averaging the remaining ones. Use the jackknife to estimate the bias and the standard error of $\hat{\theta}$ by hand.

Exercise 4

Let $y_1, \ldots, y_n$ be iid with variance σ^2. Show that, when using

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(y_i - \bar{y})^2$$

to estimate σ^2, the jackknife estimate of bias is

$$\widehat{\text{bias}}_{\text{jack}} = -\frac{1}{n}s^2,$$

where

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2.$$

Exercise 5

Salaries are often modelled by the Pareto distribution

$$f(y) = \theta k^\theta y^{-(\theta+1)}, \qquad y \geq k,$$

with $k > 0$ and $\theta > 1$. One estimator of θ is

$$\hat{\theta} = \frac{n\bar{y} - y_{(1)}}{n(\bar{y} - y_{(1)})},$$

where $y_{(1)}$ is the smallest observation. For the salary data

$$43, \quad 24, \quad 60, \quad 55, \quad 27,$$

calculate the jackknife estimates of bias and variance of $\hat{\theta}$.

Exercise 6

The following data represent independent observations of service hours between failures of air-conditioning equipment in a Boeing 720 jet:

$$3, \quad 5, \quad 7, \quad 18, \quad 43, \quad 85, \quad 91, \quad 98, \quad 100, \quad 130, \quad 230, \quad 487.$$

(a) Create a Q–Q plot using an exponential distribution with parameter $\hat{\lambda}$ estimated from the data as reference distribution. Briefly interpret the plot.
(b) Suppose the sample mean is used as estimator $\hat{\theta}$ of the expected number of service hours θ. Use the nonparametric bootstrap to estimate the bias and standard error of the sample mean. Also plot a histogram of the bootstrap replications.
(c) Assuming the data come from an exponential distribution with parameter λ, estimate the bias and standard error of the sample mean using the parametric bootstrap, and plot a histogram of the bootstrap replications. Compare the results with part (b).

Exercise 7

Income data are often positively skewed, so the median is often preferred to the sample mean. A common model is the lognormal distribution

$$f(y) = \frac{1}{y\sqrt{2\pi}} \exp\left(-\frac{(\log y - \mu)^2}{2}\right), \qquad y > 0.$$

(*Hint:* the mean of the lognormal distribution is $\exp(\mu + 1/2)$.)

Given the monthly income data

$$1550, \quad 1580, \quad 1620, \quad 1910, \quad 2180, \quad 3690,$$

use the nonparametric bootstrap to estimate the bias of the sample median. Repeat using the parametric bootstrap.

7.2 Practical 1: Jackknife & Bootstrap

Practical 1: Jackknife & Bootstrap

Exercise 1 (Jackknife). *In the lectures we saw that for the sample mean the jackknife estimate of bias is equal to zero and that the jackknife estimate of variance is the sample variance divided by the sample size. Verify this for a random sample.*

(a) *Sample $N = 10$ points from a standard Gaussian random variable and compute the sample mean $\hat{\theta}$.*
(b) *Compute the jackknife replicates $\hat{\theta}_{[i]}$ for $i = 1, \dots, N$.*
(c) *Compute the jackknife estimate of bias and variance and compare them with the theoretical ones.*
(d) *Repeat the above calculations using the built-in function in Python. See* https://docs.astropy.org/en/latest/api/astropy.stats.jackknife_stats.html *and* https://docs.astropy.org/en/latest/api/astropy.stats.jackknife_resampling.html.

Exercise 2 (Jackknife). *Two techniques for measuring blood flow, called DYE and EFP, were used on $N = 9$ dogs. A medical investigator wanted to measure the association between the two methods. The data are*

DYE	1.15	1.70	1.42	1.38	2.80	4.70	4.80	1.41	3.90
EFP	1.38	1.72	1.59	1.47	1.66	3.45	3.87	1.31	3.75

Here the parameter of interest is the correlation.

(a) *Compute the sample correlation $\hat{\theta}$.*
(b) *Compute the jackknife replicates $\hat{\theta}_{[i]}$ for $i = 1, \dots, N$.*
(c) *Compute the jackknife estimate of bias and variance and compare them with the theoretical ones.*
(d) *Repeat the above calculations using the built-in function in Python.*

Exercise 3 (Bootstrap). *Eight subjects wore medical patches designed to increase the blood levels of a certain natural hormone. Each subject had the levels of the hormone in his blood measured after wearing three different patches: a placebo patch, an old patch, and a new patch. The observations are*

placebo	92437	9671	11792	13357	9055	6290	12412	18806
old	17649	12013	19979	21816	13850	9806	17208	29044
new	16449	14614	17274	23798	12560	10157	16570	26325

The Food and Drug Administration (FDA) criterion for bioequivalence is

$$\frac{|\mathbb{E}[\text{new}] - \mathbb{E}[\text{old}]|}{|\mathbb{E}[\text{old}] - \mathbb{E}[\text{placebo}]|} \leq 0.2.$$

Denote the parameter of interest by

$$\theta = \frac{\mathbb{E}[\text{new}] - \mathbb{E}[\text{old}]}{\mathbb{E}[\text{old}] - \mathbb{E}[\text{placebo}]} = \frac{\mathbb{E}[Y]}{\mathbb{E}[X]},$$

where $Y = new - old$ and $X = old - placebo$.

(a) *Compute the plug-in estimate of θ.*
(b) *Write a function which takes the number of bootstrap replicates n as input and outputs the bootstrap replicates $\hat{\theta}_i^*$.*
(c) *Compute the number of distinct bootstrap replicates and plot the distribution of $n = 400$ bootstrap replications.*
(d) *Compute the bootstrap estimates of bias and standard error. Repeat the calculation for a second batch of bootstrap replications; are the results very different?*
(e) *Repeat the above calculations using the built-in function in Python. See* https://docs.scipy.org/doc/scipy-1.16.0/reference/generated/scipy.stats.bootstrap.html.

Exercise 4 (Parametric Bootstrap). *The lifetime Y in hours of some type of electronic component is known to have an exponential distribution with parameter λ. An engineer is interested in the fraction of components that fail in less than 24 hours. Thus we wish to estimate*

$$\theta = F_\lambda(24).$$

A series of $N = 100$ observations gave $\bar{y} = 50$ hours, and 35 components had lifetime less than a day.

(a) *Compute the plug-in estimate of θ.*
(b) *Apply the parametric bootstrap to estimate bias and variance.*
(c) *Compare the plug-in estimate and the bootstrap estimate of variance with the exact values obtained in the lectures.*

Exercise 5 (Bootstrap confidence intervals). *For a sample of $N = 15$ American law schools, Efron and Tibshirani (1993) report the following average LSAT scores:*

School	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Score	576	635	558	578	666	580	555	661	651	605	653	575	545	572	594

Suppose we are interested in estimating the standard deviation of LSAT scores in the population.

(a) *Compute the plug-in estimate of the standard deviation.*
(b) *Implement the nonparametric bootstrap with* $n = 400$, $n = 1000$, *and* $n = 5000$ *replications, and find bootstrap estimates of bias and standard error for each case.*
(c) *Compute the BCa 95% confidence interval and the percentile 95% confidence interval.*

7.3 Transformations of Random Variables & Rejection Sampling: Recap Exercises

Exercise 1

Propose a way to sample from a Gamma distribution with parameters α, β with $\alpha \in \mathbb{N}$.

Exercise 2

Show that if $Z_1, Z_2 \sim \mathcal{N}(0,1)$, then

$$R^2 = Z_1^2 + Z_2^2 \sim \text{Exp}(1/2), \qquad \theta = \arctan \frac{Z_2}{Z_1} \sim \text{Unif}(0, 2\pi).$$

Exercise 3

Let $Y \sim \text{Exp}(\lambda)$ and let $a > 0$. Consider

$$X = Y \,|\, Y \geq a.$$

Calculate $F_X(x)$ and $F_X^{-1}(u)$, and describe an algorithm to simulate X from $U \sim \mathcal{U}_{[0,1]}$.

Let $a < b$. Show that

$$X = F_Y^{-1}\big(F_Y(a)(1-U) + F_Y(b)U\big)$$

has the law of $Y \,|\, a \leq Y \leq b$.

Consider also the rejection algorithm:

(a) simulate $Y = y \sim \text{Exp}(\lambda)$;
(b) if $Y > a$, return $X = y$, otherwise restart.

Show that this is a rejection algorithm by identifying proposal, target, and optimal M. Calculate the expected number of trials to first acceptance. Why is inversion preferable when $a \gg 1/\lambda$?

Exercise 4

Let $X \sim \text{Exp}(1)$ independent of $U \sim \mathcal{U}(0,1)$.

(a) Show that the pdf of $Z := X + U$ is

$$f_Z(t) = g_1(t) + g_2(t),$$

with

$$g_1(t) = (1 - e^{-t})\,\mathbf{1}_{(0,1)}(t), \qquad g_2(t) = (e-1)e^{-t}\,\mathbf{1}_{(1,+\infty)}(t).$$

(b) Explain how to simulate from X and then from Z using the transformation method.

Exercise 5

Let

$$X \sim f_X(x) = 12x(1-x)^2, \qquad 0 < x < 1.$$

Prove that

$$Y = \frac{X}{1-X}$$

has density

$$f_Y(y) = 12\,\frac{y}{(1+y)^5}, \qquad y > 0.$$

Exercise 6

1. Let $X_1 \sim \text{Gamma}(a,1)$ and $X_2 \sim \text{Gamma}(b,1)$ be independent. Show that

$$R = \frac{X_1}{X_1 + X_2}, \qquad S = X_1 + X_2$$

are independent, with

$$R \sim \text{Beta}(a,b), \qquad S \sim \text{Gamma}(a+b, 1).$$

2. Let $U \sim \mathcal{U}_{[0,1]}$ and $a > 0$. Show that
$$X = U^{1/a} \sim \text{Beta}(a, 1).$$
3. Let $U, V \sim \mathcal{U}_{[0,1]}$ be independent, $a \in (0,1)$, and define
$$Y = U^{1/a}, \qquad Z = V^{1/(1-a)}.$$
Calculate
$$\mathbb{P}\left(\frac{Y}{Y+Z} \leq t,\ Y+Z \leq 1\right)$$
for $t \in (0,1)$ and deduce that the conditional distribution of
$$W = \frac{Y}{Y+Z} \,\Big|\, Y+Z \leq 1$$
is $\text{Beta}(a, 1-a)$.
4. In the setting above, show that the conditional distribution of TW given $Y + Z \leq 1$, where $T \sim \text{Exp}(1)$ is independent, is $\text{Gamma}(a, 1)$.
5. The following algorithm is given for $a \in (0,1)$:
   (a) simulate $U, V \sim \mathcal{U}_{[0,1]}$ independently;
   (b) set $Y = U^{1/a}$ and $Z = V^{1/(1-a)}$;
   (c) if $Y + Z \leq 1$, go on, otherwise restart;
   (d) simulate $A \sim \mathcal{U}_{[0,1]}$ independently and set $T = -\log(A)$;
   (e) return $TY/(Y+Z)$.
   Explain what this procedure is doing and why it is relevant for simulation.
6. Based on item 1, explain how to generate a $\text{Beta}(a, b)$ random variable from uniforms, for any $a > 0$ and $b > 0$.

Exercise 7

Design a rejection sampler for the standard normal density
$$p(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$
using the Laplace proposal
$$q_\alpha(x) = \frac{\alpha}{2} e^{-\alpha|x|}, \qquad \alpha > 0.$$

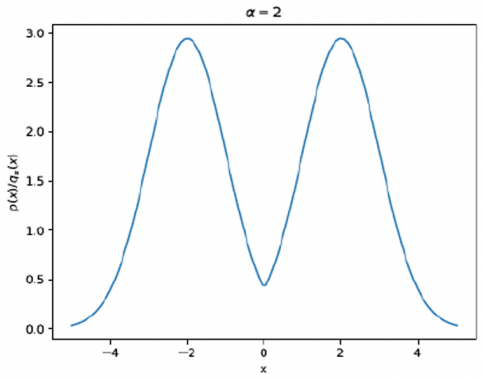


Figure 1: Ratio plot for $\alpha = 2$.

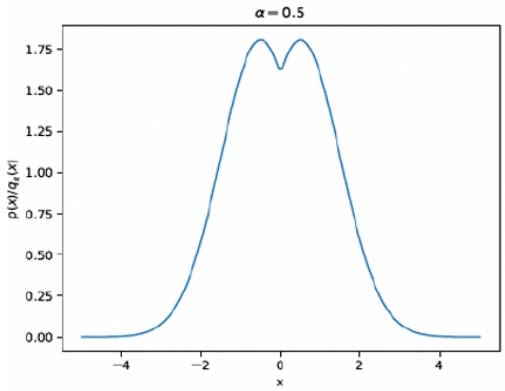


Figure 2: Ratio plot for $\alpha = 0.5$.

(a) Based on Figures 1 and 2, explain how to construct rejection samplers generating samples from p using samples from q_α for $\alpha = 2$ and $\alpha = 0.5$, and write the corresponding pseudo-code. In each case, what is approximately the optimal acceptance rate? Which value of α should be preferred, and why?
(b) Set $\alpha = 2$. Is it possible to build a rejection sampler for q_2 using the normal distribution as proposal?

Exercise 8

The aim is to design a rejection sampler for the normal distribution using a Cauchy proposal.

(a) Given a target density p and strictly positive proposal density q on $\mathbb{R}$, assume there exists $M \geq 1$ such that

$$\frac{p(x)}{q(x)} \leq M$$

for all $x \in \mathbb{R}$. Write pseudo-code to generate samples from p using rejection sampling.

(b) Determine the optimal bound M_* and explain in which sense it is optimal. Discuss the consequences of taking $M_1 < M_*$ or $M_2 > M_*$.

7.4 Monte Carlo Integration: Recap Exercises

Exercise 1

Propose a Monte Carlo estimator for each of the following integrals:

- $\int_1^4 \left(\sqrt{x} + \frac{1}{2\sqrt{x}}\right) dx$
- $\int_0^1 e^x \, dx$
- $\int_5^6 (1-x^2)^{3/2} \, dx$
- $\int_{-2}^2 e^{x+x^2} \, dx$
- $\int_{-\infty}^{+\infty} e^{-x^2} \, dx$

Compute the variance of the Monte Carlo estimator whenever possible.

Exercise 2

Let X be a t-distribution with 3 degrees of freedom. Propose two Monte Carlo estimators for

$$\mathbb{P}(X > 5),$$

and discuss which one is better.

Exercise 3

Consider the joint distribution

$$\pi(x,y,z) = \frac{e^{-x}}{(1+e^{-x})^2} \cdot \frac{e^{-\frac{(y-x)^2}{2}}}{\sqrt{2\pi}} \cdot \frac{e^{-\frac{(z-(y+x))^2}{2}}}{\sqrt{2\pi}}, \qquad (x,y,z) \in \mathbb{R}^3.$$

(a) Let

$$f(x) = \frac{e^{-x}}{(1+e^{-x})^2}.$$

Show that

$$F_X(x) = \frac{1}{1+e^{-x}}$$

and propose a method to sample from X.

(b) Explain how to sample Y given $X = x$. Write the corresponding pseudo-code with a uniform input.

(c) Propose a composition method to sample from the joint distribution.

(d) Propose a Monte Carlo estimator for

$$\iiint \pi(x,y,z)\varphi(x,y,z) \, dxdydz, \qquad \varphi(x,y,z) = z,$$

and compute its variance, knowing that $\mathbb{E}[X] = 0$ and $\mathbb{E}[X^2] = \pi/3$.

Exercise 4

Useful fact: the density of a Gamma(α, β) random variable is

$$f(z) = \frac{\beta^\alpha}{\Gamma(\alpha)} z^{\alpha-1} e^{-\beta z}, \qquad z > 0.$$

Let X have density

$$f(x) = xe^{-x^2/2}, \qquad x > 0.$$

(a) Show that

$$F(x) = 1 - e^{-x^2/2}, \qquad x > 0,$$

and explain how to simulate from F.

(b) Show that

$$\mathbb{E}[X^n] = 2^{n/2}\Gamma\left(1 + \frac{n}{2}\right).$$

(c) Write the MC estimator of

$$\theta = \int_0^\infty (1+x) x e^{-x^2/2}\, dx$$

using $X_1, \ldots, X_n \overset{iid}{\sim} f$, and compute its variance.

Exercise 5

Let

$$X \sim f_X(x) = 12x(1-x)^2, \qquad 0 < x < 1.$$

(a) Prove that

$$Y = \frac{X}{1-X}$$

has density

$$f_Y(y) = 12\, \frac{y}{(1+y)^5}, \qquad y > 0.$$

(b) Write the MC estimator of

$$\theta = \int_0^\infty \sin(y) e^{-y}\, \frac{y}{(1+y)^5}\, dy$$

using $Y_1, \ldots, Y_n \overset{iid}{\sim} Y$.

Exercise 6

Consider

$$I = 4 \int_0^1 \sqrt{1-x^2}\, dx.$$

(a) Propose a Monte Carlo estimator. Identify the target distribution π and the test function φ.
(b) Knowing that $I = \pi \approx 3.14$, compute the variance of the estimator.

7.5 Multivariate Distributions and Composition: Recap Exercises

Exercise 1

Consider the joint distribution

$$\pi(x,y,z) = \frac{e^{-x}}{(1+e^{-x})^2}\frac{e^{-\frac{(y-x)^2}{2}}}{\sqrt{2\pi}}\frac{e^{-\frac{(z-(y+x))^2}{2}}}{\sqrt{2\pi}}, \qquad (x,y,z)\in\mathbb{R}^3.$$

(a) Let

$$f(x) = \frac{e^{-x}}{(1+e^{-x})^2}.$$

Show that

$$F_X(x) = \frac{1}{1+e^{-x}}$$

and propose a method to sample from X.

(b) Explain how to sample from Y given $X = x$. Write the corresponding pseudo-code with a uniform input.

(c) Propose a composition method to sample from this joint distribution.

Exercise 2

Consider

$$X \sim p_k = \frac{1}{2k!}e^{-2}\left(4^k e^{-2} + \frac{10^k}{2}e^{-8} + 2^{k-1}\right), \qquad k = 0,1,2,\ldots$$

Is this a mixture? If yes, is it continuous or discrete? Explain how to generate from it using `if-else` clauses.

Exercise 3

Let $X \sim \text{Exp}(1)$ independent of $U \sim U(0,1)$.

(a) Show that the pdf of

$$Z := X + U$$

is

$$f_Z(t) = g_1(t) + g_2(t),$$

with

$$g_1(t) = (1-e^{-t})\,\mathbf{1}_{(0,1)}(t), \qquad g_2(t) = (e-1)e^{-t}\mathbf{1}_{(1,+\infty)}(t).$$

(b) Explain how to simulate from X and then from Z using the transformation method.

(c) Compute the integrals of g_1 and g_2, explain why they are not densities, and rewrite f_Z as

$$f_Z(t) = p_1 f_1(t) + p_2 f_2(t),$$

where f_1 and f_2 are densities.

(d) Explain how to use the composition method to simulate from f_Z. Would you use the method for discrete mixtures or for continuous mixtures? Why?

Exercise 4

Let $X_1 \sim \text{Gamma}(2,4)$ and $X_2 \sim \text{Gamma}(3,6)$ be independent. Generate samples from

$$F_X(x) = \frac{1}{3}F_{X_1}(x) + \frac{2}{3}F_{X_2}(x).$$

Exercise 5

The Student's t distribution with ν degrees of freedom can be written as a continuous mixture:

$$X \mid Y = y \sim \mathcal{N}(0, \nu/y), \qquad Y \sim \chi^2_\nu.$$

Write an algorithm that simulates T_ν from this mixture and check the result with two plots.

7.6 Practical 2: Generating Random Variables

Practical 2: Generating Random Variables

Exercise 1 (Pseudo random numbers). *We will implement a general LCG.*

(a) Write code that produces a sequence $(x_1, \dots, x_n)$ based on

$$x_{n+1} = (ax_n + c) \bmod m,$$

where $a, c, m \in \mathbb{N}$ and $x_0 \in \mathbb{N}$ are given.

(b) Although $(x_1, \dots, x_n)$ is deterministic, the rescaled sequence $(x_1, \dots, x_n)/m$ should resemble uniform random numbers in $[0, 1]$. To test this:

i) plot a histogram of $(x_1, \dots, x_n)$;

ii) plot the lag-k correlation coefficients

$$r_k = \frac{\sum_{i=1}^{n-k}(x_i - \bar{x})(x_{i+k} - \bar{x})}{\sqrt{\sum_{i=1}^{n-k}(x_i - \bar{x})^2 \sum_{i=1}^{n-k}(x_{i+k} - \bar{x})^2}}, \qquad k = 1, \dots, K.$$

Perform these tests for

$$a_1 = 1664525, \qquad c_1 = 1013904223, \qquad m_1 = 2^{32},$$

and

$$a_2 = 5, \qquad c_2 = 1, \qquad m_2 = 16,$$

with $n = 10^5$ and other lengths. Compare with Python's built-in RNG and discuss.

Exercise 2 (RANDU). *The LCG RANDU is*

$$x_i = 65539 x_{i-1} \bmod 2^{31}, \qquad u_i = x_i / 2^{31} \in (0, 1).$$

(a) Show that (u_i, u_{i-1}, u_{i-2}) always lies on the union of 15 hyperplanes.

(b) Implement RANDU, generate $2 \cdot 10^4$ points, and make a scatter plot of u_i versus u_{i-1} for indices with $0.5 \le u_{i-1} \le 0.51$. Discuss.

(c) Make a 3D plot of (u_i, u_{i-1}, u_{i-2}).

Exercise 3 (Inverse transform). *This exercise is about transforming uniforms into samples from other distributions.*

(a) Let $U \sim \text{Unif}(0, 1)$ and let X be a real-valued random variable with invertible cdf F_X. Show that $F_X^{-1}(U)$ has the same distribution as X.

(b) An exponential random variable with parameter λ has density

$$\pi(x) = \lambda e^{-\lambda x} \mathbf{1}_{[0,+\infty)}(x).$$

Compute F_X and F_X^{-1}.

(c) Using (a) and (b), devise and implement an algorithm producing exponential samples. Check the histogram and sample mean.

(d) Define the pmf on $\{1, 2, 3, 4, 5\}$ by

$$\mathbf{p} = (0.2, 0.3, 0.2, 0.1, 0.2).$$

Plot its PMF and CDF.

(e) Implement the inverse-cdf method for this pmf and compare the simulated frequencies with the target PMF.

Exercise 4 (Box–Müller). *We will check the Box–Müller algorithm discussed in the lectures.*

(a) For $U_1, U_2 \sim \text{Unif}(0, 1)$ and

$$Z_1 = \sqrt{-2 \log U_1} \cos(2\pi U_2), \tag{1}$$

$$Z_2 = \sqrt{-2 \log U_1} \sin(2\pi U_2), \tag{2}$$

show that $Z_1, Z_2 \sim \mathcal{N}(0, 1)$.

(b) Implement the Box–Müller algorithm, plot a histogram, compute mean and variance, and inspect Gaussianity with a Q–Q plot.

Exercise 5 (Geometric random variable). *Consider a geometric random variable with parameter p.*

(a) Write a function to sample from $\text{Geom}(p)$ using the inverse transform method.

(b) Propose a transformation method to sample from $\text{Geom}(p)$.

Exercise 6 (Sampling from a circle). *Consider the transformation*

$$r \sim \text{Unif}(0,1), \qquad \theta \sim \text{Unif}(-\pi, \pi),$$

and

$$x_1 = r^\alpha \cos\theta, \qquad x_2 = r^\alpha \sin\theta.$$

Implement it for $\alpha = 1/2, 1, 2$ and comment on the results.

7.7 Practical 3: Composition & Rejection Sampling

Practical 3: Composition & Rejection Sampling

Exercise 1 (Joint, Conditional and Marginal Distributions). *Consider*

$$\pi(x, y) = \mathcal{N}(x; 0, 1)\mathcal{N}(y; x, 2).$$

(a) *Implement a sampling algorithm for* π.
(b) *Show that*

$$\pi(y) = \int \pi(x, y)\, dx = \mathcal{N}(y; 0, 3).$$

(c) *Compare* $N = 1000$ *samples from* $\pi(y)$ *with samples obtained by*
- *sampling* $X \sim \mathcal{N}(0, 1)$;
- *then sampling* $Y \mid X = x \sim \mathcal{N}(x, 2)$.

What do you observe?

Exercise 2 (Gaussian Mixture). *Consider a simplified Gaussian mixture model with K Gaussian components* $\mathcal{N}(\mu_k, 1)$*, each with weight* $1/K$*. Write down the density of* π*. Simulate a dataset from this model, for example with* $N = 100$*,* $K = 2$*, and two well-separated values of* μ_1, μ_2*.*

Exercise 3 (Improved Box–Müller). *The improved Box–Müller algorithm samples two unit Gaussians as follows:*

- *sample* $U_1, U_2 \sim \text{Unif}(-1, 1)$ *until* $U_1^2 + U_2^2 \leq 1$;
- *return*

$$X = U_1\sqrt{-2(\log S)/S}, \qquad Y = U_2\sqrt{-2(\log S)/S}, \qquad S = U_1^2 + U_2^2.$$

(a) *Show that this is a rejection algorithm by identifying target, proposal, and M. Compute the acceptance rate.*
(b) *Implement the algorithm and verify the target distribution.*
(c) *Compare its performance with the standard Box–Müller algorithm.*

Exercise 4 (Sampling under the box). *Consider*

$$\pi(x) = \text{Beta}(x; \alpha, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1 - x)^{\beta-1}.$$

(a) *Design a sampler that samples uniformly under the curve* $\pi(x) = \text{Beta}(x; 2, 2)$.
(b) *Instead of the box, take*

$$q(x) = \mathcal{N}(x; 0.5, 0.25)$$

as proposal.

Exercise 5 (Laplace & Rejection).

(a) *Derive and implement a random generator for the Laplace distribution*

$$\pi(x) = \frac{1}{2}e^{-|x|}.$$

(b) *Propose a rejection sampler for* $\mathcal{N}(0, 1)$ *using the Laplace distribution as proposal. Implement it and test it.*
(c) *Give the acceptance rate of the algorithm. Find a way to simplify the acceptance condition.*
(d) *Is it possible to do the reverse, i.e. sample from the Laplace distribution using a rejection sampler based on the normal distribution?*